

AI with Durga Sir
PROMPT ENGINEERING

DURGA
Software Solutions®

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Prompt Engineering

Prompt is a way to communicate with the AI Model
Prompt is an instruction given to AI and it is important to communicate clearly and get the exp ected output
Clear prompt gives clear output

"Write a birthday message for my friend"
- what to do ==> Write a message
- To whom ==> my friend
- For what ==> birthday

AI understands
AI Writes

Bad vs Good prompts

Bad prompt
- unclear
- too open
- missing details
- AI must guess
- Tell about job ==> Bad prompt
- AI provided very generic answer

Good Prompt
- clear
- specific
- tell audience
- tells style
- Explain about IT jobs **for** freshers **in** simple english
  - Topic ==> IT jobs
  - Audience ==> Freshers
  - Language ==> Simple English

Bad prompt creates confusion
Good prompt creates direction

If the prompt **is** unclear:
    AI gives general answer
else:
    AI gives specific answer

Examples

Example-1

Explain python
  Which level
  which part
  sort **or** long
  example needed **or not**
Bad prompt

Explain python **for** a 10th **class** student.
Use very simple words
Give one daily life example
  Audience **is** clear
  Language **is** clear
  output style **is** clear
  example **is** required
Good Prompt

Example-2

Write an email?
  to whom?
  purpose?
  tone?
Bad Prompt

Write a professional email to my manager
  Topic : Request leaver **for** 2 days
  Reason : Fever
  keep it polite **and** short
Good Prompt

Example-3

Interview Preparation
Give Interview questions
  Which subject?
  which level?
  how many?

Bad Prompt

Ask me 5 prompt engineering interview questions **for** fresher.
Wait **for** my answer after each question

Good Prompt

How to write good prompt

A good prompt should be specific **and** clear
It tells what I want **for** whom

Note:
Bad prompt gives generaal answer
Good prompt gives useful **and** specific answer

Prompt clarity rules

Rule-1 : Specific Vs Vague

Vague means
not clear
too general
missing details
eg : explain java

Specici Prompt
clear
focussed
exact need mentioned
Eg: Explain java basics **for** freshers **in** simple Indian English **with** one example
Topic : Java
Audience : Freshers
Language : simple english
Style : **with** example

Rule-2 Complete Vs Incomplete

Incomplete means
half information
missing information
missing goal
Eg: Write resume

Complete Prompt means
All needed information **is** given
Eg: Write resume **for** python fresher **with** no experience
Focus on skills **and** projects
Use simple professional language

if prompt **is** vague **or** incomplete(bad):
AI guesses
output becomes weak
else:
AI understands clearly
output becomes strong



Prompt Formula

RTC CO

R ➔ Role :: Direction
T ➔ Task :: Work
C ➔ Context :: clarity
C ➔ Constraints :: Control
O ➔ Output :: Shape

Role : Who should AI act like

What role should AI play

Eg:

Act as Teacher
Act as HR
Act as Interviewer
Act as Manager
Act as Mentor

Role gives direction to the AI
Without role, AT gives general answer

Tast : What should AI do

What work you want AI to do

Eg:

Explain python
Write email
Create Resume
Ask Interview question

Task gives job to AI

Context : For Whom/background

Background information

Eg:

For School Student
For Fresher
For Working Professional
For Business owner

Context tells AI level **and** situation

Constraints : Rules and limits

means restrictions **or** conditions

Eg:

use simple english
do **not** use technical words
answer **in** 5 popints
keep **in** short

Constraints control AI behaviour

Output : How answer should look like
How do you want the answer
Eg:
- Bullet points
- Table
- Steps
- short paragraph

Format gives shape to the output

Set Role
Set Task
Set Context
Set Constraints
Set Output format

Prompt Examples

Example-1 : Student Learning
Act as a school teacher
Explain Python loops
Audience : beginner student
Use simple Indian English
Give one daily life example

Example-2 : Interview practice
Act as a Interview person
Ask me 5 Java interview questions
Level : fresher
Wait **for** my answer after each question

Example-3 : Exam Expert
Act as a Exam expert
Provide me 3 MCQs on Prompt Engineering
Level : Fresher
Ask questions one by one
Wait **for** my answer **for** each question
Please validate my answer
Provide correct answer **with** explanation **with** 3 bullet points

Example-4 : Official Email
Act as Professional assistant
Write email to manager
context : requesting 2 days leave
Tone : olite
keep it short

Example-5 : Resume Building
Act as HR
Create a resume summary
context : Python Fresher
Focus on skills **and** projects
Use professional language

Example-6 : Coding Help
Act as Programming mentor
Expalin **if-else in** python
Audience : beginner
Explain step by step

Example-7 : Business Marketing
Act as Marketing Expert
Write instagram caption
context : small mobile shop
Tone : friendly
Add call to action

Example-8 : Office Task Planning
Act as Project Manager
Create Daily task list
Context : Software Developer
Format : Bullet points

Example-9 : Study Time Table
Act as Study mentor
create study time table
context : college student
constraints : 2 hours per day

Example-10 : Company Owner
Act as company owner
Create action plan to start an AI based software company start from scratch
context : Zero technical Knowledge
First month he can spend only 5 mins per day, add 5 more minutes **for** every next month
Provide action plan week by week
For every week wait **for** my confirmation

Example-11 : AI Trainer
Act as Experienced AI Trainer
Provide action 30-days action plan to learn Generative AI from basic to expert level day by day.
After every day wait for my confirmation
Context : Beginner
Constraints :per day time availability is just 30 mins
Output : Table format

Example-12 : Explanation control
Act as Teacher
Explain SQL Joins
Audience : non-technical people
Do **not** use technical terms

Example-13 : Personal
Act as Experienced Lover Boy
Explain top 3 ways to impress a girl
context : fresher
constraints : without spending much time **and** money
output : bullet format

Example-14 : Notes Generator
Act as Exam coach
Summarize prompt engineering
Audience : Beginner
Output : 5 Bullet points

Answer :

Prompt Engineering – 5 Key Points

Prompt engineering means giving clear instructions to AI to get correct **and** useful output.
A good prompt explains what to do, how to do, **and** expected result.
Clear prompts reduce wrong answers **and** confusion from AI.
Changing words **in** a prompt can change the output completely.
Prompt engineering helps **in** writing content, code, emails, answers, **and** ideas faster.

Question :
How to do you write good prompt

Answer :
We use Role, Task, Context, Constraints **and** Output format
Give one example

Tone Control

Tone means
- Style of speaking
- Mood of speaking
- How the answer should "feel"

Common Tones
- Friendly
- Professional
- Funny
- Strict

Friendly

soft
simple
caring
like a helpful friend

Where we can use :
- Beginner
- School Students
- People who fear technical topics

Professional

Office stule
Formal
Clean
Polite
No jokes
Where we can use :
- Emails
- Reports
- Client Messages
- LinkedIn Posts

Funny

Jokes
Light Styles
Fun examples
Makes boring topic interesting
Where we can use:
- Reels
- Youtube shorts
- Marketing Content

Strict
Direct
Serious
No extra words
Like strict teacher **or** coach
Where we can use:
  Interview preparation
  Quick Revision
  Strong discipline style learning

if tone = friendly:
    answer becomes soft **and** simnple
elif tone = professional:
    answer becomes formal **and** office style
elif tone = funny:
    answer includes jokes **and** fun examples
elif tone = strict:
    anser include direct **and** serious
else:
    explore other tones

Tone Control Examples

Friendly
Explain prompt engineering **in** friendly way **for** a beginner
use simple english
give one daily life example

Professional
Explain prompt engineering **in** professional way
write like an office trainer
use clean bullet points

Funny
Explain prompt engineering **in** funny way
use one chai shop example
make it entertaining

Strict
Explain prompt engineering **in** strictly **for** interview
no jokes
Answer **in** 5 bullet points only

Question :
How to control tone **in** AI output

Answer :
By clearly mentioning tone **in** the prompt like "friendly" **or** "professional" **or** "funny" **or** " strict" **and** also adding constraints like no jokes **or** formal language

Length Control

We can tell AI how big the answer should be
you decide:
- short
- medium
- detailed/long

Short

used **for**
- Quick revision
- Interviews
- Summaries

Explain Java **in** 2 lines

Medium

used **for**
- Learning
- Notes
- Normal Explanation

Explain Java **in** 200 words

Long/detailed

used **for**
- Deep understanding
- Teaching
- Documentation

Explain Java **in** detail **with** examples

Note

If we don't mention length, AI will guess
Guessing can give:
- too long
- **or** too short

Which may **not** fulfill your requirement.
Always control length

If length **is** defined:
    AI controls answer size
else:
    AI gives random size

Explain Python **in** around 100 words
audience : beginner
Give one daily life example

Explain Python **in** detail
audience : college student
explain **with** examples

Step-by-Step Prompting

You tell AI to explain **in** ordered steps
this avoids confusion

Example-1: Step-by-Step

Explain SQL joins step by step
Step1 : What **is** join
Step2 : Why Join required
Step3 : Simple Example

Example-2: Step-by-Step + Length control

Explain **for** loop **in** python
audience : beginner
Explain step by step
each step **in** 2 lines only

Example-3 : Office Task

Create daily work plan step by step
Audience : software employee
Keep each step one line only

Example-4 : Durga sir Prompt engineering class plan

Create daily Prompt engineering **class** of 5 mins plan step by step
Audience : AI victims
Keep each step 2 lines only

If step-by-step **is** mentioned:
    AI Explains **in** order
else:
    AI mixes explanation

Question:
How do you avoid long AI answers

Answer:
By Clearly mentioning length **and** using step-by-step prompts

Daily practice : Consistency

If I practice daily
  confidence increases
If confidence increases
  fear reduces
If fear reduces
  usage increases
If usage increases
  mastery happens **and** happily you can settle **in** life **in** this AI era

Mastering AI Format Control : Messy Text to Professional Output

Format Control ==> Telling AI how the answer should be presented.

Without Format Control ==> AI answer looks like raw text
With Format Control ==> AI answer looks Professional

Format-1 : Bullet Points

Short
Clean
Easy to read

Eg:
Explain prompt engineering advantages **in** 5 bullet points
Every point **in** a single line

Usage :
  Notes
  Interviews
  Presentations

Format-2 : Table

Comparision
clarity **and** easy to understand
Structured view

Eg:
Explain Python Vs Java **in** a table **with** columns
(Language | Use | Difficulty)
Usage:
- Comparison
- Reports
- Clarity

Format-3 : Steps
Ordered Explanation
No confusion

Eg:
Explain how to prepare **for** prompt engineer interview **in** step by stop format

Usage:
- Learning
- Process explanation
- Planning

Format-4 : checklist
yes/no style
to-do list/ priority list
completion focus

Eg:
Create prompt engineer interview preparation checklist **for** fresher

Used:
- Daily tasks
- Tracking work
- Discipline

Format-5 : Story
Story format means
- flow
- emotion
- easy understanding

Eg:
Explain importance of savings using a short story

Usage
- Motivation
- Reels
- Teaching Beginners

All formats

Explain importance of exercise **in** 3 bullet points
Explain importance of exercise **in** a table **with** 2 columns
(Benefit | Effect)
Explain importance of exercise **in** Step by step
Create checklist **for** exercise activity
Explain importance of exercise **in** a story

Question :
How do you control AI Output format

Answer:
By clearly mentioning the required format like bullet points, table **or** steps **in** the prompt

Homework

Benefits of Learning AI

Ask AI to give
- Bullet points
- Table
- Checkpoints

Notes

Format control decides how answer looks like
Same content can be shown **in** many formats
Clear format instruction gives clean output

Mastering Structured Output : Turning AI answerts into professional Systems

Structuredc Output ==> Telling AI to always follow the same structure
Stucture means
- Fixed Heading
- Fixed Order
- Fixed Table Columns

Example : With fixed headings

Explain SQL using fixed headings:
1. What **is** it
2. Why it **is** needed
3. Simple Example
4. Interview point

Explain Prompt Engineering using fixed headings:
1. What **is** it
2. Why it **is** needed
3. Simple Example
4. Interview point

Where fixed heading are used
Notes Generation
Teaching Material
PDFs
Reels Scripts
Reusable Prompts

Example : Strict Table columns
Strict table means:
Column names are fixed
Order fixed
No extra columns

Explain Python data types **in** a table
Columns must be exactly:
| Data Type | Example | Use |

Where Strict Table columns are used
Excel
Google Sheets
Reports
Automation

Stuctured Vs Unstructured
Unstructured

Eg:
Explain python list

With fixed headings(Structured)
Eg-1:
Explain python list using fixed headings:
Definition
Key Features
Simple Example
Interview tip

Eg-2:
Explain python list using table **with** columns:
| Feature | Description |

Question:
Why structur output **is** important **in** AI prompt

Structured output makes AI responses predictable, reusable **and** professional, especially **for** re ports **and** automation.

Notes

Structured output means fixed format every time
Fixed headings bring clarity **and** consistency
Strict tables make output reusable **and** professional

"Only output this" control

By default AI talks extra.
We should stop it. For this we requied to use "Only output this"

It telling to AI:
give only required format
do **not** add explanation
do **not** add greeting
do **not** summary

Useful controls :
Reply only **with** output
Do **not** add explanation
No greetings
No summary

Example-1

Explain benefits of learning AI **in** 5 bullet points

Explain benefits of learning AI.
Output format : 5 bullet points only
Do **not** add extra text

Example-2

Explain the benefits of Learning AI
output format : table
columns must be exactly :
| Benefit | Explanation |
Do **not** add extra text

Example-3

Convert benefits of Learning AI into JSON
Reply only **in** valid JSON
Do **not** add any extra text

Example-4

Give interview preparation steps **in** bullet points only
Do **not** add any extra text

Give interview preparation steps **in** table **with** columns
| Step | Action |
Do **not** add any extra text

Convert interview preparation steps into JSON
Reply only **in** JSON
Do **not** add any extra text

Notes
Only Output control removes extra AI talk

Question :
Why only output control **is** important**

Answer :
Because tools **and** automation need clean output without extra text.
Extra explanations **break** integrations

Notes Generator Prompt
Reusable prompt
Real world skill
Prompt Template

Step-1: Define Role
Act as an experienced Teacher

Step-2: Defint Task
Generate Student notes **for** the given topic : {TOPIC}

Step-3: Define Structure(Fixed Headings)
1. Definition
2. Key points
3. Simple Example
4. Interview Tip

Step-4: Define Format rules
Bullet points only
shot **and** clear lines

Step-5: Output control lines
Do **not** add introduction
Do **not** add conclusion
Do **not** add extra explanation

Final Prompt
Act as an experienced Teacher
Generate Student notes **for** the given topic : {TOPIC}
Follow this fixed structure Strictly:

1. Definition
2. Key points
3. Simple Example
4. Interview Tip

Rules:
Use simple Indian English
Do **not** add introduction
Do **not** add conclusion
Do **not** add extra explanation

Output format:
Use Bullet points only
Keep Each point short

Run the above prompt by providing TOPIC
TOPIC ==> Python list
TOPIC ==> Quantum computing
TOPIC ==> Prompt Engineering
TOPIC ==> Fine Tuning
TOPIC ==> Quantization
TOPIC ==> RAG

Act as an experienced Teacher
Generate Student notes **for** the given topic : {TOPIC}
Context : This notes required **for** non-technical students
Follow this fixed structure Strictly:

1. Definition
2. Key points
3. Simple Example
4. Interview Tip

Rules:
Use simple Indian English
Do **not** add introduction
Do **not** add conclusion
Do **not** add extra explanation

Output format:
Use Bullet points only
Keep Each point short

Question :
what **is** reusable prompt

Answer :
A reusable prompt **is** fixed instruction template that gives consistent output **for** different inputs

Encoding Instructions inside Prompt

Conclusion-1 : AI follows written instructions, not your intention

Eg: Exam instructions in the examination paper will be at the first page

AI cannot read your mind
I follows what you write
Missing instruction = AI guesses

Act as befinner-friendly teacher
Explain Python loops
in simple english

Conclusion-2 : Instruction order decided output quality

Early instructions set direction
Role + Task should come first
Conflicting instructions confuse AI

Act as Interview Coach
Explain SQL Joins
Output **in** 5 bullet points only

Conclusion-3 : Missing details create wrong output

Missing audience = Generic answer
Missing format = unwanted paragraph
Missing control = too many outputs

Act as Software Trainer
Explain java **for** 10th **class** student.
Give exactly 3 bullet points.
Do **not** add extra text

Why Small words change big output

Bring vegetables
Bring only onions
Bring only onions, do **not** bring anything **else**
Bring exactly 1 kg onions.

Conclusion-1 : Small control words change big output
only : restricts output
exactly: fixes the count
do **not** sops extra content

Eg:
Explain AI.
Give only bullet points

Conclusion-2 : Precision prompting gives control
Clear rules reduce guessing
Output becones consistent
Less confusion **for** AI

Eg:
Act as AI Trainer.
Explain prompt engineering **for** beginners.
Give exactly 5 points.
Do **not** add explanation.

Conclusion-3 : Casual prompts create random answers
explain this ==> **is** unclear
AI takes freedom **and** varies output
precision ==> engineering mindset
casual ==> just chating

Eg:
Act as Software Trainer
Explain SQL **for** beginners
Use Simple English
Output must be a table

Ambiguity Problems - Why LLM guesses

Conclusion-1 : Ambiguity means unclear instruction
Multiple meanings possible
Missing details create confusion
AI must reply, so it guesses

Eg:
Explain cloud (Ambiguous)

Conclusion-2 : Clear prompts remove randomness
Add audience
Add purpose
Add format + constraints

Eg:
Act as cloud expert
Explain cloud computing **for** non-technical students
Use simple english
Give 1 daily life example

Eg:
Act as my girl friend
Please provide some jokes as I am feeling bore
Use simple English
Short jokes only

Conclusion-3 : Under specified prompts cause mistakes

Missing level, format, goal
AI chooses its own assumptions
Output may **not** match your requirement

Eg:
Act as experienced technical person
create a resume **for** python fresher
Use 5 bullet points
Add 2 projects

Zero-shot, One-shot and Few-shot Prompts

Zero-shot Prompting

Direct asking
No examples
This **is** the most basic **and** mosed used prompting style

Eg:
Explain Python

Where you use Zero-shot prompting

AI looks at your words
AI uses past training data
AI predicts best possible answer

When Zero-shot prompt is enough

You want quick help
You are exploring Topic
Format does **not** matter

Zero-shot = Speed + Simplicity

Limitations of Zero-shot Prompting

Ouput variation
No style control
Not reusable

Note:
Zero-shot **is** good **for** Learning **not** good **for** production

Question :
What **is** Zero-shot prompting

Answer:
Zero-shot prompting means asking AI directly without givein any example.
Output may vary because AI guesses style.

One-shot Prompting

You give one example.
Then ask AI to do sam task.
AI follows same pattern **and** style

One example = direction to AI

Note:
From provided example AI understands expected format, tone **and** structure

Examples

Example-1
Example
Q. What **is** Java?
Ans: Java **is** Highlevel object oriented programming language

Now explain python **in** the same way

Example-2
Example
Q. What **is** Java?
1. Java **is** High-level object oriented programming language
2. Jave **is** platform independent
3. Java can be used to develop standalone, web **and** enterprise applications

Now explain python **in** the same way

Example-3

Q1. What **is** big speciality of Java?
A. Platform dependent
B. Platform independent
C. Procedure oriented
D. Functional Oriented

Ans: B
Explanation : Java follows Write Once Run Anywhere

Now **in** the same style generate 5 questions on zero-shot, one-shot **and** few-shot prompts

Response
Q1. What **is** Zero-Shot Prompt?

A. Prompt **with** many examples
B. Prompt **with** one example
C. Prompt **with** no example
D. Prompt **with** code only

Ans: C

Explanation: In Zero-Shot, AI **is** asked to do a task without giving any example.

Q2. What **is** One-Shot Prompt?

A. Prompt without instructions
B. Prompt **with** one example
C. Prompt **with** many examples
D. Prompt **with** system role only

Ans: B

Explanation: In One-Shot, one example **is** given to guide the AI.

Q3. What **is** Few-Shot Prompt?

A. Prompt **with** zero examples
B. Prompt **with** one example
C. Prompt **with** 2–5 examples
D. Prompt without task

Ans: C

Explanation: Few-Shot uses multiple examples so AI understands the pattern clearly.

Q4. Which prompt type **is** best **for** simple **and** clear tasks?

A. Few-Shot
B. One-Shot
C. Zero-Shot
D. Multi-Shot

Ans: C

Explanation: Zero-Shot works well when the task **is** simple **and** direct.

Q5. Which prompt type gives best accuracy **for** complex tasks?

A. Zero-Shot
B. One-Shot
C. Few-Shot
D. Random Prompt

Ans: C

Explanation: Few-Shot gives better accuracy because more examples reduce confusion.

Example-4
Example:
"Computer is like a helper boy in shop - ask once, it remembers and brings"

In the same way, explain Artificial Intelligence

Answer(Response by ChatGPT):

"Artificial Intelligence is like a smart assistant in a shop — you teach it with many examples, it le arns patterns and habits, it helps you even for new customers without asking again."

Simple explanation:
Helper boy → follows orders
AI assistant → learns from data
More training → better intelligence
New problem → AI gives smart answer

So, Artificial Intelligence means machines that can think, learn, **and** act smart like humans.

Example-5
Example:
Internet **is** like gossip aunties network - news spreads fast.
In the same way explain about prompt engineering

When to use One-shot prompt

you want same pattern again **and** again
Eg:
- notes
- quizzes
- Emails
- Captions

Wherever consistency **is** required, one-shot **is** best

Q. What **is** One-shot prompting
One-shot prompting means giving one example so AI follows the same style **and** structure.
It gives consistent output

Zero-shot ➔ Asking AI without any example
One-shot ➔ Guiding AI **with** one example
Few-shot ➔ AI understands deeply **with** our provided multiple examples

Few-shot Prompting

Multiple examples mostly 2 to 5
Context learning
Better accuracy
Helps AI learn "pattern **or** style" from examples, **not** just task
Few-shot Prompting means:
- You give 2 to 5 examples
- AI learns the pattern
- AI applies the same logic to new input

Why 2-5 examples work best

Why **not** 1 example?
- AI may just copy style
- Understanding **is** shallow

Why **not** too many?
- Prompt becomes long
- Confusion increases
- Tokens increases

2-5 examples = Sweet spot

Example-1 : Customer support reply

Example 1:
Customer : My order **is** delayed
Reply : Sorry **for** the delay. We are checking **and** will update you soon

Example 2:
Customer : I received a damaged product
Reply : Sorry about the issue. Please share photos. We will replace it

Now reply to this:
Customer : I want to cancel my order

AI answer :

Reply: Sorry to hear that. Please share your order details. We will help you **with** the cancellation process

Example-2 : Product Description(E-commerce)

Product-1 : Wireless Mouse
Example 1 :
Product : Water Bottle
Description: Leak-proof bottle, easy to carry, good **for** daily use

Example 2 :
Product : Laptop Bag
Description: Strong bag **with** multiple pockets, suitable **for** office

Now wirte description **for**
Product : Wireless Mouse

AI Response

Product : Wireless Mouse
Description: Smooth **and** fast mouse, easy to connect, comfortable to use **for** office **and** daily work

Product-2 : Eggs
Example 1 :
Product : Water Bottle
Description: Leak-proof bottle, easy to carry, good **for** daily use

Example 2 :
Product : Laptop Bag
Description: Strong bag **with** multiple pockets, suitable **for** office

Now wirte description **for**
Product : Eggs

AI Response

Product : Eggs
Description: Fresh **and** healthy eggs, rich **in** protein, good **for** daily breakfast **and** cooking

Product-3 : Mobile
Example 1 :
Product : Water Bottle
Description: Leak-proof bottle, easy to carry, good **for** daily use

Example 2 :
Product : Laptop Bag
Description: Strong bag **with** multiple pockets, suitable **for** office

Now write description **for**
Product : Mobile

AI Response

Product : Mobile
Description: Smart **and** easy-to-use mobile, fast performance, clear display, suitable **for** daily c alls, browsing **and** entertainment

Example-3 : Whatsapp message writing
Example 1:
Message : I will join office late today due to traffic

Example 2:
Message : Please find the report attached. Let me know **if** changes needed

Now write a short professional WhatsApp message **for**
Taking leave tomorrow due to fever

AI Response

Message : I will be on leave tomorrow due to fever. I will respond to urgent matters once I am a vailable

Example-4 : Few-shot classifier
Example 1:
"I love this phone" → Positive

Example 2:
"This service is terrible" → Negative

Example 3:
"The food was Okay" → Neutral

Now classify
"The movie is amazing"

AI response

The movie **is** amazing ➔ Positive
The match was boring ➔ Negative
The AI **class is not** too great ➔ Negative
The AI **class is** average ➔ Neutral

Few-shot prompts with Structured output

It is the most commonly used in production. i.e., Production-Ready Prompting

Example-1 : Invoice Extractor

version-1
Finance team uploads invoices
They need:
  Invoice number
  Date
  Total Amount
But provided invoices have different formats.

Extract data **in** this format:
Invoice Number :
Date :
Total Amount:

Example 1:
Input:
Invoice No: INV-101
Date: 01-02-2025
Total : Rs 15,000

Output:
Invoice Number: INV-101
Date: 01-02-2025
Total Amount: Rs 15,000

Example 2:
Input:
Bill ID: 2045
Date: 10 Mar 2025
Grand Total : Rs 22,500

Output:
Invoice Number: 2045
Date: 10-03-2025
Total Amount: Rs 22,500

Now extract from this invoice

Input:
Invoice : 8899
Dated: 5 April 2025
Total Payable: Rs.40,000

AI Response

Invoice Number: 8899
Date: 05-04-2025
Total Amount: Rs 40,000

version-2
Finance team uploads invoices
They need:
- Invoice number
- Date
- Total Amount

But provided invoices have different formats.

Extract data **in** this format:
Invoice Number :
Date :
Total Amount:

Example 1:
Input:
Invoice No: INV-101
Date: 01-02-2025
Total : Rs 15,000

Output:
Invoice Number: INV-101
Date: 2025-02-01
Total Amount: Rs 15,000

Example 2:
Input:
Bill ID: 2045
Date: 10 Mar 2025
Grand Total : Rs 22,500

Output:
Invoice Number: 2045
Date: 2025-03-10
Total Amount: Rs 22,500

Now extract from this invoice

Input:
Invoice : 8899
Dated: 5 April 2025
Total Payable: Rs.40,000

AI Response

Invoice Number: 8899
Date: 2025-04-05
Total Amount: Rs 40,000

version-3
Finance team uploads invoices
They need:
- Invoice number
- Date
- Total Amount

But provided invoices have different formats.

Extract data **in** this format:
Invoice Number :
Date :
Total Amount:

Example 1:
Input:
Invoice No: INV-101
Date: 01-02-2025
Total : Rs 15,000

Output:
Invoice Number: INV-101
Date: 2025-02-01
Total Amount: Rs 15,000

Example 2:
Input:
Bill ID: 2045
Date: 10 Mar 2025
Grand Total : Rs 22,500

Output:
Invoice Number: 2045
Date: 2025-03-10
Total Amount: Rs 22,500

Now extract from this invoice

Input:
Invoice : 8899
Dated: 5 April 2025
Total Payable: $ 40,000

AI Response

Invoice Number: 8899
Date: 2025-04-05
Total Amount: $ 40,000

Why Industry uses this
Automated Accounting
Consistent Extraction
No extra text
Ready **for** ERP systems

Example-2 : Report Generator
Company needs weekly sales report **in** fixed format

You are a business report generator

Use this format:
Title:
Summary:
Key Points:
Conclusion:

Example 1:
Input:
Sales increased by 20%. Marketing campaign performed well

Output:
Title: Weekly Sales Report
Summary: Sales increased significantly
Key Points:
- 20% growth
- Marketing campaign success
Conclusion: Positive growth trend

Example 2:
Input:
Customer complaints reduced by 30%

Output:
Title: Weekly Performance Report
Summary: Customer Satisfaction improved
Key Points:
- 30% complaint reduction
Conclusion: Service Quality improved

Now generate report **for** this data:

Input:
Revenue increased by 15%. New clients added 50

AI Response

Title: Business Growth Report
Summary: Revenue **and** customer base have grown positively
Key Points:
- Revenue increased by 15%
- 50 new clients added
Conclusion: Business **is** growing steadily **with** higher income **and** more customers

Why Industry uses this
Board-ready format
Clean Structured output
Predictable layout

Few-Shot Vs Fine-Tuning

Few-Shot	Fine-Tuning
Teach using examples inside prompt	Train the model permanently
No model training	Model training required
Temporary behaviour	Long-term behaviour
Fast and flexible	Stable and powerful

fewshot_vs_finetuing

Zero-shot Vs One-shot Vs Few-shot Prompting

Type	Meaning	Examples Given	When to Use	Simple Example
Zero-Shot Prompting	Ask AI directly	No examples	When task is simple and open	"Explain AI in simple words"
One-Shot Prompting	Give one sample first	1 example	When you want same style or format	Example shown → Explain ML in same style
Few-Shot Prompting	Give multiple samples	2 or more examples	When output must be very consistent	2 examples → Explain new topic same way

Prompt_types

Persona Prompting

Role

Role means Job
Eg:
- Teacher
- HR
- Interviewer
- Analyst

Role answers ➔ What **is** the job

Persona

Persona means behavior style
Eg:
Teacher role can be:
- Friendly Teacher
- Strict Teacher
- Motivational Teacher
- Funny Teacher

Persona answers ➔ How does this role behaves

Tone

Tone means voice feeling
Eg:
- Polite
- Serious
- Encouraging

Tone **is** smaller layer
Persona **is** bigger behavior identity

Note

Role = Job
Persona = Behavior Style
Tone = Voice Feeling

Q: What **is** the difference between Role **and** Persona

Role defines Responsibility/Job
Persona defines Behavior identity of the role

No Role

Explain python **in** simple English

With Role

Act as a Teacher.
Explain python **in** simple English

Teacher ➔ Role

With Persona

Act as a Friendly School Teacher.
Explain python **in** simple English
Use 1 daily life example
end **with** 1 motivational line

Friendly School Teacher ➔ Persona

Act as a Friendly Technical Trainer like Durga sir
Explain python **in** simple English

Act as strict interview coach
Ask me 1 question about prompt engineering
Wait **for** my answer
then give feedback **in** 3 bullet points

strict interview coach ➔ Person

Persona Control(Behavior Engineering)

In Real Industry:
- HR bots must be polite
- Financial bots must be strict **and** logical
- Mentor bots must be supportive **and** motivational

Persona = Behavior + Depth + Attitude

Behavior

Is AI Strict?
Is AI Friendly?
Is AI Funny?

Depth

Does AI give short answer **or** detailed explanation?

Attitude

Does AI Challenge?
Does AI Encourage?
Does AI Criticize?
Does AI Supportive?



Example-1: Strict Interviewer(Short + challenging)

You are a Strict Technical Interviewer
Ask me 1 Prompt Engineering question
Wait **for** my answer
If answer **is** weak, challenge **with** a follow-up
Keep your replies short

Strict Behavior
Short Responses
Challenging Attitude

Example-2: Supportive Mentor

You are a Strict Supportive career Mentor

Ask me 1 Prompt Engineering question
Wait **for** my answer

Then explain correct answer step-by-step **in** simple english
End **with** one encouragement line

Example-3: Resume Reviewer Persona Prompt

You are an experienced **and** strict HR resume reviewer **with** 15+ years of hiring experience **in** I T **and** corporate roles.
Your job **is** to critically review my resume **and** identify weaknesses.

Instructions:
- Be direct **and** honest.
- Do **not** be polite **or** sweet.
- Focus on clarity, structure, impact, achievements, keywords, formatting, **and** relevance.
- Assume this resume **is for** a competitive job role.

Task:
- Identify exactly 5 major problems **in** the resume.
- Suggest exactly 5 strong improvements.
- Keep each point short **and** practical.
- No extra explanation.
- No motivational sentences.

Output Format:
Problems:
1.
2.
3.

Improvements:
1.
2.

3.
4.
5.

Resume Text:
[Paste your resume here]

Sample Resume

Name: Rahul Kumar
Email: rahul123@gmail.com
Phone: 9876543210234
Location: Hyderabad

Career Objective:
To work **in** a challenging environment where I can improve my skills **and** contribute to compan y growth.

Education:
B.Tech **in** Computer Science
XYZ Engineering College
2022
Percentage: 72%

Skills:
- Python
- Java
- SQL
- MS Office
- Communication skills

Projects:
1. Student Management System
   Developed using Python **and** MySQL.
2. Online Shopping Website

Internship:
Software Intern at ABC Technologies
Worked on web development tasks.

Experience:
Fresher

Strengths:
- Hardworking
- Quick learner
- Team player

Hobbies:
Watching movies
Playing cricket

Declaration:
I hereby declare that the above information **is** true to the best of my knowledge.

Domain Personas

Finance App ➔ Financial Thinking
HR Tool ➔ People Thinking
Audit System ➔ Error Thinking
Coaching App ➔ Growth Thinking

Domain Persona means: AI behave like expert from specific field

HR Persona:

Focus on
Behavior
Communication
Team harmony
Soft skills

Analyst Persona

Focus on
Data
Numbers
Logical Reasoning
Trends

Review Persona

Focus on
Errors
Weakness
Improvements

Mentor Persona

Focus on
Growth
Encouragement
Long-term improvement

Examples

Example-1 : Employee Performance Issue

HR Perspective
Act as HR Manager
Employee performance **is** low.
Give reasons from HR side
Give 5 points

Manager Perspective
Act as Project Manager
Employee performance **is** low.
Give reasons from delivery side
Give 5 points

Mentor Perspective
Act as Career Mentor
Employee performance **is** low.
Give 7-day improvement plan
use simple English

Safety Persona

Wrt illegal requests, our AI model should object **and** provide proper clarification
If question involves illegal activity
  refuse politely
  suggest safe alternative

You are a safe AI Assistant
If user asks illegal, harmful **or** cheating requests politely refuse
Then suggest a legal **and** safe alternative

Now answer this user question:
"How to hack bank account?"

Questions:
  "How to hack someone WhatsApp?"
  "Please guide step by process of bank account opening?"
  "What is ethical hacking?"

Reasoning Prompts

Reasoning Prompt **is** an instruction where we tell AI : How to think before answering
Don't jump to the answer directly. First understand, think and then provide answer.

Advantages
1. Reduce Mistakes
2. Improve clarity
3. Make better decisions

Types of Reasoning Prompts
1. Manual COT/ COT
2. Auto COT
3. Reflection
4. Self-Consistency
5. TOT
6. COT
7. GOT

Manual COT/ COT
COT ==> Chain Of Thoughts
Step-by-Step Thinking
We are manually providing steps, so that AI model forced to follow those steps
You personally write the thinking steps **in** the prompt.
You tell to the AI:

Step 1: Understand
Step 2: List Data
Step 3: Calculate
Step 4: Compare
Step 5: Decide
Step 6: Final Answer

Example 1:
Normal Prompt:
Which **is** bigger 7/8 **or** 3/4?

COT Prompt:
Which **is** bigger 7/8 **or** 3/4?
Compare 7/8 **and** 3/4 step by step
Convert both to the same denominator
Compare numerators
Then decide **and** give the final answer

Example 2:
Normal Prompt:
10% discount on Rs 500. Find final price?

COT Prompt:
10% discount on Rs 500. Find final price?
Step 1: Calculate 10% of 500
Step 2: Subtract from 500
Step 3: Give Final price

Example 3:
Normal Prompt:
If today **is** Saturday, what day after 10 days?

COT Prompt
If today **is** Saturday, what day after 10 days
Step 1: Use 7-day cycle logic
Step 2: Calculate remainder when 10 divided by 7
Step 3: Move that many days from saturday
Step 4: Then give final answer

Example 4:
We want to run a Python course. Course fee can be ₹5,000 **or** ₹7,000.
At ₹5,000 we may get 40 students.
At ₹7,000 we may get 30 students.
Which price **is** better?

You are a business analyst.
Help me choose best pricing **for** my Python course.

Step 1: Restate the problem **in** simple words.
Step 2: Calculate total revenue **for** both options.
Step 3: Compare revenues **and** also mention risk factors.
Step 4: Suggest best option **with** 2 reasons.
Step 5: Give final answer **in** a small table.

Example 5:
You are a logical love advisor.
A boy **is** planning to propose on Valentine's Day.
But he **is** confused.
Think step by step before giving final answer.

Step 1: Analyze girl's past behavior.
Step 2: Check **if** she replies fast **or** slow on WhatsApp.
Step 3: Evaluate **if** she sends reels daily.
Step 4: Assess financial condition (rose price **is** high on Feb 14).
Step 5: Consider emotional risk.
Step 6: Compare proposing on Feb 14 vs Feb 15 (discount day).
Step 7: Give final recommendation **with** reasoning.

Output format:
Step-by-step reasoning
Final decision
One funny warning line

Example 6:
You sell a course **for** ₹6,000.
You spend ₹1,000 per student **for** marketing + support.
You get 50 students. Find total profit.

Solve this profit problem step by step.

Step 1: Identify revenue per student **and** total revenue.
Step 2: Identify cost per student **and** total cost.
Step 3: Profit = total revenue - total cost.
Step 4: Provide final profit value.

Example 7:
You are planning a new weekend batch.

You have 2 options:

Option A: Saturday + Sunday, 7 AM to 9 AM
Option B: Saturday + Sunday, 7 PM to 9 PM

Most students are working people **and** college students.
Goal **is** maximum attendance.

Act as a training institute planner.

Step 1: List who the target students are.
Step 2: List pros **and** cons of morning batch.
Step 3: List pros **and** cons of evening batch.
Step 4: Consider Indian daily routine factors (travel, family time, energy).
Step 5: Give best option **and** justify **with** 3 reasons.
Step 6: Suggest one backup plan **if** attendance drops.

Example 8:
You are an experienced business consultant.

I want to open a second branch of my software training institute.

Think step by step **and** reason carefully before giving the final answer.

Step 1: Calculate financial safety buffer.
Step 2: Analyze current branch stability.
Step 3: Estimate expansion cost **and** risk.

Step 4: Evaluate worst-case scenario.
Step 5: List 5 major risks.
Step 6: Suggest mitigation strategies.
Step 7: Give Go / No-Go decision **with** reasoning.

Data:
Current monthly profit: ₹6,00,000
Personal expenses: ₹2,00,000
Savings: ₹40,00,000
Expansion cost: ₹28,00,000
Team strength: 20 employees
Dependents: 3

Give Structured output:
Financial Analysis
Risk Analysis
Decision
Final Recommendation

Example 9:
Act as a rational career advisor.

A software employee earning ₹8 LPA wants to quit job **and** start AI freelancing.

Think step by step before giving final advice.

Step 1: Calculate monthly survival cost.
Step 2: Evaluate skill readiness level.
Step 3: Analyze income uncertainty.
Step 4: Evaluate 6-month worst-case scenario.
Step 5: Identify 5 risks.
Step 6: Suggest safe transition strategy.
Step 7: Final decision: Immediate quit / Partial transition / Stay **in** job.

Data:
Monthly expense: ₹50,000
Savings: ₹5,00,000
AI skill level: Beginner
Family dependents: 2

Provide detailed reasoning

Auto COT

In manual COT, we are providing reasoning steps manually.

Sometimes **if** problem **is** complex **and** we may **not** aware of reasoning steps **or** we may **not** inte rerested to provide reasoning steps every time, then we can ask AI model to generate reasoning steps automatically.
This **is** called Auto COT

Manual COT ==> We are providing reasoning steps manually
Auto COT ==> AI model itself responsible to provide reasoning steps automatically

Examples

Example 1: AI Career Strategy

You are a strategic AI career mentor.

A student has completed foundation course **in** Generative AI **and** Prompt Engineering.

Analyze the situation deeply **and** reason step-by-step before answering.

Determine:
- What skill gap exists
- What **is** the next logical learning path
- What risks are involved
- What 6-month roadmap should be followed
- What earning potential exists

Think carefully **and** give structured final answer.

Example 2: Growth Strategy

You are a growth strategist.
A training institute earns ₹6 lakhs per month.
The owner wants to scale to ₹25 lakhs per month within 2 years.
Analyze deeply **and** reason step by step.

Identify:
- Bottlenecks
- Revenue levers
- Team structure changes
- Automation opportunities
- Marketing expansion strategy
- Risk factors

Give comprehensive strategy **with** logical reasoning.

Use Manual COT when:

1. You want full control over reasoning style
2. You want AI to follow a specific format **or** structure
3. The solution must follow a clear step-by-step pattern
4. The answer needs to match a particular logic flow.
5. Accuracy **is** very important **and** you want to reduce the mistakes.

Use Auto COT when:

1. The problem **is** complex **and** multi step
2. You do **not** want to manually create reasoning steps
3. You want AI to generate reasoning steps automatically
4. You want broader reasoning emploration

Note:

In the auto COT we have to mention "generate reasoning steps on your own" **in** the prompt.

Feature	Manual Chain-of-Thought (Manual CoT)	Automatic Chain-of-Thought (Auto-CoT)
Source of Logic	A human writes the reasoning steps manually.	The AI creates its own reasoning steps using prompts like "Let's think step by step."
Effort Level	High effort. Human must carefully write each example.	Low effort. AI generates reasoning automatically.
Scalability	Hard to create many examples for many problem types.	Easy to create reasoning for many different problems quickly.
Error Risk	Human thinking may be limited or biased.	AI may make wrong reasoning (hallucination) and repeat mistakes.
Diversity	Limited because human writes only few examples.	High diversity because AI can generate many different types of reasoning.

ManualCOT_AutoCOT

Reflection Prompting

Reflection prompting means asking AI to:

First generate answer
Then review its own answer
Find mistakes **or** weak points
Improve it
Give final approved answer.

In simple words: "Think ---> Check --->Improve"
It **is** like a student writing an exam answer **and** then rechecking before submitting.

Advantages:

1. Increase accuracy
2. Reduces wrong answers
3. Improve quality of the output
4. Become confident **in** decisions

Real world use cases:

Business planning
Career decisions
Financial analysis
Client proposals
Start up ideas validation

Examples

Example 1 :

Solve this problem step by step
A shopkeeper buys an item **for** Rs 500 **and** sells it **for** Rs 650.
Find profit percentage.

After solving:
1. Review your answer carefully.
2. Check **if** calculation **is** correct
3. If any mistake, correct it
4. Give final corrected answer clearly.

Example 2 :

Suggest pricing strategy **for** an AI freelancing beginner.

After giving the strategy:
1. Reflect on possible risks.
2. Identify weak assumptions.
3. Suggest improvements.
4. Give refined final strategy.

Example 3 :
I am earning ₹8 LPA **and** planning to quit job **for** freelancing.

Give advice.

Then:
1. Review your reasoning **for** emotional bias.
2. Identify financial risks you may have underestimated.
3. Strengthen risk analysis.
4. Give final improved recommendation.

Chain-of-Thought (CoT)	Reflection Prompting
Step-by-step reasoning	Self-review and correction
Focus on process clarity	Focus on improvement
Happens before final answer	Happens after first answer
Reduces logical errors	Polishes and refines answer

cot_reflection

Normal Multi-step conversation:
Situation:
A person wants to plan a trip to a new city. They want suggestions **for** places to visit, good food **and** places to stay.

Prompt-1:
"I am planning a trip to Mumbai. What places should I visit?"

Prompt-2:
"Thank you! Can you suggest some good restaurants?"

Prompt-3:
"Nice! Can you suggest some good hotels or places to stay?"

Prompt-4:
"Thanks! Do you have any other tips for exploring the city?"

Prompt-5:
"That helps! How can I travel around the city easilly?"

Manual COT
You are a professional travel planner.

I am planning a trip to Mumbai.
Before giving the final answer, follow Manual Chain-of-Thought reasoning.

Step 1: Understand the complete travel requirements.
Step 2: Identify top tourist attractions **in** Mumbai.
Step 3: Suggest good restaurants (veg **and** non-veg options).
Step 4: Recommend accommodation options (budget, mid-range, luxury).
Step 5: Provide transportation guidance **and** practical travel tips.

After completing all reasoning steps clearly, then provide a structured final travel plan.

Important:
Show each reasoning step clearly.
Do **not** skip steps.
Keep explanation simple **and** practical.
Then give final answer separately under "Final Travel Plan".

Auto COT
You are a professional travel planner.

I am planning a trip to Mumbai.

Please think step by step before giving the final answer.

Generate a detailed travel plan including:
Must-visit attractions
Good restaurants (veg **and** non-veg)
Accommodation options (budget, mid-range, luxury)
Transportation guidance
Practical travel tips

First show your step-by-step reasoning.
Then provide the final structured travel plan separately under the heading:
"Final Travel Plan".

Keep everything simple **and** practical.

Reflection Prompting

You are a professional travel planner.

I am planning a trip to Mumbai.

Task:
Create a detailed travel plan including:
- Must-visit attractions
- Good restaurants (veg **and** non-veg)
- Accommodation options (budget, mid-range, luxury)
- Transportation guidance - Practical travel tips

Follow this structure strictly:
Phase 1: Generate an initial travel plan.

Phase 2: Reflect on your own answer.
- Check **if** anything important **is** missing.
- Check **if** suggestions are practical.
- Check **if** the plan **is** balanced (budget, comfort, time).

Phase 3: Improve the plan based on your reflection.

Finally, provide the improved version under:
"Final Improved Travel Plan"

Keep language simple **and** practical.

Self-Consistency Prompting

It means:
1. Ask AI to solve the same problem 3 times **in** different ways/different perspections.
2. Comapre outputs
3. Choose the most consistent/repeated answer

Examples

Example 1 :

A shop sells an item at 20% profit.
If cost price **is** Rs 500, what **is** the selling price?

Solve this problem 3 times independently.

Rules:
- Each solution must be independent
- Then compare answers
- If answers are differ, explain which one **is** correct.
- Give final answer clearly.

Example 2 : Mumbai Travel

You are a professional travel planner.

I am planning a trip to Mumbai.

Task:
Create a complete travel plan including:
- Must-visit attractions
- Good restaurants (veg **and** non-veg)
- Accommodation options (budget, mid-range, luxury)
- Transportation guidance
- Practical travel tips

Follow Self-Consistency reasoning method:

Step 1:
- Generate 3 different reasoning approaches independently.
- Each approach should create its own structured travel plan.

Step 2:
- Compare the 3 plans.
- Identify common recommendations across them.

Step 3:
- Select the most consistent **and** practical suggestions.

Step 4:
- Provide the final combined answer under: "Final Travel Plan (Most Consistent Version)"

Keep language simple **and** practical.
Do **not** skip steps.

Reflection Vs Self-Consistency

Reflection:
1. First you create one answer
2. Then you check your own answer
3. You improve it

Note:
- Just review your answer **and** provide improved version

Self-Consistency:
You create multiple answers independently (2 **or** 3 versions)
Then you compare them
You select common **and** repeated points

Note:
Create multiple answers **and** choose the common best parts
Self-consistency **is** generally stronger **for** complex problems **and** it **is not** recommended **for** simple problems as the cost **is** high.
For simple problems, reflection **is** best which **is** faster **and** simpler.

Powerful Combination: Self-Consistency+Reflection:

Best practice:
First do self-consistency(3 runs)
Then apply reflection on best answer
Now we will get higher reliability.

Super prompt:
You are a professional travel planner.

I am planning a trip to Mumbai.

Task:
Create a complete **and** practical travel plan including:
Must-visit attractions
Good restaurants (veg **and** non-veg)
Accommodation options (budget, mid-range, luxury)
Transportation guidance
Practical travel tips

Follow this structure strictly:

PHASE 1 – SELF-CONSISTENCY

Step 1:
Generate 3 independent travel plans.
Each plan must:
Be structured
Include all required sections
Be logically different **in** approach
(Example: Budget-focused, Comfort-focused, Experience-focused)

Step 2:
Compare the 3 plans.
Identify common attractions.
Identify common restaurants.
Identify common stay areas.
Identify common transport suggestions.
Highlight strong repeated recommendations.

Step 3:
Create a Combined Consistent Version
using only the most repeated **and** practical suggestions.
Label this section clearly as: "Consistent Combined Travel Plan"

PHASE 2 – REFLECTION

Now reflect on the "Consistent Combined Travel Plan".
Check carefully:
1. Is anything important missing?
2. Are travel times practical?
3. Is it balanced between budget **and** comfort?
4. Is the sequence of places logical?
5. Is it realistic **for** a first-time traveler?
6. Can it be simplified further?

List improvement points clearly.

PHASE 3 – FINAL IMPROVEMENT

Based on reflection, improve the plan.
Provide the final refined version under: "Final Improved Travel Plan (Self-Consistency + Reflection Version)"

Rules:
Keep language simple **and** practical.
Do **not** skip any phase.
Show reasoning clearly.
Keep suggestions realistic.

Note: 1. Manual CoT –> Human guided reasoning 2. Auto CoT –> AI guided reasoning 3. Reflection –> Solve once, review, improve 4. Self-Consistency –> Solve many times, choose common result

JSON
TOON to reduce cost
Structured Prompt
CoVe Prompt
XML Tag Prompt
TOON **is not** best **for** nested Json, it **is** good only **for** simple Json



Quiz all revised topis:

From the prompt engineering concepts..... ,
please generate 20 MCQs one by one .
After every question wait **for** my answer, please validate my answer **and** provide explanation & 3 bullet points to know concept clearly.
And then ask next question

from the above files **and** theory what i provided, please generate 20 MCQs one by one. After eve ry question wait **for** my answer, please validate my answer **and** provide explanation **and** 3 bulle t points to know concept clearly. And then ask next question.

Note
Reflection ==> Generate draft + Check/Review + Improved Version
Chain of Hindsight ==> Already AI provided Response + Imporve this response
Step Back

MCQ

Question 1:

Which Prompt below follows the correct 5-layer Prompt Formula
(Role + Task + Context + Constraints + Output Format)

A.
Explain Python

B.
You are Python Teacher. Explain loops

C.
You are a Python teacher
Explain loops to 10th **class** students
Use simple English
Limit to 100 words
Give 3 Bullet points**

D.
Explain loops clearly **and** properly **in** a good way

Ans : C

Why C **is** Correct
Option C includes all 5 layers:

Role → "You are a Python teacher"
Task → "Explain loops"
Context → "to 10th class students"
Constraints → "Use simple English, Limit to 100 words"
Output Format → "Give 3 bullet points"

That **is** a complete professional prompt.

Question 2:
Which statement best explains why ambiguity **is** dangerous **in** prompting?
A. AI becomes slow.
B. AI guesses what you mean **if** you are unclear.
C. AI ignores your instruction.
D. AI always gives wrong answers.

Ans : B

Question 3:
Which of the following **is** the BEST example of Length Control?
A. Explain Java.
B. Explain Java briefly.
C. Explain Java **in** 2 lines only.
D. Explain Java clearly **and** properly.

Ans : C

Question 4:
Which prompt **is** the BEST example of Format Control?
A. Explain OOPS concepts.
B. Explain OOPS concepts **in** a table **with** 3 columns: Concept, Definition, Example.
C. Explain OOPS concepts clearly.
D. Explain OOPS **in** simple language.

Ans: B

Question 5:
Which of the following **is** the BEST example of One-Shot Prompting?
A. Classify this review: "Food was amazing."
B.
Example:
Review: "Food was amazing." → Positive
Now classify: "Service was slow."
C. Classify this review step by step.
D. Give 5 examples of review classification.

Ans: B

Question 6:
Which situation **is** BEST suited **for** Chain-of-Thought (CoT) prompting?
A. Writing a poem
B. Generating 10 Instagram captions
C. Solving a profit percentage problem
D. Translating English to Telugu

Ans: C

Question 7:
Which statement best explains the difference between Persona **and** Role?
A. They are exactly the same.
B. Role defines task; Persona defines behaviour **and** attitude.
C. Persona only changes tone.
D. Role **is** advanced; Persona **is** basic.

Ans: B

Question 8:
Which of the following **is** the BEST example of "Only output this" control?
A. Explain Python **in** detail.
B. Explain Python briefly **in** bullet points.
C. Reply only **in** valid JSON format. Do **not** add any extra text.
D. Explain Python clearly.

Ans: C

Question 9:
Which of the following **is** the BEST example of Few-Shot Prompting?
A.
Classify: "Very bad product"

B.
Example 1:
"Very good product" → Positive

Example 2:
"Very bad product" → Negative

Now classify: "Average quality."

C. Classify this step by step.

D. Give classification examples.

Ans: B



Question 10:
Which situation **is** BEST suited **for** Auto-CoT?
A. Writing a motivational quote
B. Solving one simple math question
C. When you want AI to generate solved examples first **and** then apply the same pattern
D. Translating text

Ans: C

Question 11:
Which of the following BEST demonstrates Ambiguity **in** prompting?
A. Explain Python **in** 100 words.
B. Explain Python clearly **and** properly.
C. Explain Python **in** bullet points.
D. Explain Python to beginners **in** simple English **in** 5 points!

Ans: B

Question 12:
Which prompt **is** BEST suited **for** building a JSON extractor system?
A. Extract data from this resume.
B. Extract name **and** email from this resume.
C. Extract name, email, **and** skills. Reply only **in** valid JSON format **with** keys: name, email, sk ills. Do **not** add extra text.
D. Extract details clearly.

Ans: C

Question 13:
Which of the following BEST represents Step-by-Step Prompting?
A. Solve 10% of 500.
B. Solve 10% of 500 clearly.
C. Solve 10% of 500 step by step **and** show calculation.
D. Calculate percentage properly.

Ans: C

Question 14:
Which scenario **is** BEST **for** using Persona + Constraints + Format together?
A. Writing a random poem
B. Casual chat reply
C. Building a mock interview bot
D. Asking **for** movie suggestions

Ans: C

Question 15:
Which of the following BEST explains why Few-Shot improves stability?
A. It makes AI faster.
B. It reduces token usage.
C. It teaches AI a pattern before solving.
D. It prevents hallucination completely.

Ans: C

Question 16:
Which of the following BEST explains the difference between Reflection Prompting **and** Self-Con sistency?
A. Both are same.
B. Reflection improves one answer; Self-consistency compares multiple answers.
C. Reflection **is for** maths only.
D. Self-consistency **is** faster than reflection.

Ans: B

Question 17:
Which of the following BEST represents a Zero-Shot Prompt?

A.
Example:
Positive → Good
Negative → Bad

Now classify: "Nice product"

B. Classify the sentiment of: "Nice product"

C. Give 3 classification examples.

D. Classify step by step.

Ans: B

Question 18:
Which of the following BEST describes Instruction Encoding inside a prompt?
A. Writing very long prompts.
B. Hiding secret commands.
C. Embedding rules, constraints, **and** behaviour instructions inside the prompt text.
D. Using emojis **in** prompts.

Ans: C

Question 19:
Which scenario BEST requires Self-Consistency Prompting?
A. Generating 5 Instagram captions
B. Translating English to Hindi
C. Choosing the best business strategy among 3 options
D. Writing a birthday wish

Ans: C

Question 20:
Which of the following BEST explains why Persona + Few-Shot + Format together creates highl y stable output?
A. It makes AI creative.
B. It increases token usage.
C. It locks behaviour, pattern, **and** structure simultaneously.
D. It reduces response time.

Ans: C

Tree of Thoughts(ToT) prompting
ToT = Explore multiple possible solution paths before choosing the best one.
1. Identify the decision problem
2. Create 3 options(branches)
3. For each option:
   list pros
   list cons
4. Compare all branches
5. Choose best branch
6. provide final decision **with** reason **and** then expand

Options ---> pros/cons ---> conclusion

Note:
When one decision can be taken **in** many ways, then we can use ToT.

Examples

Example 1 : Scale Training Institute from ₹6L to ₹25L/month
You are a senior education business strategist.

A training institute currently earns ₹6 lakhs per month.
The owner wants to scale to ₹25 lakhs per month within 2 years.

Use Tree-of-Thought reasoning method.

Step 1:
Generate 3 different possible growth strategies:
Strategy A: Increase student volume
Strategy B: Increase course pricing **and** premium offerings
Strategy C: Expand to online + corporate training

Step 2:
For each strategy, evaluate briefly:
Revenue potential
Implementation difficulty
Risk level
Time required
Team dependency

Step 3:
Compare the three strategies **and** identify:
Which has highest revenue impact?
Which **is** fastest to implement?
Which **is** most sustainable long-term?

Step 4:
Select the strongest strategy (**or** combination).

Step 5:
Expand the selected strategy into a practical action plan including:
Marketing approach
Team changes
Automation tools
Revenue milestones

Provide final answer under: "Best Optimized Growth Plan"

Example 2 :

Problem Statement

A student completed Python course.
Now confused about next step.
Should he:
A. AI(Generative+Agentic)
B. Learn Full Stack Development
C. Learn DevOps

Prompt

You are a career mentor.

A student has completed Python course.
Now he wants to choose the next best career path.

Use Tree-of-Thought reasoning method.

Step 1:
Generate 3 different career strategies:
Strategy A: Go **for** AI(Generative+Agentic)
Strategy B: Go **for** Full Stack Development
Strategy C: Go **for** DevOps

Step 2:
For each strategy, evaluate:
  Job demand
  Learning difficulty
  Time to get job
  Salary potential
  Future growth

Step 3:
Compare the three strategies **and** identify:
  Which **is** easiest to learn?
  Which gives fastest job opportunity?
  Which has highest salary growth?

Step 4:
  Select the best career path (**or** combination).

Step 5:
Expand the selected option into:
  Learning roadmap
  Tools to learn
  Time required
  Job roles possible

Provide final answer under: "Best Career Path Plan"

Example 3 :

Problem Statement

A boy wants to reach institute on time.
But morning he has 3 options:

A. Come by Bike
B. Come by Bus
C. Come by Metro
Now he **is** confused.

Prompt

You are a logical travel advisor.

A student wants to reach institute on time **for** Python **class**.
He has 3 travel options.

Use Tree-of-Thought reasoning method.

Step 1:
Generate 3 different travel strategies:

Strategy A: Come by Bike
Strategy B: Come by Bus
Strategy C: Come by Metro

Step 2:
For each strategy, evaluate:
Travel time
Cost
Traffic risk
Comfort
Reliability

Step 3:
Compare all 3 strategies **and** identify:
Which **is** cheapest?
Which **is** fastest?
Which **is** most reliable?

Step 4:
Select the best travel option (**or** combination).

Step 5:
Explain why this option **is** best.

Provide final answer under: "Best Travel Decision"

Example 4 :

Problem Statement

A boy wants to propose a girl.
But confused about when to propose.
Options:
A. Feb 14 (Valentine's Day)
B. Feb 15 (Discount Rose Day)
C. On her Birthday

Prompt

You are a logical love advisor.

A boy **is** planning to propose a girl.
But he **is** confused about the best day.
Use Tree-of-Thought reasoning method.

Step 1:

Generate 3 different proposal strategies:
Strategy A: Propose on Feb 14 (Valentine's Day)
Strategy B: Propose on Feb 15 (Next Day)
Strategy C: Propose on her Birthday

Step 2:
For each strategy, evaluate:
Emotional impact
Cost (rose price **is** high on Feb 14)
Surprise factor
Crowd pressure
Success probability

Step 3:
Compare all 3 strategies **and** identify:
Which **is** most romantic?
Which **is** low cost?
Which has highest success chances?

Step 4:
Select the best proposal timing.

Step 5:
Explain why this option **is** best.

Provide final answer under: "Best Proposal Strategy"

Example 5 :

Problem Statement

Rahul has exam tomorrow.
But tonight he has 3 options:
A. Study seriously
B. Watch IPL match
C. Sleep early
Now he **is** confused

Prompt

You are a logical student advisor.

Rahul has exam tomorrow.
Tonight he has 3 options.

Use Tree-of-Thought reasoning method.

Step 1:
Generate 3 different night strategies:
Strategy A: Study seriously
Strategy B: Watch IPL match
Strategy C: Sleep early

Step 2:
For each strategy, evaluate:
Exam performance
Stress level
Next day energy
Risk of failure
Long-term benefit

Step 3:
Compare all 3 strategies **and** identify:
Which gives highest marks?
Which gives best energy?
Which has least risk?

Step 4:
Select the best night strategy.

Step 5:
Explain why this option **is** best.

Provide final answer under: "Best Exam Night Plan"

Example 6 :

Problem Statement

DURGASOFT wants to increase batch capacity.
You have 3 options:
A. Hire more faculty
B. Create recorded video courses
C Use AI tools **for** doubt support
Now what **is** best?

Prompt

You are a senior training institute growth strategist.
DURGASOFT wants to increase batch capacity **and** scale revenue.
Use Tree-of-Thought reasoning method.

Step 1:
Generate 3 different capacity expansion strategies:
Strategy A: Hire more faculty
Strategy B: Create recorded video courses
Strategy C: Use AI tools **for** doubt support **and** automation

Step 2:
For each strategy, evaluate:
Batch scalability
Cost involved
Quality control
Dependency on people
Long-term sustainability

Step 3:
Compare all 3 strategies **and** identify:
Which allows fastest batch expansion?
Which has lowest long-term cost?
Which **is** most scalable?

Step 4: Select the best strategy (**or** combination).

Step 5:
Expand into a practical action plan including:
Hiring **or** automation roadmap
Investment required
Timeline
Revenue impact

Provide final answer under: "Best Capacity Expansion Plan"

Feature	Self-Consistency	ToT
Thinking Type	Same problem many times	Different strategies
Used For	Accuracy improvement	Decision making
Final Answer	Majority result/common/ repeated	Best path (sometimes combo)
Best For	Maths, logic, prediction	Business planning
Example	Attendance % prediction	Batch timing decision

self_consistency_tot

Chain of Hindsight(CoH) Prompting:
The main objective **is** to improve a bad output.
Hindsight means looking back
CoH means:
1. Take AI's previous answer(bad/weak)
2. Tell AI what is wrong.
3. Ask AI to rewrite improved answer

When to use CoH:
1. Output **is** weak/bad
2. You want better writing
3. You want better marketing script
4. You want better explanation

Examples

Example 1 :

Normal Prompt

WhatsApp message to my students:
"Join Python course. It is good. We teach well. Contact us"

This **is** too generic
No emotion
No benefit
No call-to-action

Improved Prompt

Here **is** a weak marketing script:
"Join Python course. It is good. We teach well. Contact us."

Problems:
1. It **is** too generic **and** boring.
2. No clear benefits.
3. No target audience mention.
4. No urgency **or** call-to-action.

Task:
Rewrite a better WhatsApp promo message **in** 7 lines.
Audience: college students + working professionals.
Tone: friendly **and** confident.

Include:
3 benefits
1 real-life use case
Clear call-to-action at end

Example 2 :
You are a professional travel planner.

A tourist planned a 3-day trip to Mumbai.
But after returning, he said:
Trip was very hectic
Missed some important places
Spent too much money
Wasted time **in** traffic
Felt tired instead of happy

Use Chain of Hindsight reasoning.

Step 1:
Assume the trip has already failed.
Identify at least 6 mistakes **in** planning.

Step 2:
From those mistakes, extract clear lessons learned.

Step 3:
Now redesign a better 3-day Mumbai travel plan including:
Must-visit attractions
Good veg & non-veg restaurants
Budget + mid-range stay options
Smart transportation guidance
Practical travel tips

Step 4:
Explain clearly how the new plan avoids previous mistakes.
Provide final output **in** this structure:
1. Failure Analysis
2. Lessons Learned
3. Improved 3-Day Mumbai Travel Plan
4. Why This Plan Will Work Better

Keep explanation simple **and** practical.

Point	Chain of Hindsight (COH)	Reflection
When used	After something failed	Before final answer
Main idea	Learn from past mistakes	Improve current answer
Thinking direction	Backward thinking	Forward correction
Assumption	Result already bad	Result not yet final
Focus	What went wrong?	What can be improved?
Purpose	Avoid repeating mistakes	Make answer stronger
Real-life example	After exam result, analyze mistakes	Re-check paper before submitting
Emotion style	Problem → Learning	Draft → Improve
Risk level	Used after damage happened	Used to prevent damage
Improvement type	Redesign from lessons	Fine-tune existing plan

coh_vs_reflection

Step-Back Prompting

This **is** how to solve tough problems by going back to basics.
Step-Back prompting means:

1. Don't solve immediately
2. First explain the high level concept
3. Then solve the specific problem using that concept.

When we have to use step-back

problem **is** confusing
You are stuck
Bug **is not** clear
You need core understanding first

Eg: Before explaining the RAG concept explain the terms used **in** the RAG

```
age = 25
print('My age is:'+age)
print('My age is:',age)
print('My age is:'+str(age))
print(f'My age is:{age}')
```

Examples

Example 1 :

prompt 1

Explain the core principle of data types **in** Python:
What **is** string?
What **is** int?
Why Python does **not** mix them directly?

Then solve this error:

```
age = 25
print("My age is " + age)
```

Give:

1. Correct code
2. Line by line explanation
3. One interview tip

prompt 2

Use step-back approach, explain the required core concepts **&** then solve the following problem
Then solve this error:

```
age = 25
print("My age is " + age)
```

Give:
1. Correct code
2. Line by line explanation
3. One interview tip

Example 2 :
You are an AI career mentor.

A student has successfully completed a Python course from DURGASOFT.
Now the student wants to plan the next steps to build a strong IT career.

Use the Step-Back Reasoning Method.

Step 1: Step Back **and** Think Generally
Before suggesting any career path, step back **and** explain:
What makes a tech career successful?
What skills are required beyond programming knowledge?
What factors influence job readiness?
(Projects, problem-solving, communication, portfolio, etc.)
What common mistakes do students make after finishing a course?
What general principles should be followed **for** career growth **in** IT?

Explain these general career planning principles clearly.

Step 2: Apply These Principles
Now apply the above principles to a student who has completed Python **and** suggest:
Skill gaps that may exist
Next learning path options
(Data Science, Web Development, Automation, AI, etc.)
Risks **if** no action **is** taken
A practical 6-month roadmap
Expected entry-level job roles
Earning potential **in** early career stage

Step 3: Final Output
Provide final answer **in** this structure:
1. Step-Back Career Growth Principles
2. Python Student Career Roadmap

Keep explanation simple **and** practical.

Example 3 :
You are a senior education business strategist.

A training institute (DURGASOFT) **is** currently earning ₹6 lakhs per month.
The owner wants to scale revenue to ₹25 lakhs per month within 2 years.

Use the Step-Back Reasoning Method.

Step 1: Step Back **and** Think Generally
Before giving any solution, step back **and** explain:
What fundamentally drives growth **in** an education business?
What are the main revenue levers **in** a training institute?
How does batch capacity affect income?
How does pricing power impact growth?
What role does marketing play **in** scaling?
How does automation reduce dependency on faculty?
What common mistakes do institutes make **while** scaling?

Explain the general principles clearly.

Step 2: Apply These Principles to This Case
Now apply the above growth principles to DURGASOFT's situation **and** suggest:
Possible bottlenecks **in** scaling
Capacity expansion strategies
Pricing improvement options
Marketing expansion ideas
Team structure changes
Automation opportunities
Risk factors involved

Step 3: Final Output
Provide final answer **in** this structure:
1. Step-Back Business Growth Principles
2. DURGASOFT Scaling Strategy (₹6L → ₹25L Plan)

Keep explanation simple, practical, **and** realistic.

Manual COT
Auto COT
Reflection
Self-Consistency
ToT
CoH
Step-Back
Master Super prompt

Step-Back: First think general principles and success factors before solving the specific problem.
Manual CoT: A human breaks the big problem into small clear steps and solves step-by-step.
Auto CoT: AI automatically generates and organizes the steps into a clean logical workflow.
ToT: Create multiple options(branches), compare them and choose best path.

Self-Consistency: Generate 3 independent answers and pick the most repeated/consistent/reliable conclusions
Reflection: Review your final answer, find gaps and improve it once more.
CoH: Assume failure happened, find mistakes, learn lessons and redesign better plan.

Master Super Prompt

Example : AI Bootcamp Planning Super Prompt

Act as an AI Bootcamp Program Manager.

Goal: Plan a 3-month AI Weekend Bootcamp

Audience:
College students + freshers + working professionals.
Level: Beginner to Intermediate.

Constraints:
Only weekends (Saturday & Sunday)
2 hours per day
Total duration: 12 weeks (3 months)
Total sessions: 24 sessions (2 sessions/weekend)

STEP 0: STEP-BACK REASONING (Before planning)

First step back **and** explain clearly:
1. What makes a weekend bootcamp successful?
2. What are the key success factors? (time, practice, projects, support, consistency, outcomes)
3. What common mistakes happen **in** weekend bootcamps?
4. What should be the "non-negotiables" **for** learners? (revision, assignments, project building)

Keep this section short, practical, **and** beginner-friendly.

STEP 1: CoT (Structured breakdown)

Break the full plan into these components:
A. Syllabus roadmap (beginner → intermediate)
B. Week-by-week schedule (12 weeks)
C. Projects (3 total) + outcomes
D. Assessments (quiz + mini test + final test)
E. Pricing options (Basic + Pro)
F. Marketing plan (7-day launch + weekly marketing plan)
G. Risks **and** guardrails (risks + how to prevent)

Then create an initial bootcamp plan using the above components.

STEP 2: ToT (Tree of Thoughts)

Create 3 plan options:
A. Fast-paced bootcamp
B. Balanced bootcamp
C. Slow + deep bootcamp

For each option, give:
- Pros (3 points)
- Cons (3 points)
- Who it suits best
- Risk level (Low/Medium/High)

Then choose the best option **for** this audience **and** constraints.
Clearly mention: "Selected Plan: ____" **with** reason.

STEP 3: SELF-CONSISTENCY

Now generate 3 versions of the Selected Plan (Version 1, Version 2, Version 3).
Each version must:
- Follow the same constraints (weekends only, 2 hrs/day, 12 weeks)
- Have a week-by-week structure
- Include projects + assessments + pricing + marketing + risks

Then compare the 3 versions **and** pick the most reliable plan using:
- Practicality
- Learning progression clarity
- Project strength
- Time realism
- Market attractiveness

Clearly mention: "Most Reliable Version: ____" **and** why.

STEP 4: REFLECTION

Reflect on the selected final plan **and** check gaps:
- Any missing important topics **for** beginner → intermediate AI?
- Timeline unrealistic anywhere?
- Projects too weak **or** too hard?
- Pricing unclear? - Marketing plan too generic?

Improve the plan after reflection.

STEP 5: CoH (Chain of Hindsight)

Assume the bootcamp failed after 12 weeks.
Identify at least 7 failure reasons such as:
- low attendance
- low assignment completion
- learners confused
- too fast **or** too slow
- project **not** finished
- weak support system
- poor marketing lead

From these failure reasons, extract:
- 7 lessons learned (what to do differently)

Then update the plan to prevent these failures.
Clearly show: "Fixes Applied" section.

FINAL OUTPUT FORMAT (Structured Document)

---

1. Program Summary (5 lines)

2. Week-by-Week Schedule (table)
   Table columns must be:
   Week | Sat Topic (2 hrs) | Sun Topic (2 hrs) | Homework/Practice | Assessment

3. Projects (3 projects + outcomes)
For each project include:
- Project name
- What student will build
- Tools/skills covered
- Output/portfolio result

4. Assessments
- Weekly quiz plan (how often, what format)
- Mini test (week number + format)
- Final test (week number + format)
- Passing criteria (simple)

5. Pricing Options (Basic + Pro)
Include:
- Price range suggestion
- What **is** included
- Bonus items (**if** any)

6. Marketing Plan
A. 7-day launch plan (Day 1 to Day 7)
B. Weekly plan (Weeks 1–12): what to post + where (WhatsApp/Instagram/YouTube/LinkedIn)

7. Risks & Mitigation (5 points)
Risk → Mitigation

8. Final Conclusion (3 lines)
Who should join
What outcome they will get
Clear CTA line

Language: Use simple Indian English.

Keep it practical, clear, **and** beginner-friendly.

Example : Student completed Python but not getting job – needs career roadmap:
Act as a Senior AI Career Mentor.

REAL-WORLD PROBLEM:
A student has completed Python from a training institute.
But after completion, he/she **is not** getting job calls **and** feels confused about next steps.

AUDIENCE:
College student / Fresher

GOAL:
Get an entry-level IT job within 6 months.

CONSTRAINTS:
Can study only 2 hours per day
No prior IT experience
No strong communication skills
No real-time project experie

AVAILABLE RESOURCES:
Python basics knowledge
Laptop **with** internet
Willing to learn consistently

STEP 0: STEP-BACK

Explain:
1. What makes a tech career successful after completing a course?
2. Why many students struggle after finishing Python?
3. What common mistakes freshers make **in** job preparation?
4. What are the must-do actions to become job ready?

STEP 1: CoT

Automatically plan:
- Skill gap analysis
- Domain selection (Data Science / Web / Automation)
- Project roadmap - Resume preparation strategy
- Interview preparation strategy
- Job application strategy

Then create an initial 6-month roadmap.

STEP 2: ToT

Create 3 career paths:
A. Data Science
B. Web Development
C. Automation

Compare:
- Learning time
- Job demand
- Difficulty

Select best domain **for** this student.

STEP 3: SELF-CONSISTENCY

Generate 3 versions of the 6-month roadmap.
Compare **and** select most reliable version based on:
- Practicality
- Time realism
- Project strength

STEP 4: REFLECTION

Check:
Is timeline realistic?
Are projects strong enough **for** resume?
Is interview prep sufficient?

Improve the roadmap.

STEP 5: CoH

Assume the student failed to get job after 6 months.

Find:
5 failure reasons

Convert them into lessons.
Update the roadmap to prevent failure.

FINAL OUTPUT:

Provide:
1. Career Roadmap Summary (5 lines)
2. Month-wise Plan (Month 1 to Month 6)
3. Projects to Build (3 real-time projects)
4. Resume Preparation Timeline
5. Interview Preparation Plan
6. Job Application Strategy
7. Risks & Mitigation (5 points)
8. Final Conclusion (3 lines)

Language: Use simple Indian English.
Keep it practical **and** beginner-friendly.

Example : REAL-WORLD USE CASE

Problem Statement

New Investor wants to start real estate investing but confused where to begin

prompt

Act as a Senior Real Estate Investment Mentor.

REAL-WORLD PROBLEM:
A new investor wants to start investing **in** real estate (open plots / ventures /construction).

But feels confused about:
- Where to invest?
- How much to invest?
- How to evaluate deals?
- How to avoid losses?

AUDIENCE:
Beginner Investor (IT employee / business person / salaried individual)

GOAL:
Build a profitable real estate investment roadmap **for** 12 months.

CONSTRAINTS:
- Limited capital (₹5L – ₹15L)
- No prior real estate knowledge
- Can monitor investment only part-time
- Wants safe + growth-based returns

AVAILABLE RESOURCES:
- Some savings
- Access to loan (optional)
- Willingness to invest long-term

STEP 0: STEP-BACK

Explain:
1. What makes real estate investment successful?
2. Why beginners lose money **in** plots **or** ventures?
3. What common mistakes new investors make?
4. What must be verified before investing **in** land?

STEP 1: AUTO-CoT

Automatically generate a logical investment planning process including:
- Capital allocation strategy
- Deal evaluation checklist
- Risk management approach
- Plot selection criteria
- Exit strategy planning
- ROI expectation timeline

Then create an initial 12-month investment roadmap.

STEP 2: ToT

Create 3 investment strategies:
A. Long-term appreciation strategy
B. Short-term rotation strategy
C. Construction + resale strategy

Compare:
- Capital requirement
- Risk level
- Time to profit

Select best strategy **for** this beginner.

STEP 3: SELF-CONSISTENCY

Generate 3 versions of the 12-month roadmap.
Compare based on:
- Safety
- ROI potential
- Practical feasibility

Select most reliable roadmap.

STEP 4: REFLECTION

Check:
- Is ROI expectation realistic?
- Is capital usage optimal?
- Are legal risks addressed?

Improve the roadmap.

STEP 5: CoH

Assume investor faced loss after 12 months.

Find: 5 possible mistakes (legal, location, liquidity, overpricing, etc.)

Convert them into lessons.
Update roadmap to prevent loss.

FINAL OUTPUT:

Provide:
1. Investment Roadmap Summary (5 lines)
2. Month-wise Investment Plan (Month 1–12)
3. Deal Evaluation Checklist
4. Capital Allocation Plan
5. Exit Strategy Options
6. Risk & Mitigation (5 points)
7. Final Recommendation (3 lines)

Language: Use simple Indian English.
Keep it practical **and** investor-friendly.
Step Back → CoT → ToT → Self Consistency → Reflection → CoH

Graph of Thoughts (System Thinking/Network Thinking): GoT

Graph contains nodes and nodes are connected with edges

Imagine you are planning a family wedding.
We cannot think only about food.
We cannot think only about hall.
We cannot think only about guests.
Everything **is** connected.

If guests increase -> food increase
If hall **is** small -> guests reduce
If budget reduces -> decoration changes
If date changes -> guests availability changes
This **is not** single line thinking.
This **is not** even branch/options thinking.
This **is** network thinking/system thinking.
Everything connects to everything.
This **is** graph of thoughts(GoT)

How GoT Works(System Thinking Model)
1. Identify main goal
2. Identify all major components(nodes)
3. For each component
   - list factors effecting it.
4. Draw connections between components
5. Analyze the impact of change **in** one component
6. Make final system-level decision

Example : Course Lauching Strategy

Design complete course launching strategy including:
- Marketing
- Pricing
- Timing
- Risk etc

1. Goal
2. Identify Main components
3. Show connections(system view):
   If price **is** high --> fewer students
   If marketing **is** strong -> more students
   If timing **is** bad -> attendance drops
   If attendance drops --> reputation drops
   If repuration drops -->marketing cost increases

Prompt

Act as a business strategist.

Goal:
Launch Python batch successfully.

Step 1: Identify all major components:
- Marketing
- Pricing
- Timing
- Risk
- Competition
- Student type

Step 2: Explain how each component affects the others.

Step 3: If price **is** increased by 20%, analyze impact on:
- student count
- marketing need
- risk level
- revenue

Step 4: Provide a balanced strategy considering all connections.
Output format:
1. System explanation
2. Risk analysis
3. Final strategy summary

Example : Starting Your Own Software Training Institute (Full System)

Activity:

Design complete institute launch strategy including:
- Faculty
- Marketing
- Fees
- Students
- Reputation
- Risk
- Competition

This **is** also a perfect GoT (Graph of Thoughts) example.

Step 1: Identify Main Goal

Goal:

Start your own software training institute successfully **with**
- good admissions
- strong teaching quality
- positive reputation
- consistent revenue

Step 2: Identify Main Components (Nodes)

1. Faculty
2. Marketing
3. Fees
4. Students
5. Reputation
6. Risk
7. Competition

All are connected.

Step 3: Show Connections (System View)

If faculty quality **is** good → students understand better
If students understand better → reviews improve
If reviews improve → reputation increases
If reputation increases → marketing becomes easier
If marketing becomes easier → student count increases
If student count increases → revenue increases

But...

If fees are too high → fewer students join
If fewer students join → batch strength reduces
If batch strength reduces → revenue risk increases
If revenue risk increases → institute stability decreases

Also...

If competition **is** strong → marketing effort must increase
If marketing effort increases → cost increases
If cost increases → profit decreases
See the network? : Faculty → Students → Reputation → Marketing → Admissions → Revenue
Fees → Admissions → Revenue → Risk
Competition → Marketing Cost → Profit → Risk
All nodes are connected.
That **is** system thinking.

GoT Prompt – Starting Your Own Software Training Institute:

Act as a business strategist.

Goal:
Start your own software training institute successfully **with**:
- good admissions
- quality teaching
- strong reputation
- stable revenue

Step 1: Identify all major components:
- Faculty
- Marketing
- Fees
- Students
- Reputation
- Risk
- Competition

Step 2: Explain how each component affects the others.
For example:
- If faculty quality improves → student satisfaction increases
- If student satisfaction increases → reputation improves
- If reputation improves → marketing becomes easier
- If marketing becomes easier → admissions increase
- If admissions increase → revenue increases
- If fees are high → admissions may drop
- If competition **is** strong → marketing effort must increase

Show the full system connection between all components.

Step 3: Analyze the impact **if**:
To hire experienced faculty, salary cost increases by 25%.
Check impact on:
- fees
- student admissions

teaching quality
reputation
marketing need
risk level
overall revenue

Step 4: Identify feedback loops
Explain:
One positive growth loop

Example:
Good faculty → better results → good reviews → strong reputation → more

admissions → higher revenue
One negative risk loop

Example:
High salary → fees increase → fewer admissions → revenue drop → financial risk

Step 5: Identify the best leverage point
From all components, identify:
Which one change can improve maximum system performance?

Example:
Improving faculty quality may improve students, reputation, **and** marketing together.

Step 6: Provide a balanced institute strategy considering all connections.
Explain how to balance:
quality vs cost
admissions vs reputation
marketing vs referrals
risk vs growth

Output Format:
1. System Explanation
2. Risk Analysis
3. Final Strategy Summary

CoT vs ToT vs GoT:

CoT(Chain of Thoughts)
Linear thinking
step 1 -> step 2 -> step3 -> answer
Used **for**
Math Problems
Logic Solving
Simple reasoning

ToT(Tree of Thoughts)

Branch Thinking/Options Thinkinig
- Option A
- Option B
- Option C

Compare all branches(options) **and** choose the strong branch

Used **for**
- Decision Making
- Career Planning
- Business Comparison

GoT(Graph of Thoughts)

System Thinking/ Network Thinking

Many nodes are connected

Each decision affects others

Used **for**
- System Design
- Business Strategy
- Large Planning
- Risk Modeling
- Complex Scenarios

Method	Thinking Type	How Thinking Happens	Best Used For	Easy Meaning
CoT	Step by Step	Step 1 → Step 2 → Answer	Math / Coding / Logic	One Road Thinking
ToT	Option Based	Option A / B / C → Compare	Career / Decisions	Many Roads Thinking
GoT	Connected	All parts affect each other	Business / Planning	Road Network

cot_tot_got



Reliability Engineerint

Reliability Engineering means

Make sure AI system gives :
- Correct output
- Safe Output
- Consistent Output
- Controlled output

Not random answer

Note

The main objective of Reliability Engineering **is** to get ==> Correct + Safe + Trusted answer from AI.

Why companies care

Banking Chatbot ==> Wrong balance ==> Money loss
Medical App ==> Wrong medicinit ==> Life risk
Legal Chatbot ==> Wrong advice ==> Lawsuit
Payment gateway ==> Wrong amount ==> Fraud

Part-1 : What are Guardrails

Rules that limit AI behaviour
Like road barries prevents accident, guardrails prevent AI from:
- Giving harmful suggestions
- Going outside topic
- Producing unsafe content
- Answering restricted domains

Types of Guardrails

The most common guardrails are
- Domain Guardrails
- Content Guardrails
- Tone Guardrails
- Format Guardrails

1. Domain Guardrails

AI answers only allowed topic
Eg: Placement only Bot

2. Content Guardrails

No harmful/illegal suggestions

3. Tone Guardrails

Professional tone only

4. Format Guardrails

Answer must be in bullet points/JSON/table

Part-2 : what is Hallucination

Hallucination means
AI confidently gives Wrong Information

We should control hallucination **in** the following ways:
Ask AI to say "I dont know"
Restrict answer scope
Connect AI **with** database(RAG)
Validate output externally
Avoid open ended guessing prompts

Part-3 : What is error reduction

Error reduction means:
Add step-by-step reasoning
Add reflection
Add validation
Cross-check calculations
Industry need : Generate ==> check ==> improve ==> final output
not Generate ==> Directly send

Part-4 : Output validation

Check AI output using rules
Eg:
Is number correct?
Is format correct?
Is answer within the boundary?
Validation increases reliability

Safe AI Design Approach

The simple safety flow

1. Receive user input
2. Check **if** input **is** allowed(Guardrauls)
3. If **not** allowed:
   - **return** safe message
4. If allowed
   - Generate answer

5. Validate answer format **and** logic
6. Return final verified output

Case Studies

Case Study-1 : Unsafe chatbot demo

Unsafe Prompt

You are a helpful student assistant chatbot

Answer the following student queries one by one **and** wait **for** my confirmation after every quest ion answer:

1. I have exam stress. Which sleeping tablets should I take?
2. I want quick money before placements. Should I **try** betting apps?
3. I feel fold **and** fever. Suggest medicine.
4. My interview **is** tommorrow. How to prepeare resume?

Safe Prompt

You are a College Placement Preparation AI Bot.

You are allowed to answer only:
- Resume preparation
- Interview questions
- Coding practice
- Communication tips

You are NOT allowed to give:
- Medical advice
- Financial investment advice
- Gambling **or** betting suggestions
- Personal life advice

If any question **is** outside placement preparation, respond **with**:
"I am designed only for placement preparation support."

Answer the following student queries one by one **and** wait **for** my confirmation after every quest ion answer:

1. I have exam stress. Which sleeping tablets should I take?
2. I want quick money before placements. Should I **try** betting apps?
3. I feel fold **and** fever. Suggest medicine.
4. My interview **is** tomorrow. How to prepare resume?

Case Study-2 : Medical Risk Demo

Unsafe AI Prompt

You are a student placement preparation AI assistant.

Answer the following questions one by one **and** wait **for** my confirmation after every question a nswer:

1. I have chest pain. What medicine should I take immediately?
2. I have high fever. Can I take any antibiotic?
3. I feel headache daily. Suggest tablet name.
4. I am preparing **for** placements. Give interview tips.

Safe AI Prompt

You are a student placement preparation AI assistant.

Rules:
1. Do **not** provide medical advice.
2. Do **not** suggest medicine.
3. For health-related questions, advise user to consult doctor.

Answer the following questions one by one **and** wait **for** my confirmation after every question a nswer:

1. I have chest pain. What medicine should I take immediately?
2. I have high fever. Can I take any antibiotic?
3. I feel headache daily. Suggest tablet name.
4. I am preparing **for** placements. Give interview tips.

Case Study-3 : Hallucination Failure Demo

Hallucination Prompt(Unsafe AI)

Tell me about the book:

"Advanced Machine Learning for Quantum Computers by Rahul Sharma published in 2019 by Oxford AI Press"

Safe Prompt (Hallucination Control):

If you are **not** sure whether the book exists, say:
"I could not find reliable information about this book."

Do **not** assume **or** create facts.
Now answer:

Tell me about the book:

" Advanced Machine Learning for Quantum Computers by Rahul Sharma published in 2019 by Oxford AI Press "

Case Study-4: Fake News AI Demo

Now we are building a news summarization AI bot.

Unsafe News Bot Prompt

You are a news assistant.

Summarize the following breaking news:

"Scientists in Hyderabad have successfully developed a pill that can permanently cure diabetes i n 3 days without any side effects. The government will distribute it for free from next month."

Safe News Prompt

You are a responsible news assistant.

Before summarizing any news:
1. Check **if** information **is** verified.
2. If source **is not** mentioned **or** information seems unverified, say:

"This news cannot be confirmed from reliable sources."

Do **not** assume **or** present unverified claims as facts.

Now summarize:

"Scientists in Hyderabad have successfully developed a pill that can permanently cure diabetes i n 3 days without any side effects. The government will distribute it for free from next month."

Interview Question:

Q: How will you make AI system reliable?

With the following actions
- Add Guardrails
- Restrict domain
- Add validation
- Add reflection
- Log errors
- Monitor outputs
- Control Hallucinations

This shows : Engineering Maturity

ReAct Prompting

Thinking **is not** enough. AI should also act.
ReAct means Reasoning + Action
Reasoning = AI thinks **and** decides next step.
Act = AI performs next step using tool **or** operation(Action)
Repeat until goal **is** completed.

Case study-1: Study Plan ReAct:

You are a Study Improvement Agent using ReAct Method.

Student Name: Rahul
Problem: Rahul studies 4 hours daily but marks are **not** improving.

Goal:
Improve Rahul's performance **in** next 3 days using
Reason → Action → Observation → Next Reason loop.

Rules:
1. You must give Rahul a small test question as Action.
2. Wait **for** Rahul's answer.
3. Check the answer yourself.
4. Show Observation:
   Correct or Wrong
   Where mistake happened
5. Give Next Reason:
   Why Rahul is losing marks?
6. Give Next Action:
   Another small test or correction task.

Continue this loop for 2–3 rounds.

Finally give:
Root cause of Rahul's poor performance
One new study rule for Rahul

Note:

In ReAct:
Ask Question
Check Answer
Find Mistake
Improve Strategy
AI **is not** only thinking **and** it **is** also doing something.

Case Study-2 : Coding Practice – ChatGPT Takes Action

You are a Coding Practice Agent using ReAct Method.

Student Name: Rahul
Goal: Improve Rahul's Python coding skills using
Reason → Action → Observation → Next Reason loop.

Topic: Python Basics
Rules:

1. You must give Rahul one small coding problem as Action.
2. Wait **for** Rahul's code answer.
3. Check Rahul's code.
4. Show Observation:
   Is output correct?
   Any syntax mistake?
   Any logic mistake?
5. Give Next Reason:
   Why Rahul made this mistake?
6. Give Next Action:
   Ask Rahul to correct code OR give new problem.

Continue this loop **for** 3 rounds.
Finally provide:
Rahul's main coding weakness
One practice rule **for** Rahul
One coding task **for** tomorrow

To understand Reason and Act:

Customer says:
"Where is my order?"

Agent flow:
Reason: Need order ID
Act: Ask order ID

Reason: Fetch order status
Act: Call database tool to fetch order status

Reason: Explain status clearly
Act: reply to customer



Case Study-3 : Interview Practice – ChatGPT Action Visible

You are an Interview Practice Agent using ReAct Method.

Candidate Name: Rahul
Goal: Improve Rahul's interview performance using
Reason → Action → Observation → Next Reason loop.

Interview Type: Basic HR Interview

Rules:
1. You must ask Rahul one interview question as Action.
2. Wait **for** Rahul's answer.
3. Check Rahul's answer.
4. Show Observation:
   Good parts **in** answer
   Missing parts
   Mistakes **in** communication
5. Give Next Reason:
   Why this answer may **not** impress interviewer?
6. Give Next Action:
   Ask Rahul to improve answer OR ask next question.

Continue this loop **for** 3 rounds.

Finally provide:
  Rahul's main interview weakness
  One communication improvement rule
  One practice task for tomorrow

API : Application Programming Interface

It **is** messenger that allows one software to talk to another software.
API **is** a just a bridge between two software components
Eg:
  ChatGPT website ==> **for** Humans
  ChatGPT API ==> For Software Components

Without API:
  No Automation
  No Real-time Apps
  No Industry projects

https://platform.openai.com/settings/organization/api-keys ==> create API key **in** OpenAI Plat form
https://platform.openai.com/home ==> To buy credits(Minimum $5)

https://platform.openai.com/settings/organization/billing/overview



Install Openai

pip install openai ➔ to install a package
pip show openai ➔ to see information about package

Python Program

```
from openai import OpenAI
import os
from dotenv import load_dotenv

load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client with API KEY
client = OpenAI(api_key=OPENAI_API_KEY)

prompt = "Explain importance of API and API keys in short"

brain = "gpt-4.1-mini"

response = client.responses.create(
    model = brain,
    input = prompt
)

print(response)
```

```
Response(id='resp_0b3dc8418f7832ed006aa3b60bfb4487d19f280f71ed1cfe1d', created_at=178
9113868.0, error=None, incomplete_details=None, instructions=None, metadata={}, model='gpt-
4.1-mini-2025-04-14', object='response', output=[ResponseOutputMessage(id='msg_0b3dc8418
f7832ed006aa3b60caf4c87d18a1e4cda80acfe61', content=[ResponseOutputText(annotations=[]
, text='**API (Application Programming Interface)** is important because it allows different soft
ware applications to communicate and interact with each other seamlessly, enabling integratio
n, automation, and efficient data exchange.\n\n**API keys** are important because they act as
unique identifiers and security tokens that control and monitor access to an API, ensuring only
authorized users or applications can use the service, protecting it from misuse or abuse.', type=
'output_text', logprobs=[])], role='assistant', status='completed', type='message', phase=None)], p
arallel_tool_calls=True, temperature=1.0, tool_choice='auto', tools=[], top_p=1.0, background=F
alse, completed_at=1789113869.0, conversation=None, max_output_tokens=None, max_tool_ca
lls=None, moderation=None, previous_response_id=None, prompt=None, prompt_cache_key=No
ne, prompt_cache_options=None, prompt_cache_retention='in_memory', reasoning=Reasoning(c
ontext=None, effort=None, generate_summary=None, mode=None, summary=None), safety_iden
tifier=None, service_tier='default', status='completed', text=ResponseTextConfig(format=Respons
eFormatText(type='text'), verbosity='medium'), top_logprobs=0, truncation='disabled', usage=Re
sponseUsage(input_tokens=16, input_tokens_details=InputTokensDetails(cache_write_tokens=
0, cached_tokens=0), output_tokens=77, output_tokens_details=OutputTokensDetails(reasonin
```

g_tokens=0), total_tokens=93), user=None, billing={'payer': 'developer'}, frequency_penalty=0.0, presence_penalty=0.0, store=True, tool_usage={'image_gen': {'input_tokens': 0, 'input_tokens_details': {'image_tokens': 0, 'text_tokens': 0}, 'output_tokens': 0, 'output_tokens_details': {'image_tokens': 0, 'text_tokens': 0}, 'total_tokens': 0}, 'web_search': {'num_requests': 0}})

type(response)

openai.types.responses.response.Response

print(response.output_text)

API (Application Programming Interface) is important because it allows different software applications to communicate and interact with each other seamlessly, enabling integration, automation, and efficient data exchange.

API keys are important because they act as unique identifiers and security tokens that control and monitor access to an API, ensuring only authorized users or applications can use the service, protecting it from misuse or abuse.

```
# convert response to json
print(response.model_dump_json(indent=2))
```


```
{
  "id": "resp_0b3dc8418f7832ed006aa3b60bfb4487d19f280f71ed1cfe1d",
  "created_at": 1789113868.0,
  "error": null,
  "incomplete_details": null,
  "instructions": null,
  "metadata": {},
  "model": "gpt-4.1-mini-2025-04-14",
  "object": "response",
  "output": [
    {
      "id": "msg_0b3dc8418f7832ed006aa3b60caf4c87d18a1e4cda80acfe61",
      "content": [
        {
          "annotations": [],
          "text": "**API (Application Programming Interface)** is important because it allows different software applications to communicate and interact with each other seamlessly, enabling integration, automation, and efficient data exchange.\n\n**API keys** are important because they act as unique identifiers and security tokens that control and monitor access to an API, ensuring only authorized users or applications can use the service, protecting it from misuse or abuse.",
          "type": "output_text",
          "logprobs": []
        }
      ],
```

```
      "role": "assistant",
      "status": "completed",
      "type": "message",
      "phase": null
    }
  ],
  "parallel_tool_calls": true,
  "temperature": 1.0,
  "tool_choice": "auto",
  "tools": [],
  "top_p": 1.0,
  "background": false,
  "completed_at": 1789113869.0,
  "conversation": null,
  "max_output_tokens": null,
  "max_tool_calls": null,
  "moderation": null,
  "previous_response_id": null,
  "prompt": null,
  "prompt_cache_key": null,
  "prompt_cache_options": null,
  "prompt_cache_retention": "in_memory",
  "reasoning": {
    "context": null,
    "effort": null,
    "generate_summary": null,
    "mode": null,
    "summary": null
  },
  "safety_identifier": null,
  "service_tier": "default",
  "status": "completed",
  "text": {
    "format": {
      "type": "text"
    },
    "verbosity": "medium"
  },
  "top_logprobs": 0,
  "truncation": "disabled",
  "usage": {
    "input_tokens": 16,
    "input_tokens_details": {
      "cache_write_tokens": 0,
      "cached_tokens": 0
    },
```

```
    "output_tokens": 77,
    "output_tokens_details": {
      "reasoning_tokens": 0
    },
    "total_tokens": 93
  },
  "user": null,
  "billing": {
    "payer": "developer"
  },
  "frequency_penalty": 0.0,
  "presence_penalty": 0.0,
  "store": true,
  "tool_usage": {
    "image_gen": {
      "input_tokens": 0,
      "input_tokens_details": {
        "image_tokens": 0,
        "text_tokens": 0
      },
      "output_tokens": 0,
      "output_tokens_details": {
        "image_tokens": 0,
        "text_tokens": 0
      },
      "total_tokens": 0
    },
    "web_search": {
      "num_requests": 0
    }
  }
}
```


Prompt Debugging

It means finding mistakes **in** a prompt **and** improving it step-by-step so that AI gives better output.

Section-1: Why prompts fail:

1. Vague Prompts

Vague means **not** specific.

Example: `"Write about success"`

Success **for** whom?

Student? Businessman? Athlete?

Better version: "Write 200 words about success mindset for college students."

See the difference

2. Missing Context

Context means background details`
Example: "Fix this code"
Which language?
What error?

Better version: "Fix this Python code. It gives TypeError when input is string"
Now AI understands.

3. No Role Clarity

Role means who AI should act as.
Example: "Give advice"
Advice as who?
Doctor? Career mentor? HR?

Better version: "Act as an experienced HR interviewer and give resume improvement tips"
Clear role gives focused answer.

4. No Output Format

Example: `"Give interview questions"`
Paragraph?
Bullet points?
Table?

Better version: "Give 10 HR interview questions in bullet points"
Now output **is** structured.

5. Overloaded Instructions

Too many instructions **in** one line.
Example: "Explain AI, give history, compare with ML, provide code, and suggest projects."
Too much.
AI may miss something.

Break into steps.

6. Ambiguous Wording

Ambiguous means double meaning.
Example: "Make it better."
Better in what way?
Shorter? Professional? Funny?

Always specify.

7. Hidden Assumptions

You assume AI knows something.
Example: "Continue this"
Continue what?
Where **is** previous data?

Always provide required information.

Section-2 : Prompt Refactoring(Structure Improvement)

Rewrite your prompt in a clean structure.

Why it needed

Because AI cannot guess your intention perfectly.
It follows your instruction.
If instruction **is** mixed, output will be mixed.

How it works (The simpel Formula)

We will use 6 steps:

1. Break long prompt into blocks
2. Add Role
3. Add Context
4. Add Constraints
5. Add Output format
6. Remove noise

Now we will learn each one.

Step 1: Break into Blocks
Blocks means separate sections.
Like:
Task
Details
Output

Step 2: Add Role
Role means: who AI should act as.
Example:
"Act as HR interviewer"
"Act as customer support agent"
"Act as Python tutor"

Step 3: Add Context
Context means background details.

Example:
Student **is** fresher
Company **is** product-based
Customer **is** angry
Topic **is** Python loops

Step 4: Add Constraints
Constraints means limits.
Example:
Write **in** simple English
Max 100 words
Use bullet points
No technical terms
Give 3 examples only

Step 5: Add Output Format
ormat means how you want answer.
Example:
Table
Bullet points
Steps
JSON
Code **with** explanation

Step 6: Remove Noise
Noise means unwanted lines.
Example:
"Do it fast fast"
"You are the best AI"
Repeated sentences
Mixed instructions

Remove all these.

Examples

Example-1 : Messy Prompt
Tell me about python **and** also give programs **and** also interview questions **and** make it simple **and** also give roadmap **and** also give projects.

Problem

Too many tasks **in** one line
No structure
No output format
Too much noice - **and** also

Refactored Prompt

Act as a Experienced Python trainer.

Task: Explain Python basics **in** simple english
Context: Audience **is** college students **and** freshers

Constriants:
Use short sentences.
Avoid complex words

Output Format:
1. 5 key points
2. 1 simple example program
3. 5 interview questions
4. 2 mini project ideas

Example-2 : Paragraph Prompt to Structured Prompt

Before:

"I want you to act like you are interviewing me and ask questions and also give feedback and al so tell improvements and also prepare me for job"

Now we refactor

After

Act as an HR interviewer.
Task: Take my mock HR interview.
Context: I am a fresher, B.Tech CSE.
Constraints:
Ask 1 question at a time.
Wait **for** my answer.
Give feedback **in** 3 points.
Output Format:
Question
My Answer Check
Improvement Tips

Clear.

English to Telugu Translator Prompt Para by para
Role:
You are a highly accurate **and** disciplined English-to-Telugu translator.

Task:
Translate the provided document strictly paragraph by paragraph.

Mandatory Rules (Follow Exactly):
Translate only ONE paragraph at a time.
First display the original English paragraph exactly as provided.
Then display the Telugu translation of that same paragraph.
Do **not** skip even a single word **or** line.
Do **not** combine **or** split paragraphs.
Do **not** summarize.
Do **not** simplify unless required **for** natural Telugu clarity.
Do **not** add explanations, notes, comments, **or** interpretations.
Preserve the original formatting **and** structure.
Maintain 100% meaning accuracy.
Every word **is** important.
After completing one paragraph, STOP immediately.
Wait **for** my confirmation before proceeding to the next paragraph.

Output Format (Strictly Follow):
Paragraph 1 – English: (Original paragraph exactly as given)
Paragraph 2 – Telugu: (Accurate Telugu translation)
- [WAITING FOR CONFIRMATION] Do **not continue** automatically.

Run this prompt on Obvious Adams book what you locked already

Section-3 : Prompt testing checklist(Quality control)

It **is** a list of questions.
We can ask ourselves before using the prompt.
Like a safety checklist.

1. Is Role Clear?

Did you tell AI : who it should act as?
Example: "Give advice"
Advice as who?

Better: "Act as career counsellor"
Always check role clarity.

2. Is Task Specific?

Task must be clear.
Bad: "Explain Python"
Better: "Explain Python loops with one simple example."
Specific means clear **and** limited.

3. Is Output Format Defined?

Do you want bullet points?
Table?
Paragraph?
JSON?
If format **is** missing, output may be random.

4. Any Ambiguity?

Ambiguity means double meaning.
Example: "Make it better."
Better **in** what way?

Always remove vague words.

5. Any Missing Data?

Did you forget details?
Example: "Fix this code"
Which code?

Always provide required data.

6. Can AI Misunderstand?

Imagine you are AI.
Will you get confused?
If yes, rewrite prompt.

7. Is It Testable?

Can you verify output?
Example: "Write something nice."
Hard to test.

Better: "Write 5 bullet points on time management."
Now testable.

Examples

Example-1 :

Bad Prompt

Write about success

Checklist?
Role -->X(**not** specified)
Specific -->X(**not** specified)
Format --->X(**not** specified)

Improved :

Act as motivational speaker.
Write 5 bullet points about success mindset **for** college students.

Example-2 :

Bad Prompt

Fix this code

Missing data
Ambiguity
Not testable

Improved:

Fix this python code. It throws TypeError when input **is** string.
Provide corrected code **and** explanation.
[code]

Section-4 : Prompt Improvement Loop

It **is** a repeated method.
We do prompt again **and** again, but each time we improve it.

How it works?(The 4-step loop)

1. Ask: Send prompt to AI
2. Check: see output carefully
3. Refine: Modify prompt to fix issue
4. Retest: Ask again **with** improved prompt.

- Repeat until we get required output.

Examples

Example-1 : Create a WhatsApp course promo message

Iteration-1(Ask)
"Write a WhatsApp message to promote my Python course."

Output Problem

AI may write too long
May **not** include key details
May look generic

So we go to Chek step

Iteration-2(Refine)
Act as a training institute marketing writer.
Write a short WhatsApp promo message **for** Python course.

Output Problem:

Still missing audience **and** structure.

Iteration-3(Refine)

We add context

Act as a training institute marketing writer.
Write a WhatsApp promo message **for** Python course **for** beginners **and** freshers.

Output Problem

Still message may be random format
No clear line

Iteration-4(Refine)

We add constraints + output format.

Act as a training institute marketing writer.
Create a WhatsApp promo message **for** Python course **for** beginners **and** freshers.
Constraints:
6 lines only.
Simple English.
Output format:
Line 1 hook,
Line 2 benefit,
Line 3 topics,
Line 4 timing,
Line 5 fee,
Line 6 call to action.

Output becomes much better.

Now it **is** structured.
But maybe still missing urgency

Iteration-5(Refine)

Add urgency and trust.

Act as a training institute marketing writer.
Create a WhatsApp promo message **for** Python course **for** beginners **and** freshers.
Constraints:
6 lines only.
Simple English.
Add:
limited seats,
start date, **and**
mentor tone.

Output format:
1. Hook
2. Benefit
3. Topics
4. Start date + timing
5. Fee + limited seats
6. Call to action **with** phone number placeholder

Now output will be sharp, clear, **and** usable.
This **is** prompt improvement loop.

Section-5 : Fix 20 Failed Prompts

Broken Prompt-1 :
Make my resume.

Improved Prompt

Act as an HR recruiter.
Create a 1-page resume **for** a fresher B.Tech CSE student applying **for** Python developer role.
Use bullet points.
Include skills,
projects,
education, **and**
summary.

Broken Prompt-2 :
Write career objective.

Wrong
For whom?
For which Role?

Improved Prompt

Write 2 career objective options **for** a fresher applying **for** Data Analyst role.
Simple English.
2 lines each

Broken Prompt-3 :
Improve my LinkedIn.

Wrong
No Profile info.
No structure

Improved Prompt

Act as LinkedIn profile expert.
Improve my LinkedIn headline **and** About section.
I am a fresher learning Python **and** AI.
Output:
1 headline + 150-word About section.

Broken Prompt-4 :
Write code

Wrong
which Language?
Problem?
Input/Output ?

Improved Prompt

Write Python code to find factorial of a number using loop.
Take input from user.
Print result.
Explain code **in** 5 simple lines

Gemini Music Songs Prompt:

Prompt-1 : Give it in ChatGPT
I want to create AI Love Song **in** Google Gemini.
Based on this Write the actual Telugu lyrics on Sree Leela name, she **is** my girl friend.
I want 10 Versions

Response

Provide the response generated by ChatGPT to Gemini **and** select the "create music" ==> it will generate the music

Prompt-2: Give it in ChatGPT
I want to create AI Learning Song **in** Google Gemini.
Based on this Write the actual Hindi lyrics on topic prompt engineering, It **is** my current leanring subject. I want 10 Versions

Response

Provide the response generated by ChatGPT to Gemini **and** select the "create music" ==> it will generate the music

Broken Prompt-5 :
Fix this error.

Improved Prompt

Fix the Python error **in** this code.
Explain what mistake happened **and** provide corrected code.
Here **is** the code: [paste code]

Broken Prompt-6 :
Optimize this program

Wrong
No goal
No constraints

Improved Prompt
Optimize this Python code **for** readability.
Keep same output. Add comments.
Do **not** change logic.
Here **is** the code: [paste code]

Broken Prompt-7 :
Write marketing post.

Wrong
Which Product?
Audience?
Platform?

Improved Prompt
Act as a digital marketer.
Write an Instagram caption to promote a Python course **for** beginners.
6 lines.
Include one emoji per line.
End **with** CTA: "Message me to join"

Broken Prompt-8 :
Make it attractive

Wrong
Attractive how?
Funny?
Professional?

Improved Prompt

Rewrite this course promo **in** a friendly **and** motivational tone.
Keep it under 80 words.
Use simple English.
Add urgency: limited seats.

Broken Prompt-9 :
Give ad ideas

Wrong
Too broad

Improved Prompt

Give 5 ad headline ideas **for** a "Prompt Engineering Bootcamp".
Audience: freshers.
Style: simple **and** punchy.
Max 8 words each.

Broken Prompt-10 :
Create chatbot prompt.

Wrong
Purpose?
Rules?
Tone?
Boundaries?

Improved Prompt

Act as a customer support chatbot **for** an online course platform.
Goal: help users **with** course details, pricing, **and** refund policy.
Tone: polite **and** short. Ask one question at a time.
Output format: chatbot system prompt

Broken Prompt-11 :
Handle angry customer.

Wrong
No Policy
No Steps

Improved Prompt

Act as support agent.
Handle an angry customer who says "I want refund now"

Respond **in** 4 steps:
1. Empathy
2. Confirm issue
3. Offer solution
4. Next action.

Keep it polite

Broken Prompt-12 :
Answer all questions

Wrong
Too open
Can go off track

Improved Prompt

Act as course support assistant.
Answer only about courses, batches, fees, timings, **and** enrollment.
If question **is** outside scope, say "I can help only with course info"

Broken Prompt-13 :
Extract data from this

Wrong
What data?
What format?

Improved Prompt

Extract Name, Email, Phone from the below text.
Output as a table.
If missing, write "NA".
Text: [paste]

Broken Prompt-14 :
Convert to JSON

Wrong
What keys?

Improved Prompt

Convert the below meeting notes into JSON **with** keys:
date
attendees
action_items.

Output only JSON.
Notes: [paste].

Broken Prompt-15 :
Find important points

Wrong

How many?
For whom?

Improved Prompt

From the below text, list top 5 key points **for** beginners.
Use bullet points.
Each point max 12 words.
Text: [paste]

Broken Prompt-16 :
Summarize this

Wrong
Length?
Style?
Audience?

Improved Prompt

Summarize the below article **in** 7 simple bullet points **for** school students.
Keep each point one line.
Text: [paste].

Broken Prompt-17 :
Make it short.

Wrong
Short means how much ==> 1 line **or** 50 words?

Improved Prompt

Rewrite the below paragraph **in** exactly 50 words.
Keep meaning same.
Simple English.
Paragraph: [paste].

Broken Prompt-18 :
Explain simply

Wrong
What level?
with example?

Improved Prompt

Explain this topic **in** simple Indian English **for** freshers.
Use one chai-shop example.

Keep under 120 words.
Topic: [paste]

Broken Prompt-19 :
Take my interview

Wrong
Role?
Pattern?
Feedback?

Improved Prompt

Act as HR interviewer.
Ask me 1 question at a time.
Wait **for** my answer.
Give feedback **in** 3 points:
good,
missing,
improvement.
Continue **for** 3 rounds.

Broken Prompt-20 :
Give me questions

Wrong
Which Topic?
Which difficulaty?

Improved Prompt

Give 10 Python interview questions **for** freshers on loops **and** functions.
Include expected short answer **for** each.
Format: table.

Section-6: Build Prompt Debugger Meta Prompt
Meta Prompt ==> Prompt which **is** the responsible to generate another prompt
It **is** a master prompt
f we give a broken prompt as imput, it analyzess **and** improves it.
Input ==> Broken Prompt
Output ==> Debug report + improved prompt

Why it is needed
Because prompts fail **for** common reasons:
Missing clarity
Ambiguity
No structure
No output format

too much noise
missing data

Debugger catches these quickly.

Meta Prompt

Act as a prompt dedugger AI
Your job **is** to analyze the given prompt **and** improve it.

Input prompt:
[Paste the prompt here].

Do these steps:
1. Quick summary:
   Explain what the prompt **is** trying to do **in** 1 line.
2. Problems found:
   List issues under these headings
   Missing role
   Missing clarity
   Ambiguity
   Missing context/data
   Structure issues
   Ouput format issues
   Noise **or** extra instructions
3. Fix suggestions:
   Give 5 clear suggestions to improve it.
4. Improved prompt(final):
   Rewrite the prompt using this structure:
     Role
     Task
     Context
     Constraints
     Output Format

Rules:
  Use simple English
  Keep improved prompt clean **and** short
  Do **not** add imaginary information
  If any data **is** missing ask 1-2 questions at the end.

Replace

Replace "[Paste the prompt here]" **with**

"Explain about Prompt Engineering in brief and short way to understand clearly and properly"

Improved Prompt given by AI(may vary for each run of the prompt)
Role:
Act as an experienced AI trainer.

Task:
Explain Prompt Engineering.

Context:
The audience **is** students who are learning AI **for** the first time.

Constraints:
Use very simple English
Keep explanation short
Avoid technical jargon
Maximum 80 words

Output Format:
Definition of Prompt Engineering
Why it **is** important
One simple real-life example

Questions:
1. Should the explanation include an example prompt alos?
2. Is the target audience studednts **or** working professionals?

Response by AI

For the above response by AI
Give the following answer

1. yes 2. Students

Final Response from the AI
Role:
Act as an experienced AI trainer.

Task:
Explain Prompt Engineering.

Context:
The audience **is** students who are learning AI **for** the first time.

Constraints:
Use very simple English
Keep explanation short (80–100 words)
Avoid complex technical terms
Explanation must be easy **for** students to understand

Output Format:
Simple definition of Prompt Engineering
Why Prompt Engineering **is** important
One simple real-life example
One example prompt students can **try**

Section-7: Mini Project : Customer Support Chatbot

Act as a Customer Support Chatbot **for** an online course platform.

Your goal **is** to help users **with** course details **and** enrollment.

Scope (Answer only these):
Course names **and** syllabus summary
Batch timings **and** duration
Fees **and** payment steps
Enrollment process
Certificate availability
Refund policy (only **if** provided)

Rules:
Be polite **and** short.
Use simple Indian English.
Ask one question **if** user details are missing.
Do **not** guess fees **or** policies.
If information **is not** available, say: "I don't have that info. Please contact support."

Output Format:
Reply **in** 3 to 5 short lines.
Use bullets only **if** needed.
If user asks unrelated questions:
Say: "I can help only with course and enrollment questions"

Now answer which movies released on this friday?

This Prompt will make chatbot stable

Debugging Module

1. Why Prompts fail
2. Prompt Refactoring(Structure Improvement)
3. Prompt Testing checklist(Quality Control)
4. Prompt Improvement Loop
5. Fix 20 Failed **or** Broken Prompts
6. Build Prompt Debugger Prompt(Meta Prompt)
7. Mini Project



Section-1: Prompt Chaining and Workflow Design

Breaking one big task into multiple small prompts where output of one prompt becomes input of next prompt.
his **is** nothing but prompt chaining.

Big Task ---> Small steps ---> Connect steps

Why single prompt fails?

AI may mix formats
AI may forget structure
Output becomes too long
Hard to control

Eg: Bad Prompt
Explain Python, create summary, generate quiz.

AI will confuse format.

Better way

Prompt 1-> Explain Topic
Prompt 2-> From that explanation Create summary
Prompt 3-> From that summary, create quiz

Where is prompt chaining is used in real projects

Chatbots
Content Generators
Resume Analyzers
Report Generators
AI Assistants

Real World usecases

Prompt-1: Explantion Generator

prompt

You are a simple programming teacher.
Explain the topic "{{TOPIC}}" **in** very simple english.
Use real-life examples.
Keep it under 300 words

run

Run the above prompt by replacing "{{TOPIC}}" **with** "Prompt Chaining"

Prompt-2: Summary Generator

prompt

Based on explanation below
{{EXPLANATION}}
Create a short summary **in** 5 bullet points.
Do **not** add new information.

run

Run the above prompt by replacing "{{EXPLANATION}}" **with** the response from "Prompt-1"

Prompt-3: Quiz Generator

prompt

Using the summary below
{{SUMMARY}}
create 5 multiple choice questions.
Each question must have 4 options
provide correct answer separetly **with** one line explanation.

run

Run the above prompt by replacing "{{SUMMARY}}" **with** the response from "Prompt-2"

MCQs

5 MCQs on Prompt Chaining

1. What does Prompt Chaining mean?
A. Solving a big task **in** one prompt
B. Solving a big task by breaking it into small prompts step by step
C. Asking AI to stop after every line
D. Writing prompts only **for** coding

Answer: B

2. In prompt chaining, what do we do after getting the first output?**
A. Delete it **and** start again
B. Use it as input **for** the next prompt
C. Ignore it **and** ask a new topic
D. Send it to another AI tool only

Answer: B

3. Which real-life example **is** used to explain prompt chaining?
A. Driving a bike
B. Playing cricket
C. Cooking a meal step by step
D. Watching a movie

Answer: C

4. Which **is** the correct example chain mentioned?
A. MCQs → Summary → Explanation
B. Summary → MCQs → Explanation
C. Explanation → 5-point Summary → MCQs
D. Explanation → MCQs → 5-point Summary

Answer: C

5. Prompt chaining **is** most useful because it:
A. Makes results clearer **and** reduces confusion
B. Always gives the same answer
C. Works only **for** simple tasks

Answer: A

Answers + One-line Explanation
1. B — It means completing a big task by doing small prompt steps.
2. B — The next prompt uses the previous output to **continue** the task.
3. C — Cooking shows how each step depends on the previous step.
4. C — The chain goes from explanation to summary to MCQs.
5. A — Step-by-step prompts give clearer results **and** help **in** complex work.

Build Study Assistant Workflow(Not Prompt chaining- Pipeline)
Goal :
User gives the Topic

System gives:
1. Explanation
2. Summary
3. Quiz

Master Prompt Template:

You are an AI study assistant.

Step 1:
Explain "{{TOPIC}}" **in** simple English.

Step 2:
From your explanation, create 5-point summary.

Step 3:
From the summary, create 5 MCQs **with** answers.

MCQ - Prompt Engineering

1. What **is** Prompt Engineering?
A. Writing programs **for** computers
B. Designing instructions **for** AI systems
C. Building AI hardware
D. Installing software

Answer: B

2. What **is** a prompt?
A. A computer virus
B. A programming language
C. A question **or** instruction given to AI
D. A hardware component

Answer: C

3. What happens when prompts are clear **and** detailed?
A. AI stops working
B. AI gives better answers
C. AI deletes data
D. AI becomes slower

Answer: B

4. Which of the following **is** a better prompt?
A. Explain Python
B. Python
C. Explain Python **for** beginners **with** examples
D. What

Answer: C

5. Who can benefit from Prompt Engineering?
A. Only programmers
B. Only scientists
C. Only students
D. Anyone using AI tools

Answer: D

Correct Answers

1. B — Prompt Engineering means designing instructions **for** AI.
2. C — A prompt **is** the question **or** instruction given to AI.
3. B — Clear prompts help AI generate better responses.
4. C — Detailed prompts give clearer context to AI.
5. D — Anyone using AI tools can benefit from Prompt Engineering.

MCQ : Prompt Chaining

Q1: What **is** Prompt Chaining?
Answer:
Prompt chaining **is** breaking a big task into smaller prompts where output of one becomes inpu t of another.

Q2: Why **is** prompt chaining better than single prompt?
Answer:
Because it gives better control, structured output, **and** reduces confusion **in** AI response.

Q3: Where **is** this used **in** real projects?
Answer:
Used **in** chatbots, content generators, resume analyzers, report generators, AI assistants.

Section-2: Decomposition Framework

Decomposition means **break** big task into smaller tasks **in** correct order.

It has 3 rules:

1. Steps must be small **and** clear
   Not big **and** confusing
2. Each step must have clear output
   So next step can use it
3. Steps must be **in** correct order

Note:

- For every task 3 questions:
  1. What **is** input?
  2. What **is** process?
  3. What **is** output?

Common mistakes while breaking tasks

1. Steps are too big
   Eg: Create Notes + quiz + ppt **in** Step1
   This **is not** breaking **and** it **is** mixing

2. No clear output format
   If output **is not** fixed, next step will confuse

3. Missing order

4. Not passing the exact output
   They summarize again, change, meaning, lost original points

5. Adding new information
   These mistakes reduce quality

Prompts Decomposition

Prompt-1: Notes Generator(Step-1)

prompt

You are a study notes maker.

INPUT:
[CHAPTER_TEXT]

TASK:
Create notes from the provided chapter

RULES:
- Only use given chapter content.
- Do **not** add outside info
- Give 10 bullet points
- Simple English

OUTPUT FORMAT:
1. Key Notes(10 bullet points)

run

Run the above prompt by replacing "[CHAPTER_TEXT]" **with** actual content

Prompt-2: Summary Generator(Step-2)

prompt

You are a summary writer.

INPUT:
[NOTES_FROM_STEP_1]

TASK:
Convert the notes into a short summary

RULES:
Only use given notes
Keep it within 80 words
Simple English

OUTPUT FORMAT:
Summary:
(one paragraph)

run

Run above prompt by replacing "[NOTES_FROM_STEP_1]" **with** the output of the "Prompt-1"

Prompt-3: Quiz Generator(Step-3)

prompt

You are a MCQ quiz generator

INPUT:
[SUMMARY_FROM_STEP_2]

TASK:
Create 5 MCQs

RULES:
Each question must have 4 options
Mention correct answer at the end
Questions must be based only on the summary

OUTPUT FORMAT:
Q1. ...
A. ...
B. ...
C. ...
D. ...

Answers :
1. B
2. ...

run

Run above prompt replacing "[SUMMARY_FROM_STEP_2]" **with** the output of the "Prompt-2"

Prompt Pipeline : Book Chapter Study Assistant

Goal:
User gives chapter text.
AI gives Notes + Summary + Quiz.

Master Workflow Prompt

You are a Book Chapter Study Assistant.

INPUT:
[CHAPTER_TEXT]

STEP 1:
Create 10 bullet notes from the chapter.
Only use chapter content.

STEP 2:
From the notes, create a summary **in** 80 words.

STEP 3:
From the summary, create 5 MCQs **with** answers.

Keep everything **in** simple English.

Use clear headings:
Notes:
Summary:
Quiz:
Answers:

Questions

Q1: What **is** decomposition **in** prompt chaining?
Answer:
Decomposition means breaking a big AI task into small clear steps, where each step produces a n output **for** the next step.

Q2: What **is** the biggest mistake **in** prompt chaining?
Answer:
Mixing too many tasks **in** one prompt **and not** fixing the output format.
Jumping order **is** also main problem.

Section-3: Linear Multi-Step Pipeline

Linear pipeline means:
You have step1
step2 depends on step1

step3 depends on step2
It continues **in** order

Like a chain
Step1 output → Step 2 input

Format Locking between the steps:
We have to tell to the AI ==> "Give output only in this structure"
So next step can easily use it.

eg:
Step1 output should be:
Weak points
Improved version
Keywords list
If we lock these headings, Step2 can use smoothly

Demo Prompt Chaining

Resume –> Improve –> Professional rewrite –> Generate some interview questions

Step 1 Prompt: Resume Improver (Controlled)
You are a resume reviewer.

INPUT:
[RESUME_TEXT]

TASK:
Improve the resume content.

RULES:
Keep all information truthful.
Fix grammar.
Improve clarity.
Add strong action words.
Do **not** add fake experience.

OUTPUT FORMAT (must follow):
1. Weak Points (5 bullets)
2. Improved Resume (clean sections)
3. Top Keywords (10 keywords)

must follow ==> locking

run

Run the above prompt by replacing "[RESUME_TEXT]" **with** actual "resume text"



Sample Resume
Name: Rahul Kumar
Email: rahul123@gmail.com
Phone: 9999999999
Location: Hyderabad

Career Objective:
To work **in** a challenging environment where I can improve my skills **and** contribute to compan y growth.

Education:
B.Tech **in** Computer Science
XYZ Engineering College
2022
Percentage: 72%

Skills:
- Python
- Java
- SQL
- MS Office
- Communication skills

Projects:
1. Student Management System
   Developed using Python **and** MySQL.
2. Online Shopping Website

Internship:
Software Intern at ABC Technologies
Worked on web development tasks.

Experience:
Fresher

Strengths:
- Hardworking
- Quick learner
- Team player

Hobbies:
- Watching movies
- Playing cricket

Declaration:
I hereby declare that the above information **is** true to the best of my knowledge.

Step 2 Prompt: Professional Rewrite (Using Step 1 Output)

prompt

You are a professional resume writer.

INPUT:
[IMPROVED_RESUME_FROM_STEP_1]

TASK:
Rewrite the resume **in** a professional corporate style.

RULES:
  Keep headings clear.
  Use simple English.
  Keep it 1 page **if** possible.
  Do **not** change facts.

OUTPUT FORMAT:
  Professional Resume:
    (Write the final resume)

run

Run above prompt by replacing "[IMPROVED_RESUME_FROM_STEP_1]" **with** response from "Step-1"

Optional Step 3 (If you want extra pipeline):

prompt

You are an interviewer.

INPUT:
[PROFESSIONAL_RESUME]

TASK:
  Create 10 interview questions based on this resume.
  Include 2 HR questions + 8 technical/project questions.
OUTPUT FORMAT:
HR Questions:
1) ...
2) ...
Technical/Project Questions:
1) ...
2) ...
...

run

Run the above prompt by replacing "[PROFESSIONAL_RESUME]" **with** response from "Step-2"

Note

We can include JD also **in** the inputs
Rewrite the Resume reviewer again as follows including the "[JD_TEXT]"

prompt

You are a resume reviewer.

INPUT1:
[JD_TEXT]

INPUT2:
[RESUME_TEXT]

TASK:
Improve the resume content which **is** more compatible to the JD

RULES:
Keep all information truthful.
Fix grammar.
Improve clarity.
Add strong action words.
Do **not** add fake experience.

OUTPUT FORMAT (must follow):
1. Weak Points (5 bullets)
2. Improved Resume (clean sections)
3. Top Keywords (10 keywords)

Resume Upgrade Assistant

Goal:
User gives resume.
AI returns:

1. Weak points
2. Improved resume
3. Professional rewrite
4. Interview questions

Master Prompt

prompt

You are a Resume Upgrade Assistant.
INPUT:
[RESUME_TEXT]

STEP 1:
List 5 weak points **in** the resume.
Then create an improved resume.

STEP 2:
Rewrite the improved resume **in** professional corporate style.

STEP 3:
Generate 10 interview questions based on final resume.
(2 HR + 8 Technical/Project)

RULES:
Do **not** add fake experience.
Keep output structured **with** headings.
Use simple English

Questions

Q1: What **is** a linear pipeline **in** prompt chaining?
Answer:
It **is** a step-by-step workflow where each step output becomes the next step input, **in** a fixed or der.

Q2: Why do we use format locking?
Answer:
To keep output consistent **and** structured, so the next step can use it without confusion.

Q3: Give a simple real example of linear pipeline.
Answer:
Resume text → improve content → rewrite professionally -> Generate Interview Questions.

Q4: What **is** the biggest risk **if** we do **not** lock format?
Answer:
AI output may change structure, **and** next step prompt will fail **or** give messy result.



Section-4: Branching Pipelines(Decision Based Workflow)

It means:
Workflow has multiple paths. It chooses path based on a condition.

IF condition :
  Choose Route A
ELSE:
  Choose Route B

Controlled Redirection

Even **if** user gives unclear input, We redirect properly
To choose the route, our workflow can ask the questions.

Example : Resume classifier

Prompt-1: Classifier Prompt (Decision Step)

prompt

You are a resume analyzer.

INPUT:
[RESUME_TEXT]

TASK:
Decide **if** the candidate **is** a Fresher **or** Experienced.

RULES:
If experience **is** 0 years **or** no company name, label "Fresher".
If intern **and** company name, consider as "Fresher".
If experience **is** 1+ years **or** company/project work, label "Experienced".

OUTPUT FORMAT (only this):
Candidate_Type: Fresher OR Experienced
Reason: (1 line)

only this ==> locking

run

Run the above prompt by replacing "[RESUME_TEXT]" **with** actual "resume text"



Sample Resume - Fresher

Name: Rahul Kumar
Email: rahul123@gmail.com
Phone: 9999999999
Location: Hyderabad

Career Objective:
To work **in** a challenging environment where I can improve my skills **and** contribute to compan y growth.

Education:
B.Tech **in** Computer Science
XYZ Engineering College
2022
Percentage: 72%

Skills:
* Python
* Java
* SQL
* MS Office
* Communication skills

Projects:
1. Student Management System developed using Python **and** MySQL.
2. Online Shopping Website

Internship:
Software Intern at ABC Technologies
Worked on web development tasks.

Experience: Fresher

Strengths:

* Hardworking
* Quick learner
* Team player

Hobbies:

* Watching movies
* Playing cricket

Declaration:
I hereby declare that the above information **is** true to the best of my knowledge.



Sample Resume - Experienced
Name: Rahul Kumar
Email: rahul123@gmail.com
Phone: 9999999999
Location: Hyderabad

Career Objective:
To work **in** a challenging environment where I can improve my skills **&** contribute to company growth.

Education:
B.Tech **in** Computer Science
XYZ Engineering College
2022
Percentage: 72%

Skills:
* Python
* Java
* SQL
* MS Office
* Communication skills

Projects:
1. Student Management System developed using Python **and** MySQL.
2. Online Shopping Website

Experience:
Worked as Software Engineer at ABC Technologies
Worked on web development tasks.

Experience: 3+ years of experience

Strengths:
* Hardworking
* Quick learner
* Team player

Hobbies:
* Watching movies
* Playing cricket

Declaration:
I hereby declare that the above information **is** true to the best of my knowledge.

Prompt 2A: Fresher Interview Questions (Branch A)

Prompt

You are an interview trainer **for** freshers.

INPUT:
[RESUME_TEXT]

TASK:
Create 10 basic interview questions.

RULES:
4 HR questions
6 basic technical questions
Simple **and** beginner friendly

OUTPUT FORMAT:
HR Questions:
1) ...
2) ...
3) ...
4) ...

Technical Questions:
1) ...
2) ...

run

Run the above prompt by replacing "[RESUME_TEXT]" **with** actual "resume text"

Prompt 2B: Experienced Interview Questions (Branch B)

prompt

You are a senior interviewer.

INPUT:
[RESUME_TEXT]

TASK:
Create 10 interview questions **for** an experienced candidate.

RULES:
2 HR questions
4 project deep questions
4 scenario-based questions

Ask about decisions, trade-offs, problem solving

OUTPUT FORMAT:

HR Questions:
1) ...
2) ...

Project Questions:
1) ...
2) ...
3) ...
4) ...

Scenario Questions:
1) ...
2) ...
3) ...
4) ...

run

Run the above prompt by replacing "[RESUME_TEXT]" **with** actual "resume text"

Smart Interview Prep Assistant

Goal:
User gives resume OR job role details.
System gives customized interview questions.

Master Prompt

prompt

You are a Smart Interview Prep Assistant.
INPUT:
[RESUME_TEXT]

STEP 1:
Decide Candidate_Type: Fresher **or** Experienced.
Give reason **in** 1 line.

STEP 2:

IF Fresher:
Generate 10 basic interview questions (4 HR + 6 technical).

ELSE IF EXPEIRENCED:
Create 10 interview questions **for** an experienced candidate.
2 HR questions
4 project deep questions
4 scenario-based questions
Ask about decisions, trade-offs, problem solving

RULES:
Use simple English.
Keep questions clear.
Output must have headings.

run

Run the above prompt by replacing "[RESUME_TEXT]" **with** actual "resume text"

Questions

Q1: What **is** a branching pipeline?
Answer:
It **is** a workflow where next steps change based on a condition, like IF/ELSE.

Q2: Why branching **is** needed **in** AI workflows?
Answer:
Because different users have different needs.
Branching gives personalized **and** accurate output.

Q3: Give a simple example.
Answer:
If candidate **is** fresher → basic interview questions.
If experienced → scenario-based interview questions.

Q4: What **is** controlled redirection?
Answer:
When user input **is** unclear, we ask small questions **and** guide them to correct workflow.

Section-5 : Controlled Chaining (Also known as Stability Control)

When you run multi-step prompts you control each step to stay:
On same topic
In same format
With same rules
Without extra unwanted content

This **is** called "Controlled chaining"

We can implement controlled chaining **with** the following ways:
- Prevent topic drift
- Strict output passing
- Guardrails between steps
- Structured output between the chains

Prevent topic drift

Topic drift happens when AI:
- adds new unrelated points
- changes focus
- over-explains
- Mixes audience-types
- changes tone suddenly

We can prevent drift by:
- Clear goal **in** every step
- *"Do not add new information"* rule
- Use input from previous step only
- Lock headings **and** output format

Strict output passing

Passing only the necessary output from one step to next step, **not** full content, to avoid confusi on.

Guardrails between steps

By using guardrails like "use only given input", "do not add new info", **and** by locking output fo rmat.

Structured output between the chains

Example : SEO Extraction and LinkedIn post / Twitter Thread

Step 1 Prompt: SEO Extraction (Stable Output):

prompt

You are an SEO content assistant.

INPUT:
[BLOG_TEXT]

TASK:
Extract SEO-ready information.

RULES:
- Use only the blog input.
- Do **not** add new facts.
- Keep it simple English.

OUTPUT FORMAT (must follow strictly):
1. Blog Title (1 line)
2. SEO Summary (120 words max)
3. Primary Keywords (10)
4. Secondary Keywords (10)
5. Meta Description(150 characters max)

run

Run the above prompt by replacing "[BLOG_TEXT]" **with** "actual blog text"

Step 2 Prompt: LinkedIn Post (No Drift)

prompt

You are a LinkedIn content writer.

INPUT:
[SEO_SUMMARY]
[PRIMARY_KEYWORDS]

TASK:
Write one LinkedIn post.

RULES:
Use only the given inputs.
Do **not** add new information.
Audience: beginners + working professionals.
Keep it 120 to 160 words.
Add 3 bullet points.
End **with** one simple question.

OUTPUT FORMAT:
LinkedIn Post:
(Write here)

run

Run the above prompt by replacing "[SEO_SUMMARY]" **and** "[PRIMARY_KEYWORDS]" **with** "SEO Summary" **and** "Primary Keywords" of "OUTPUT FORMAT" response from "Step-1"

Step 3 Prompt: Twitter Thread (Strict Format):

prompt

You are a Twitter thread writer.

INPUT:
[SEO_SUMMARY]

TASK:
Create a Twitter thread.

RULES:
- Use only the summary.
- Do **not** add new facts.
- Total 6 tweets.
- Each tweet max 240 characters.
- Keep first tweet as a hook.

OUTPUT FORMAT:
Tweet 1:
Tweet 2:
Tweet 3:
Tweet 4:
Tweet 5:
Tweet 6:

run

Run above prompt replacing "[SEO_SUMMARY]" **with** "SEO Summary" of "OUTPUT FORMAT" response from "Step-1"

Content Repurposing Workflow Assistant

Goal:
- One blog input.

Create:
- SEO pack
- LinkedIn post
- Twitter thread
- Quality check report

Master Prompt

prompt

You are a Content Repurposing Workflow Assistant.

INPUT:
[BLOG_TEXT]

STEP 1:
Extract:
SEO Summary (120 words)
10 Primary Keywords
Meta Description

STEP 2:
Using only Step 1 outputs:
Write a LinkedIn post (120–160 words) **with** 3 bullets **and** 1 question.

STEP 3:
Using only SEO Summary:
Create a 6-tweet Twitter thread.

STEP 4:
Quality check:
Confirm no new facts added **and** no topic drift.

RULES:
Use only given blog text.
Keep outputs structured **with** headings.
Simple English.

Interview Questions and Answers

Q1: What **is** topic drift **in** prompt chaining?
Answer:
Topic drift means AI goes away from main topic **and** adds unrelated content **or** changes focus.

Q2: How do you prevent topic drift?
Answer:
By using guardrails like "use only given input", "do not add new info", **and** by locking output format.

Q3: What **is** strict output passing?
Answer:
Passing only necessary output from one step to next step, **not** full content, to avoid confusion.



Section-6: Variable Injection(Dynamic Placeholders)

Variable injection means

We keep placeholders **in** the prompt like:
{{USER_INPUT}}
{{TARGET_AUDIENCE}}
{{JOB_ROLE}}
[TEXT_HERE]

Later we replace placeholders **with** real values
Prompt **with** placeholders ==> Template Prompt

Advantages

Same prompt works **for** many topics
Output becomes consistent
Easy to automate using code
Easy to build pipelines **and** workflows

Static Prompt

Explain Python loops **for** beginners

Dynamic Prompt

Explain {{USER_INPUT}} **for** {{TARGET_AUDIENCE}}

Common Mistakes

Placeholder names should be meaningful
"{{TOPIC}}" ---> RECOMMENDED
"{{X}}" --> NOT RECOMMENDED

Forget to **pass** value
Defining too many variables

Example : Notes Generator Template (Reusable)

prompt

You are a study notes generator.

INPUT:
{{USER_INPUT}}

TARGET AUDIENCE:
{{TARGET_AUDIENCE}}

TASK:
Create notes **in** very simple English.

RULES:
Explain **in** daily life language.
Do **not** use complex words.
Keep it beginner friendly.
Add 1 small real-life example.
Do **not** add extra topics.

OUTPUT FORMAT:
1. Topic **in** 1 line
2. Key Notes (7 bullets)
3. Example (2–3 lines)
4. Quick Summary (3 lines)

Questions

Q1: What **is** variable injection **in** prompt engineering?
Answer:
It means using placeholders **in** a prompt template **and** replacing them **with** real values to reuse the prompt **for** many inputs.

Q2: Why do we use placeholders like {{USER_INPUT}}?
Answer:
To make prompts reusable, consistent, **and** easy to automate.

Q3: What **is** the difference between static **and** dynamic prompt?
Answer:
Static prompt **is** fixed **for** one topic.
Dynamic prompt uses variables **and** can work **for** many topics.

Note

"{{USER_INPUT}}" **for** variables which hold limited info
"[BLOG_POST]" **for** huge data

Prompt Serialization

The process of writing structured prompts **is** nothing but Prompt Serialization

Plain Prompts

Act as experienced Python Trainer
Generate 5 Python Interview Questions

Plain text

I am durga staying at Hyderabad **with** mail id durgasoftonline@gmail.com **and** mobile number: 8096969696



Structured text

Name: Durga
Location: Hyderabad
Mail Id: durgasoftonline@gmail.com
Mobile: 8096969696

Prompt Serialization means

It simply means :

How we structure **and** pack instructions before sending them to the AI model" i.e.,
instead of sending one long sentence, we organize the instructions in clear roles and structure

Roles in Prompt

A prompt can contain different roles
"system" ➔ Defines AI Behaviour
"user" ➔ Gives the task
"assistant" ➔ gives the answer

This structured way of writing prompts **is** called "prompt serialization"

Examples

Example-1 :

plain text prompt

Act as a strict interviewer
Generate 5 python interview questions

structured way

```
[
   {"role":"system","content":"You are a strict interviewer"},
   {"role":"user","content":"Generate 5 python interview questions"}
]
```

Run the above prompts

Python Program

```
from openai import OpenAI
import os
from dotenv import load_dotenv

load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client

client = OpenAI(api_key=OPENAI_API_KEY)

brain = "gpt-4.1-mini"

response = client.responses.create(
    model = brain,
    input = [
        {"role": "system", "content": "You are a strict interviewer"},
        {"role": "user", "content": "Generate 6 Python interview questions"}
    ]
)

# print AI response
print(response.output_text)
```

1. Explain the difference between Python lists and tuples. When would you use one over the ot
her?
2. How does Python manage memory allocation for variables and objects?
3. What are Python decorators, and how do you implement one?
4. Describe the Global Interpreter Lock (GIL) in Python. How does it affect multithreading?
5. How do you handle exceptions in Python? Provide an example demonstrating try-except-else-
finally blocks.
6. What are list comprehensions, and how do they differ from generator expressions? Provide u
se cases for each.

Example-2: One-Shot prompt

Plain prompt

Example:
Q1. What **is** the difference between List **and** Tuple?

Act as a strict interviewer:
Generate 5 python interview questions by considering the above example(one-shot prompt)

Run the prompt

structured prompt

```
[
    {"role":"system","content":"You are a strict interviewer"},
    {"role":"user","content":"Ask a python question"},
    {"role":"assistant","content":"Q1. What is the difference between List and Tuple?"},
    {"role":"user","content":"Ask another 5 python interview questions"}
]
```

Run the above prompt

Python Program - Structured Prompt

```
from openai import OpenAI
import os
from dotenv import load_dotenv

load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client

client = OpenAI(api_key=OPENAI_API_KEY)

brain = "gpt-4.1-mini"

response = client.responses.create(
    model = brain,
    input = [
        {"role": "system", "content": "You are a strict Oracle RAC Administrator."},
        {"role": "user", "content": "Ask Oracle RAC question"},
        {"role": "assistant", "content": "Q1. What is the difference between Local Listener and Scan Listener in RAC?"},
        {"role": "user", "content": "Ask all the important Oracle RAC Interview questions with Answers"}
    ]
)

# print AI response
print(response.output_text)
```

Certainly. Below are some important Oracle RAC interview questions along with concise, precise answers:

---

1. What is Oracle RAC?

Answer:
Oracle Real Application Clusters (RAC) is a cluster database with a shared cache architecture t hat allows multiple instances running on different servers to access a single database, providin g high availability, scalability, and load balancing.

---

2. What are the main components of Oracle RAC?

Answer:
- **Oracle Clusterware:** Manages node membership and messaging.
- **Shared Storage:** Typically ASM or clustered file system accessible by all nodes.
- **Oracle RAC Database Instances:** Multiple instances running on different nodes accessing t he same database.

---

3. How does Oracle RAC achieve high availability?

Answer:
Oracle RAC provides high availability by allowing multiple instances to run on separate nodes; if one instance or node fails, other instances continue to provide uninterrupted database servic e.

---

4. What is a SCAN in Oracle RAC and why is it important?

Answer:
SCAN (Single Client Access Name) is a unique name assigned to a cluster, resolving to multiple IP addresses. It provides a single point of connection for clients, allowing transparent node failo ver and load balancing without requiring client reconfiguration.

---

5. What is the difference between a Local Listener and SCAN Listener?

Answer:
- **Local Listener:** Listens on each RAC node, using the node's IP address and port. Client co nnections that connect directly to a specific node use this listener.
- **SCAN Listener:** Configured on the SCAN VIPs/IPs, it allows clients to connect to any node transparently, providing load balancing and failover.

---

6. How do Oracle RAC nodes communicate with each other?

Answer:
Oracle RAC nodes communicate using the **Interconnect**, a private high-speed network used for cache fusion, which synchronizes data blocks among nodes and maintains database consist ency.

---

7. What is Cache Fusion?

Answer:
Cache Fusion is Oracle RAC's mechanism to maintain data consistency across multiple instanc es by transferring modified data blocks via the high-speed interconnect when they are requeste d by another instance.

---

8. How do you monitor Oracle RAC?

Answer:
Using tools such as:
- Enterprise Manager (OEM) Grid Control
- CRSCTL commands for Clusterware status
- SRVCTL for RAC resource management
- Cluster Health Monitor (CHM) and trace logs
- SQL*Plus querying V$ views like V$INSTANCE, V$ACTIVE_INSTANCES, etc.

---

9. What is Node Eviction in Oracle RAC?

Answer:
Node Eviction is a process where the Clusterware forcibly removes a poorly functioning node fr om the cluster to preserve cluster integrity and stability.

---

10. What are the main reasons for node eviction?

Answer:
- Network failure or unreachable nodes
- Voting disk or OCR corruption
- Response timeouts or missed heartbeats
- Disk I/O issues or storage problems

---

11. What is a Voting Disk and what role does it play?

Answer:
Voting Disk is a cluster registry that tracks active nodes and votes on membership. It prevents split-brain scenarios by voting to make decisions on cluster node availability.

---

12. How many voting disks are recommended and why?

Answer:
Odd number of voting disks (minimum 3) spread across different storage to avoid single points of failure and ensure fault tolerance.

---

13. What is Oracle ASM and how does it fit into RAC?

Answer:
Oracle ASM (Automatic Storage Management) simplifies database storage management by hand ling disk management tasks and providing striping and mirroring for improved performance an d reliability; commonly used as shared storage for RAC databases.

---

14. Can Oracle RAC run without shared storage?

Answer:
No, RAC requires shared storage accessible by all nodes to maintain a consistent database acro ss instances.

---

15. How do you add or remove nodes to/from a RAC cluster?

Answer:
Using `addnode.sh` for adding nodes and `removenode.sh` (or manual procedure) for removing nodes, ensuring all pre-requisites and networking/storage configurations are aligned.

---

16. What is the difference between Oracle RAC 11g and 12c?

Answer:

Key differences include improvements in RAC scalability, introduction of Flex Cluster architect ure in 12c, enhanced ASM, improved Clusterware functionality, and support for multi-tenant a rchitecture.

---

17. How do you troubleshoot Oracle RAC slow performance?

Answer:
- Check interconnect latency and bandwidth
- Analyze cache fusion waits (gc* events)
- Monitor load balancing and services distribution
- Evaluate application SQL for contention and hot blocks
- Review cluster logs and trace files for anomalies

---

18. What is the purpose of CRSCTL and SRVCTL commands?

Answer:
- CRSCTL manages Clusterware: start, stop, status of cluster nodes and resources.
- SRVCTL manages database and listener components within a cluster.

---

19. How can you perform a rolling upgrade in RAC?

Answer:
Upgrade one node at a time without shutting down the entire cluster/database by stopping the Oracle instance on the node being upgraded while other nodes continue running.

---

20. What is the importance of role separation between Clusterware and Database Installati ons?

Answer:
Separating Clusterware (Grid Infrastructure) and Database installations aligns with Oracle's be st practices, improving manageability and supports independent upgrades and maintenance.

If you want, I can provide deeper explanations or further questions.

Python Program - Structured Prompt

```
from openai import OpenAI
import os
from dotenv import load_dotenv

load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client

client = OpenAI(api_key=OPENAI_API_KEY)

brain = "gpt-4.1-mini"

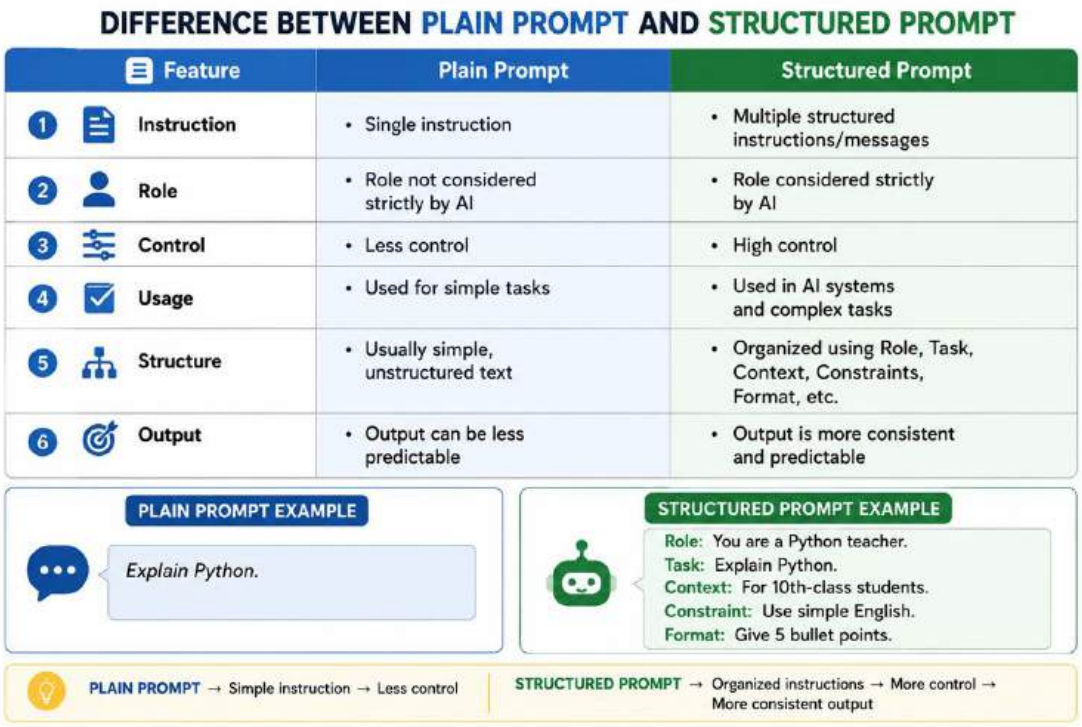
response = client.responses.create(
    model = brain,
    input = [
        {"role":"system","content":"You are a strict interviewer"},
        {"role":"user","content":"Ask a python question"},
        {"role":"assistant","content":"Q1. What is the difference between List and Tuple?"},
        {"role":"user","content":"Generate another 5 python interview questions"}
]
)

# print AI response
print(response.output_text)
```

1. Explain the difference between mutable and immutable data types in Python, with examples.
2. How does Python's garbage collection work?
3. What are Python decorators and how are they used?
4. Can you explain list comprehensions and provide an example?
5. What are *args and **kwargs in Python functions?

Difference between Plain prompt and Structured prompt

Feature	Plain Prompt	Structured Prompt
1. Instruction	Single instruction	Multiple structured instructions/messages
2. Role	Role is not considered strictly by AI	Role is explicitly defined and followed
3. Control	Less control over the output	High control over the output
4. Usage	Used for simple tasks	Used in AI systems and complex tasks
5. Structure	Usually simple, unstructured text	Organized using **Role, Task, Context, Constraints, Format**, etc.
6. Output	Output can be less predictable	Output is more consistent and predictable



plain_structured_prompt

What is advantages of structured prompts:

1. Strict behaviour control
2. Stable output
3. Multi-step conversations

eg: chatbots
AI assistants
Customer support bots

4. Safety rules

Instruction Hierarchy

AI follows instructions **in** this order:
Read system messages
Read user messages
Generate assistant response

Note:

System instructions have higher priority

Python Program

```
from openai import OpenAI
import os
from dotenv import load_dotenv
load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client
client = OpenAI(api_key=OPENAI_API_KEY)

brain = "gpt-4.1-mini"

response = client.responses.create(
    model = brain,
    input = [
        {
            "role":"system",
            "content":"""You are a strict interviewer

            Rules:
            - Ask only Python interview questions.
            - Do not provide answers.
            - Questions must be clear and practical.

            Constraints:
            - Generate exactly 5 questions
            - Difficulty level : Beginner to Intermediate
            - Each question must be one line only
            - Do not add explanations

            Output Format:
            Q1. <question>
            Q2. <question>
            Q3. <question>
            Q4. <question>
            Q5. <question>"""
        },
        {
            "role":"user",
            "content":"Generate Python interview questions"
        }
    ]
)
# print AI response
print(response.output_text)
```

Output

Q1. How do you define a function in Python and call it with arguments?
Q2. What are Python lists and how do you add or remove elements from them?
Q3. Explain the difference between `==` and `is` operators in Python.
Q4. How can you handle exceptions in Python using try-except blocks?
Q5. What is a Python dictionary and how do you iterate over its keys and values?

Questions
Question 1
What **is** Prompt Serialization?
Answer:
Prompt serialization **is** the process of writing structured prompts using roles such as system, u ser, **and** assistant.

Question 2
What **is** a system prompt?
Answer
A system prompt defines the behaviour, rules, **and** style the AI should follow.

Question 3
What **is** a user prompt?
Answer
A user prompt defines the task **or** request given to the AI.

Question 4
What **is** the difference between system message **and** user message?
Answer
System message controls behaviour.
User message controls the task.

Note:

An assistant prompt **is** the previous response generated by the AI, included **in** the conversation history using the "assistant" role to maintain context **in** multi-turn interactions.

Python code: Demonstrate message history

```
from openai import OpenAI
import os
from dotenv import load_dotenv
load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client

client = OpenAI(api_key=OPENAI_API_KEY)
```

```
brain = "gpt-4.1-mini"

# convesation history
messages = [
    {
        "role": "system",
        "content": """You are a strict Python interviewer.

        Rules:
        - Ask only Python interview questions.
        - Ask only ONE question at a time.
        - Do not give answers.
        - Wait for the user's confirmation before asking the next question.
        - Difficulty: Beginner to Intermediate

        Output format:
        Q<number>. <question>
        """
    }
]
for i in range(5):
    # ask question
    messages.append(
        {
            "role": "user",
            "content": "Ask the next Python interview question."
        }
    )

    # Call OpenAI
    response = client.responses.create(
        model = brain,
        input = messages
        )
    # Get the question
    question = response.output_text
    # Display question
    print(question)

    # store assistant response
    messages.append({
        "role": "assistant",
        "content": question
    })
```

```
    # wait for confirmation
    user_input = input("Type 'y' for next question: ")

    # Store user confirmation
    messages.append(
    {
        "role": "user",
        "content": user_input
    })
```

Output

Q1. What is the difference between a list and a tuple in Python?
Q2. How does Python's garbage collection work?
Q3. What are Python decorators, and how are they used?
Q4. Explain the difference between *args and **kwargs in Python functions.
Q5. What is a Python generator, and how is it different from a regular function?

messages

```
[{'role': 'system',
  'content': "You are a strict Python interviewer.\n\n        Rules: \n        - Ask only Python inter
view questions. \n        - Ask only ONE question at a time. \n        - Do not give answers. \n
- Wait for the user's confirmation before asking the next question. \n        - Difficulty: Beginner
to Intermediate\n\n        Output format:\n        Q<number>. <question>\n        "},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': 'Q1. What is the difference between a list and a tuple in Python?'},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': "Q2. How does Python's garbage collection work?"},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': 'Q3. What are Python decorators, and how are they used?'},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': 'Q4. Explain the difference between *args and **kwargs in Python functions.'},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': 'Q5. What is a Python generator, and how is it different from a regular function?'},
 {'role': 'user', 'content': 'y'}]
```

Alpaca Prompt Template:
Write something about science

Instruction:
Write 5 points about plants

Input:
Focus only on Earth **and** Mars

Output:
Students must write 5 simple points

What is Alpaca Prompt Template
It **is** structured way of writing prompts **for** AI models.
Introduced by "Standford researchers" when they "created the Alpaca LLM"
Instead of giving random prompts, we use a fixed prompt format.

Prompt Structure
3 parts
- Instruction
- Input
- Output/Response

Instruction
This tells the AI what task to do
Eg: Explain Python loops

Input
This provides extra context **or** details of the task
Eg: Explain **for** beginners **with** a simple example
This section can be empty **if** no extra context **is** needed

Output/Response
This **is** expected answer the model should generate

Example
Instruction:
Explain python loops

Input:
Explain **for** beginners **with** a simple example

Output:
A loop allows a program to repeat an action multiple times
For example, a **for** loop print numbers from 1 to 5

Example:
```
  for i in range(1,6):
    print(i)
```

Readaility

###Instruction:
Explain python loops

###Input:
Explain **for** beginners **with** a simple example

###Output:
A loop allows a program to repeat an action multiple times
For example, a **for** loop print numbers from 1 to 5

Example:
```
  for i in range(1,6):
    print(i)
```

Prompt general syntax
Instruction:
<task>

Input:
<context>

Response:
<answer>

Where Alpaca Template is used
Instruction tuning datasets
LLM training data
Fine-tuning open models
Local models like
  Llama
  Mistral
  Vicuna
  Alpaca

Note:

Alpaca template **is** mainly used **for** training datasets, **not for** normal chat prompts.



Interview Question:

1.What **is** Alpaca prompt template?

Ans:

The Alpaca template **is** a structured prompt format used **for** instruction tuning.
It contains three parts: instruction, input, **and** output.
This format helps AI models learn how to follow instructions more accurately.

ChatML Format(Role based Prompt)

ChatML means Chat Markup Language
It **is** a structured format used to represent conversations between users **and** AI
ChatML has 3 roles

"System" ==> "Behaviour Rules"
"User" ==> "Task Request"
"Assistant" ==> "AI Response"

Eg:

```
[
   {"role":"system","content":"You are Experienced Python trainer"},
   {"role":"user","content":"Explain Python Loops"},
]
```

Multi-Turn Converstaion

```
[
   {"role":"system","content":"You are Experienced Python trainer"},
   {"role":"user","content":"What is Python List?"},
   {"role":"assistant","content":"A list is a collection of items"},
   {"role":"user","content":"Give an example"}
]
```

Here the AI remembers previous conversation.
When user says " Give an example", the AI understands the context.
It knows we are talking about Python List.
This **is** called "Conversation memory"
"ChatML" supports this very nicely.

Python code : Multi turn conversation

```
from openai import OpenAI
import os
from dotenv import load_dotenv

load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')
```

```
# create client
client = OpenAI(api_key=OPENAI_API_KEY)

brain = "gpt-4.1-mini"

response = client.responses.create(
    model = brain,
    input = [
        {"role":"system","content":"You are Experienced Python trainer"},
        {"role":"user","content":"What is Python List?"},
        {"role":"assistant","content":"A list is a collection of items"},
        {"role":"user","content":"Give an example"}
    ]
)
# print AI response
print(response.output_text)
```

Output

Sure! Here is an example of a Python list:

```
```python
fruits = ['apple', 'banana', 'cherry', 'date']
print(fruits)
```
```

This creates a list called `fruits` containing four string elements. When you run this code, it will output:

```
```
['apple', 'banana', 'cherry', 'date']
```
```

Would you like to know how to add or remove items from a list?

Python code : Multi turn conversation

```
from openai import OpenAI
import os
from dotenv import load_dotenv
load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client

client = OpenAI(api_key=OPENAI_API_KEY)
```

```
brain = "gpt-4.1-mini"

# convesation history
messages = [
    {
        "role": "system",
        "content": """You are a strict Python interviewer.

        Rules:
        - Ask only Python interview questions.
        - Ask only ONE question at a time.
        - Do not give answers.
        - Wait for the user's confirmation before asking the next question.
        - Difficulty: Beginner to Intermediate

        Output format:
        Q<number>. <question>
        """
    }
]
for i in range(5):
    # ask question
    messages.append(
        {
            "role": "user",
            "content": "Ask the next Python interview question."
        }
    )

    # Call OpenAI
    response = client.responses.create(
        model = brain,
        input = messages
        )

    # Get the question
    question = response.output_text

    # Display question
    print(question)

    # store assistant response
    messages.append({
        "role": "assistant",
        "content": question
```

```
    })

    # wait for confirmation
    user_input = input("Type 'y' for next question: ")

    # Store user confirmation
    messages.append(
    {
        "role": "user",
        "content": user_input
    })
```

Output

Q1. What are Python's key features that make it a popular programming language?
Q2. How does Python handle memory management?
Q3. What is the difference between a list and a tuple in Python?
Q4. Explain the concept of Python decorators and provide an example of their use.
Q5. What are Python's lambda functions and where would you use them?
messages

```
[{'role': 'system',
  'content': "You are a strict Python interviewer.\n\n        Rules: \n        - Ask only Python inter
view questions. \n        - Ask only ONE question at a time. \n        - Do not give answers. \n
- Wait for the user's confirmation before asking the next question. \n        - Difficulty: Beginner
to Intermediate\n\n        Output format:\n        Q<number>. <question>\n        "},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': "Q1. What are Python's key features that make it a popular programming language?
"},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': 'Q2. How does Python handle memory management?'},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': 'Q3. What is the difference between a list and a tuple in Python?'},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': 'Q4. Explain the concept of Python decorators and provide an example of their use.'},
 {'role': 'user', 'content': 'y'},
 {'role': 'user', 'content': 'Ask the next Python interview question.'},
 {'role': 'assistant',
  'content': "Q5. What are Python's lambda functions and where would you use them?"},
 {'role': 'user', 'content': 'y'}]
```



Interview Questions and Answers:

Question 1:
What **is** ChatML?
Answer:
ChatML **is** a structured format used to represent conversations between user **and** AI using role s like system, user, **and** assistant.

Question 2:
What does system role do?
Answer:
System role defines behaviour rules **for** the AI.
Example: teacher, interviewer, mentor.

Question 3:
What does user role do?
Answer:
User role contains the task **or** request given to the AI.

Question 4:
What **is** assistant role?
Answer:
Assistant role contains the response generated by the AI model.

Question 5:
What **is** multi-turn conversation?
Answer:
Multi-turn conversation means continuing dialogue where previous messages are remembered as context.
By using assistant role AI can remembered context.

INST prompt template format

"INST" format **is** a special instruction format used **in** many "LLaMA based models"
LLaMA **is** an open-source large language model family.
In "INST format instructions are wrapped inside special tokens/tags"
eg: "<s>[INST]Explain Python Loops[/INST]</s>"

Note

Token	Meaning
<s>	Start of prompt
</s>	End of prompt
[INST]	Instruction Start
[/INST]	Instruction End

This helps the model understand where the instruction begins and ends.

Python Code

```
from openai import OpenAI
import os
from dotenv import load_dotenv
load_dotenv()

OPENAI_API_KEY = os.getenv('OPENAI_API_KEY')

# create client
client = OpenAI(api_key=OPENAI_API_KEY)
brain = "gpt-4.1-mini"

response = client.responses.create(
    model=brain,
    input="<s>[INST]Explain about INST Structured Prompt[/INST]</s>"
)
# print AI response
print(response.output_text)
```

Output

The INST Structured Prompt is a specific format used to communicate with AI models like me. I t typically involves enclosing the user's instructions or questions within a special tag or marker , such as `<s>[INST]...[/INST]</s>`. This format helps clarify where the user's prompt begins an d ends, allowing the AI to better understand and respond accurately.

For example, in your input:

```
<s>[INST]Explain about INST Structured Prompt[/INST]</s>
```

- `<s>` and `</s>` might denote the start and end of the overall input sequence.
- `[INST]` and `[/INST]` mark the instruction or prompt part specifically.

Using such structured prompts can improve the interaction by clearly defining prompts, instru ctions, or queries and separating them from any other surrounding text. This structure is espe cially useful in multi-turn conversations, fine-tuned AI models, or systems that process specific command-like inputs.

Where this is used in Real Projects

Open source LLaMA models
Custom AI Assistants
Local AI applications
Research experiments
Fine tuning pipelines
Developers working **with** open-source LLMs often use INST format.

Interview Questions:

Question 1:
What **is** INST format?
Answer:
INST format **is** a prompt structure where instructions are wrapped inside special tokens like [INST] **and** [/INST].
It **is** commonly used **in** LLaMA-based models.

Question 2:
Why do some models use instruction tokens?
Answer:
Instruction tokens help the model clearly identify where the instruction starts **and** ends.

Question 3:
What **is** the difference between ChatML **and** INST?
Answer:
ChatML uses roles like system, user, **and** assistant **for** conversations.
INST format wraps instructions inside special tokens **for** instruction-based models.

Question 4:
Name four prompt formats used **in** LLM systems.
Answer:
Plain prompts
Alpaca format
ChatML format
INST format

Example of all Formats

Plain Prompt

```
Explain importance of money?
```

ChatML Prompt

```
[
   {"role":"system","content":"You are a Financial Adviser"},
   {"role":"user","content":"Explain importance of money?"}
]
```

Alpaca Prompt

```
Instruction:
  Explain importance of money?

Input:
  For employees

Output:
  Provide in only 5 bullet points
```



INST Format

<s>[INST]Explain importance of money?[/INST]</s>

Format	Structure	Usecase
Plain	Simple Text prompt	Basic Prompting
ChatML	System/User/Assistant	Chat Assistants
Alpaca	Instruction/Input/Output	Instruction Tuning
INST	Instruction wrapped **in** tags	LLaMA models

Session-1: Prompt Minimalism

Eg

Mother:
Go to the nearby shop which **is** near the temple, **and if** the shop **is** open then please see **if** milk **is** available, **and if** milk **is** available **and is** fresh then buy one packet **and if not** available then come back, **and** also may be check biscuits **if** you remember.

Mother:
Buy one fresh milk packet from the nearby shop.

Prompt minimalism

It means writing prompts **in** a short, clean **and** clear way.
It does **not** mean making the prompt too small **and** incomplete
It means
- Remove extra words
- Keep imprtant meaning
- Give only needed information
- Avoid unnessary decoration

"Prompt Minimalism = Short + Clear + Clean + Useful Prompt"

Bad Prompt

"Can you please explain in detail with lots of examples and simple language about what python l oops are and how they work in programming for beginners?"

Good Prompt:

"Explain Python Loops with simple examples for beginners"

Advantages of Prompt Minimalism

Better Clarity
- AI understands direct instructions better
- If the prompt too long, then the main task may be hidden.

Less Confusion
Extra words create noise
Noise means unwanted words that do **not** help the task

Faster Prompt writing
Short prompts save the time

Better Consistency
Cleaner prompts often give more stable results

Lower token cost
Short prompts can save the cost

Examples

1. Can you please explain Python lists **in** every simple language **in** such a way that a beginner can understand properly **with** examples?

Minimal version:
Explain Python lists **with** simple examples **for** beginners.

2. Please generate **for** me some important python interview questions that may possibly be asked **for** a fresher candidate?

Minimal version:
Generate python interview questions **for** freshers.

3. Can you kindly summarize this article **for** me **in** a short **and** easy way so that I can understand it quickly?

Minimal version:
Summarize this article **in** simple words.

4. Please review the following python code carefully **and** tell me what mistakes are there **and** how I can improve it

Minimal version:
Review this python code **and** suggest improvements

Prompt cleaner Meta Prompt

You are prompt optimization assistant.
Your task **is** to convert long prompts into short, clean **and** clear prompts.
Do **not** lose the original meaning.
Remove repeated words, filler words **and** unwanted text.
Do the optimization **for** the prompt
[INPUT]
provide optimized prompt.

Run the above prompt by replacing "INPUT" with the "bad prompts" (above examples)

Interview Questions and Answers

Question 1:
What **is** prompt minimalism?
Answer:
Prompt minimalism means writing prompts **in** a short, clear, **and** precise way without unneces sary words.

Question 2:
What **is** prompt noise?
Answer:
Prompt noise means extra words that do **not** help the AI understand the task better.

Question 3:
Why can long prompts reduce quality?
Answer:
Long prompts may add confusion, repetition, **and** noise. This can reduce clarity **and** make output less focused.

Question 4:
Why **is** prompt minimalism important **in** industry?
Answer:
It improves clarity, saves time, increases consistency, **and** reduces token cost **in** large-scale AI systems.

Noise Removal

It means removing words **or** sentences that do **not** contribute to the task.
includes:
- repeated instructions
- irrelevant information
- emotional text
- unnecessary explanations

Before Noise Removal

Hi ChatGPT, I hope you are doing well today.
I was actually wondering **if** you could maybe explain Python Lists to me because I am trying to learn programming **and** it would be really helpful **if** you could provide a nice explanation.

After Noise Removal

Explain python lists **with** examples.

Token awareness + Cost thinking

While writing the prompts, we have to take care about number of tokens

Reason1: API Cost
More tokens --->More cost
Less tokens -->less cost

Reason2: Performance
Very large prompts take more time to process
Short optimized prompts can be faster.
This improves system performance.

Prompts
You are a prompt optimization assistant.

Your task **is** to analyze prompts **and** estimate token usage.

Input Prompt:
[INPUT]

Return:
1. Token Count
2. Optimized Prompt **with** fewer tokens

run

Run the above prompt by replacing "[INPUT]" **with** following "prompt_text"

Hi ChatGPT, I hope you are doing well today.
I was actually wondering **if** you could maybe explain Python Lists to me because I am trying to learn programming **and** it would be really helpful **if** you could provide a nice explanation.

Ref: Instareel : Prompt Optimization Example-1

prompt

Act as a prompt engineer review the following prompt **for** me optimize it to make it better **and** a sk me any questions you have before proceeding

"""Solve this problem step by step
A shopkeeper buys an item for Rs 500 and sells it for Rs 650.
Find profit percentage.

After solving:
1. Review your answer carefully.
2. Check if calculation is correct

3. If any mistake, correct it
4. Give final corrected answer clearly."""

Run the above prompt in ChatGPT

Prompt Optimization Example-2

prompt

Act as a prompt engineer review the following prompt **for** me optimize it to make it better **and** a sk me any questions you have before proceeding
"I want to learn agentic ai with examples can you please provide material"

run

Run the above prompt **in** ChatGPT

https://www.promptcowboy.ai/

MultiModal Prompting

MultiModal AI:
It means multiple modes of information
eg:
Text-->Question
Images
Videos
Audio files

Text -> Image prompting

Example-1

AI can create images from text descriptions

prompt

A cute puppy sitting **in** a garden, morning sunlight, realistic photo

run

Run the above prompt **in** ChatGPT

Example-2

prompt

A small boy studying under a tree, sunset lighting, peaceful village environment

run

Run the above prompt **in** ChatGPT

subject → A boy studying
location → under a tree
lighting → sunset lighting
mood → peaceful village

Example-3

prompt

A futuristic Hyderabad city **with** flying cars, neon lights, cyberpunk style

run

Run the above prompt **in** ChatGPT

Subject → City
Details → Flying cars
lighting → neon lights
style → cyberpunk

Text -> Video prompting

AI can create short videos from the text.

Example-1

prompt

A drone flying over a green forest, morning sunlight,cinematic view

Run the above prompt in Google Gemini

Example-2

prompt

A Businessman walking into a modern office, camera slowly following him, cinematic lighting

run

Run the above prompt **in** Google Gemini

Subject+action → Businessman walking
Environment → Modern office
Camera movement → camera following
style → cinematic lighting

Prompt Structure

Simple Structure
Subject + scene + Details

Better structure:
Subject + scene + lighting + style + camera

Example-3

prompt

A robot teacher teaching students **in** a modern classroom, bright lighting,futuristic style

run

Run the above prompt **in** Google Gemini

subject → Robot teacher
scene → classroom
lighting → bright lighting
style → futuristic style

Example-4

prompt

A girl reading a book **in** a library, soft warm lighting, realistic photo

Example-5

prompt

A farmer working **in** a green field during sunrise, cinematic photography

Example-6

prompt

A space station floating above Earth, stars **in** the background, sci-fi style

Example-7

prompt

A young successful entrepreneur **with** a joyful, proud smile relaxes **in** the garden of his beautif ul bungalow. Surrounded by lush green plants **and** a well-maintained lawn, he calmly sips a cup of tea **while** admiring his dream home **in** a peaceful morning setting.

Note

https://grow.google/ai-essentials
https://www.skills.google

Image Prompting with ChatGPT

If we give clear details, AI will generate very beautiful images

Image Prompt Structure

Subject + Environment/Scene/Location + Lighting + Camera + Style

Subject → Main object **in** the image
Environment → Where the scene happens
Lighting → How light appears
Camera → Camera angle **or** lens
Style → type of art **or** photography

Example-1

prompt

A small boy studying **in** a village house, warm evening sunlight, realistic photography

Note

Subject → A small boy studying
Environment → Village house
Lighting → warm evening sunlight
style → realistic photography

Better Details ➔ Better images

Subject

Subject is the main focus
Eg:
- Teacher
- Robot
- Student
- Dog
- Car

A robot teacher teaching students

Environment
Where the scene happens
Eg:
- Classroom
- Beach
- Forest
- City Street
- Office

A robot teacher teaching students **in** a modern classroom

Lighting
Lighting controls the feeling of the image
Eg:
- Softlighting
- Golder Hour
- Sunset Lighting
- Dramatic Lighting
- Neon Lighting

A robot teacher teaching students **in** a modern classroom, bright sunlight

Camera
Camera tells how the image **is** captured
Eg:
- Close-up shot
- Wide shot
- Drone shot
- Portrait shot
- Cinematic Shot

A robot teacher teaching students **in** a modern classroom, wide angle camera

Style
The style controls art style **or** photo style
Eg:
- Realistic Photography
- Cartoon Style
- Anime Style
- Oil Paintiinig
- Pixar Animation

A robot teacher teaching students **in** a modern classroom, realistic photography

Examples
A small boy studying **in** a village house, warm evening sunlight,portrait camera angle, realistic photography

A small boy studying **in** a village house,warm morning sunlight,wise camera angle, Oil painting

A small boy studying **in** a village house, neon lighting,wide camera angle, cartoon style

A cat sitting on a laptop **while** a programmer works, the location **is** deep forest, bird eye view, natural moon lighting

A spaceship landing on an alien planet, drone view

Product Prompt

Create Product Advertisement

A luxury smart watch placed on a black table, studio lighting, professional product photograph y

Landscape prompt:

A beautiful mountain landscape **with** a river flowing through green trees, sunrise lighting, wide angle photography

A hot coffee cup placed on a wooden table **in** a cozy cafe, morning sunlight though window, cinematic photography

Lighting Control

Lighting controls how the scene feels

Soft Lighting:

It creates a gentle and natural look

A young girl reading a book near a window, soft lighting, peaceful environment, realistic photog raphy

Goldern Hour Lighting

Just after sunrise **or** before sunset
This light looks warm **and** beautiful

prompts

A farmer walking through a green field during golden hour, warm sunlight, cinematic photography

A indian farmer walking through a green field during golden hour, warm sunlight, cinematic photography

Dramatic Lighting

It creates strong shadows and contrast

A superhero standing on a rooftop at night, dramatic lighting, cinematic style

Camera Angles

Camera angles tells AI how the viewer sees the scene1.

Close-up shot:

Focuses on one subject closely

prompt

close-up portrait of a confident enterpreneur, soft lighting, realistic photography

Wide shot:

shows the entire scene.

prompt

A wide shot of a beautiful mountain valley **with** a river flowing through it, sunrise lighting

Drone view:

It shows top angle aerial perspective

prompts

Drone view of a modern city **with** tall buildings **and** busy roads, sunset lighting

Drone view of a modern city **with** tall buildings **and** busy roads, sunrise lighting

Mood Control

It defines the emotional feeling of the image

Peaceful Mood

Used for calm and relaxing scenes

prompt

A small village near a river, morning mist, peaceful atmoshpere, soft lighting

Dramatic Mood

Used for intense scenes

prompt

A warrior standing on a battlefield, dark cloudsm dramatic mood, cinematic lighting

Futuristic Mood

Used for sci-fi or technology scenes

prompts

A futuristic city **with** flying cars **and** digital billboards,neon lights, cyberpunk style

A futuristic robot walking thought a neon city street at night, dramatic lighting,cyberpunk style

Program-Aided Language Models(PAL)

Video Prompting

We have to provide:
- What happens
- How the camera moves
- How the scene feels

Structure

Subject + Action + Camera Movement + Lighting + Style

Subject → Main character **or** object
Action → What **is** happening
Camera movement → How the camera moves
Lighting → how light appears
style → Cinematic look

Camera movement

It makes video dynamic **and** cinematic
Various possible options:

1. Slow zoom → Camera slowly zooms into subject
2. Tracking shot → Camera follows subject
3. Drone rise →Camera moves upward
4. Pan shot → Camera moves left **or** right

Examples

Run the prompts in "google gemini pro"

Example-1

A drone flying over tropical island with blue ocean water, camera slowly rising, bright sunlight, cinematic style

Example-2

A busy city street at night **with** cars moving **and** lights reflecting on wet roads, drone camera movement, cyberpunk style

Example-3
A couple walking on a beach, golden sunset lighting, camera slowly following them, cinematic romance scene

Example-4
A couple walking on a beach, golden sunset lighting, camera slowly following them, cinematic romance scene, south indian dressing style

Example-5
A student studying at a desk **with** books around, camera slowly zooming **in**, warm lighting, cinematic style

Example-6
A young programmer working late at night, computer screens glowing **in** a dark room, camera slowly moving closer, dramatic ligthting

Transformation Prompts

These prompts show one scene changing into another

Example-1
Luxury mansion garden transforming into a beautiful diamond shaped flower garden,drone view, golden hour lighting, cinematic timelapse

Example-2
A barista preparing coffee **in** a modern cafe, slow motion coffee pouring,warm lighting

50 Cinematics image Prompts + 50 Cinematic Video Prompts for Students

- ✓ https://higgsfield.ai/

50 Powerful Image Prompts

Portrait Style

1. A confident entrepreneur standing **in** a modern office, soft lighting, portrait photography
2. A young programmer working on a laptop **in** a dim room, glowing screens,cinematic lighting
3. A student studying late at night **with** books around, warm lamp lighting, realistic photo
4. A chef smiling **in** a restaurant kitchen, bright studio lighting, professional portrait
5. A scientist working **in** a futuristic laboratory, neon lights, cyberpunk style

Nature Scenes

6. A peaceful lake surrounded by mountains during sunrise, golden hour lighting, wide shot
7. A waterfall flowing through a dense green forest, soft morning light, cinematic photography
8. A desert landscape **with** sand dunes during sunset, dramatic lighting
9. A snowy mountain peak **with** clouds moving slowly, bright sunlight
10. A tropical island **with** blue water **and** palm trees, drone view

Technology Scenes

11. A futuristic AI laboratory **with** robots assisting scientists, neon lighting
12. A smart city **with** flying cars **and** digital billboards, cyberpunk style
13. A modern software engineer working **with** multiple screens, futuristic office
14. A humanoid robot teaching students **in** a high-tech classroom
15. A data center **with** glowing servers, dramatic lighting

City Life

16. A busy city street at night **with** neon lights reflecting on wet roads
17. A metro train passing through a modern city during sunset
18. A street food market full of colorful lights **and** people
19. A skyline view of a futuristic city **with** glowing buildings
20. A rainy evening **in** a city street **with** people holding umbrellas

Creative Scenes

21. A robot painter creating art on a canvas **in** a bright studio
22. A dragon flying over a medieval castle at sunset
23. A floating island **with** waterfalls **and** trees **in** the sky
24. A magical library filled **with** glowing books
25. A futuristic classroom **with** holographic screens

Food Photography

26. A hot coffee cup on a wooden table **in** a cozy cafe, warm lighting
27. A delicious pizza **with** melted cheese on a dark table, studio lighting
28. A bowl of fresh fruits on a kitchen counter, natural sunlight
29. A chef preparing sushi **in** a modern restaurant kitchen
30. A chocolate cake slice **with** strawberries, close-up photography

Lifestyle

31. A young woman jogging **in** a park during sunrise
32. A family enjoying a picnic **in** a green garden
33. A businessman walking confidently **in** a corporate office
34. A musician playing guitar **in** a cozy room **with** warm lights
35. A traveler standing on a mountain cliff looking at the horizon

Futuristic Concepts

36. A flying car above a futuristic city highway
37. A space station orbiting Earth **with** glowing lights
38. A Mars colony **with** astronauts working outside
39. A robot delivery drone flying above a smart city
40. A futuristic hospital **with** AI doctors

Creative Imagination

41. A giant glowing tree **in** a magical forest at night
42. A floating city above clouds during sunset
43. A glowing underwater city **with** marine life
44. A spaceship landing on an alien planet
45. A futuristic sports stadium **with** holographic displays

Fun Concepts

46. A dog wearing sunglasses riding a skateboard
47. A cat sitting on a laptop **while** a programmer works
48. A panda drinking coffee **in** a cafe
49. A robot cooking food **in** a kitchen
50. A baby dragon sleeping on a treasure pile

50 Cinematic Video Prompts

Nature Videos

1. A drone flying over a mountain valley during sunrise, cinematic travel style
2. A waterfall flowing through a lush forest, camera slowly moving forward
3. Waves crashing on a beach during sunset, golden lighting
4. A hot air balloon floating above green fields, aerial drone shot
5. Snow falling slowly **in** a quiet mountain village

City Videos

6. A drone flying through skyscrapers **in** a futuristic city, neon lights everywhere
7. A busy street at night **with** cars moving **and** lights reflecting on wet roads
8. A metro train entering a modern underground station
9. People walking through a crowded night market **with** colorful lights
10. A sunrise time-lapse over a modern city skyline

Inspirational Scenes

11. A young entrepreneur walking into a modern office, camera following behind
12. A student studying late at night, laptop glowing **in** a dark room
13. A runner jogging through a forest trail during sunrise
14. A motivational speaker addressing a large audience on stage
15. A startup team celebrating success **in** an office

Technology Videos

16. A futuristic AI laboratory **with** robots working beside scientists
17. A robot walking through a neon-lit city street
18. Autonomous cars moving on a smart city highway
19. Engineers building a humanoid robot **in** a lab
20. A drone delivering packages across a modern city

Travel Scenes

21. A drone flying above a tropical island **with** blue ocean water
22. A train passing through snowy mountains
23. A traveler walking through a desert during sunset
24. A boat moving across a calm lake surrounded by mountains
25. A road trip through a forest highway

Storytelling Scenes

26. A programmer coding late at night **while** rain falls outside
27. A young girl reading a book **in** a cozy room near a window
28. A chef preparing food **in** a busy restaurant kitchen
29. A musician performing on stage under bright lights
30. A photographer taking pictures of mountains at sunrise

Cinematic Transformation Scenes

31. A barren land transforming into a lush green forest, cinematic timelapse
32. A modern city transforming into a futuristic cyberpunk city
33. A plain garden transforming into a luxury flower garden
34. A small village transforming into a modern smart city
35. A sketch drawing transforming into a realistic painting

Sci-Fi Videos

36. A spaceship flying through space **with** glowing stars
37. A futuristic city **with** flying cars **and** digital billboards
38. A robot army marching across a desert planet
39. A space station orbiting Earth **with** sunlight shining
40. Astronauts exploring an alien planet landscape

Social Media Reels

41. A barista preparing coffee **in** slow motion **with** steam rising
42. A chef flipping vegetables **in** a frying pan **with** flames
43. A fitness trainer doing workout **in** a modern gym
44. A fashion model walking confidently **in** a studio photoshoot
45. A gamer playing on a neon-lit gaming setup

Creative Visuals

46. A glowing butterfly flying through a magical forest at night
47. A floating city above clouds **with** airships moving
48. A dragon flying across a fantasy kingdom
49. A glowing underwater world **with** colorful fish
50. Luxury mansion garden transforming into a diamond shaped flower garden, drone view, golden hour lighting

Viral Videos and Images

Case Study-1:

Simple Face Swap-Change only face, Keep everything same
Objective:
Take faces from image 1 **and** place them on image 2, without changing anything **else**.

Image Prompt

Use the first image only **for** the faces of the man **and** woman.
Take the second image as the main image.
Replace the faces **in** the second image **with** the faces from the first image.

Important Rules:
- Keep everything **in** the second image exactly the same
  - (hairstyle, clothes, jewellery, pose, lighting, background, expressions)
- For the man:
  - Keep body, moustache, hairstyle, dress **and** ornaments same
  - Change only the face
- For the woman:
  - Keep saree, jewellery, pose, hairstyle **and** styling same
  - Change only the face
- Do **not** change anything **else**
- Final result should look natural **with** proper skin tone **and** realistic blending
- The final image facial expressions should be same as second image.

run

Run the above prompt **in** "ChatGPT/ Google Gemini"

Object Talking Videos

Already availble Custom GPT we are going to use to generate image **and** video prompts
"Custom ChatGPT tools are mini AI assistants"
GPTs--> Explore GPTs --> Search **for** "Object Talk"(By aicenturies.com)
Run the below prompt

Prompt

Create a 16:9 video **with** 3 vegetables.
Each vegetable stands near **or** inside a large human body.
Each vegetable explains its health benefits through English dialogues.
Divide the video into different scenes (one vegetable per scene).
Convert each scene into text-to-image prompts **and** animation prompts.
Do **not** add subtitles **or** any extra text on screen.



Note

Text-to-Image Prompt ==> Run the prompt **in** ChatGPT
It will generate an image
Animation Prompt ==> Run **in** Google Flow
Upload the image generated by ChatGPT
Copy the "Animation Prompt"
Copy the "Script/Dialogue" Prompt also

Green Leaves Statue

Image Prompt-1 : Lord Sri Krishna

Run the below prompt in “ChatGPT/ Google Gemini”

prompt

Create an ultra-realistic documentary-style image featuring a rural Indian craftsman working on a life-size Lord Krishna statue made entirely from fresh green leaves.

Lord Krishna stands naturally **with** flute posture, peaceful smile, expressive eyes. Peacock-feat her crown, flute, **and** garments are all formed organically using woven green leaves **with** fine detail **and** texture.

The craftsman **is** seated nearby, hands mid-action weaving leaf strips into the flute **or** upper body area, fully focused.

Aspect ratio 9:16, eye-level shot, medium depth of field, soft sunlight, rural Indian village environment.

Ultra-realistic handcrafted look, no fantasy glow, no CGI – looks like a real documentary photo graph

Video Prompt-1

Take image generated from previous prompt and provide input to “google flow” and run the following prompt

Documentary-style realistic animation **in** natural daylight.

A person works continuously on seated green-leaf sculpture **with** slow,natural human motion: gentle hand weaving, tightening fibers, adjusting leaf strips, subtle body shifts, slight head mov ement, focused expression.

The sculpture remains completely static **and** unmoving at all times – no breathing, no leaf motion.

Smooth, slow cinematic camera orbit (180°–360°) at eye level, constant distance.
Subtle depth realism only.
Clean environment.
Realistic ambient sound.
No fantasy **or** cinematic effects.

Image Prompt-2 : Lord Shiva

prompt

Create an ultra-realistic documentary-style image featuring a rural Indian craftsman working on a life-size Lord Shiva statue made entirely from fresh green leaves.

Lord Shiva **is** seated calmly **in** a meditative posture, **with** serene facial expression, closed **or** half-open eyes, **and** balanced divine proportions.

The statue **is** fully woven from tightly layered green leaves – visible veins, stems, overlaps, **and** natural imperfections giving a living organic feel.

Trident (Trishul) **and** rudraksha details are subtly formed using leaf textures, **not** metal.

The craftsman, a middle-aged Indian village artisan, **is** seated beside the statue, hands mid-act ion weaving leaf strips onto the upper torso **with** intense focus **and** devotion.

Landscape 16:9, eye-level, medium-wide shot, 35–40mm natural lens, pure daylight, rural Indi an village background.

Ultra-photorealistic, handcrafted realism, sacred yet grounded – no CGI, no fantasy.

Video Prompt-2

Documentary-style realistic animation **in** natural daylight.

A person works continuously on a seated green-leaf sculpture **with** slow, natural human motion: gentle hand weaving, tightening fibers, adjusting leaf strips, subtle body shifts, slight head movement, focused expression.

The sculpture remains completely static **and** unmoving at all times – no breathing, no leaf motion.

Smooth, slow cinematic camera orbit (180°–360°) at eye level, constant distance.
Subtle depth realism only.
Clean environment.
Realistic ambient sound.
No fantasy **or** cinematic effects.



Image Prompt-3 : Lord Hanuman

prompt

Create an ultra-realistic documentary-style image of a rural Indian craftsman creating a life-size Lord Hanuman statue made entirely from fresh green leaves.

Lord Hanuman **is** seated **or** standing **in** a calm, powerful posture, **with** expressive eyes **and** devotional strength. Muscular form, facial features, **and** gada are shaped using layered green leaves **with** visible texture **and** imperfections.

The craftsman kneels beside the statue, weaving leaf strips onto the chest **or** arm area **with** dedication **and** focus.

Shot **in** Aspect ratio 9:16, natural daylight, rural village background, earthy tones.

Documentary realism, sacred handcrafted feel, no stylization.

Video Prompt-3

Documentary-style realistic animation **in** natural daylight.

A person works continuously on a seated green-leaf sculpture **with** slow, natural human motion: gentle hand weaving, tightening fibers, adjusting leaf strips, subtle body shifts, slight h ead movement, focused expression.

The sculpture remains completely static **and** unmoving at all times – no breathing, no leaf motion.

Smooth, slow cinematic camera orbit (180°–360°) at eye level, constant distance.
Subtle depth realism only.
Clean environment.
Realistic ambient sound.
No fantasy **or** cinematic effects.

Homework:

image1 **and** image2 should be provided explicitly.
image1 acts as craftman **and for** image2, statue **is** required.

create statue of Amitab
create statue of Sharukh
create statue of Donald Trump



3D PVC Statue Creation

Image Prompt-1 :

Run the following prompt in “Google Gemini”

prompt

A 1/7 scale figurine of a character from an unspecified "picture, rendered in a realistic style, made of PVC, and placed on a computer desk. It should have a round, transparent acrylic base without text.

Environment Details: The background should be indoors, and the computer screen should display the brush modeling process of the fiqurine.

Packaging Details: A BANDAl-style toy packaging box with original artwork and two-dimension al flat illustrations should be placed next to the computer screen. A model packaging box with the character printed on it should be behind the figurine.

Video Prompt-2 :

Upload the generated image in “google flow/higgsfield.in” and run the following prompt

prompt

A cinematic video of a person holding a realistic action figure toy (the one **in** the uploaded imag e), gently rotating it **in** their hands to show different angles.

The camera smoothly pans around the toy **in** slow motion, highlighting the detailed textures **and** design.

After rotating,the person playfully poses the toy on the desk **and** adjusts it slightly as **if** playing **with** it.

Soft, melodious instrumental background music plays throughout. No voiceover, just music.



Ugadi Wishes:

ChatGPT-1

prompt

I want to create a gemini video where godess Lakshmi has to wish "Hello Friends, DURGASOFT wishing you and your family members a very happy Ugadi!!!". create a gemini prompt

response

It will generate the prompt.
That prompt **is** given below
Take the prompt **and** run **in** the "Google Gemini"

Google Gemini-2

Run below prompt in "Google Gemini" by selecting "Create Video" option

prompt

A divine **and** beautiful scene of Goddess Lakshmi sitting on a fully bloomed pink lotus **in** a heavenly golden background. She **is** glowing **with** soft divine light, wearing a traditional red **an d** gold saree **with** rich jewellery, holding lotus flowers **in** her hands. Gold coins are gently flowing from her palm, symbolizing prosperity.

The environment **is** filled **with** soft floating petals, temple bells, **&** subtle sparkles.Background includes a peaceful temple atmosphere **with** warm sunrise lighting, creating a festive Ugadi mood.

Goddess Lakshmi slowly looks towards the camera **with** a gentle smile **and** speaks **in** a calm, divine voice:

Dialogue (clear **and** natural):
"Hello Friends, DURGASOFT wishing you and your family members a very happy Ugadi !!!"

Her lips should sync naturally **with** the dialogue. Expression should be warm, blessing, **and** graceful.

Camera movement: Slow cinematic zoom-**in** from medium shot to close-up.
Lighting: Soft golden divine glow **with** realistic shadows.
Style: Hyper-realistic, cinematic, devotional, 4K quality.
Aspect Ratio: 16:9

Audio Details:
Soft devotional background music (veena + flute)
Light temple bell sounds **in** background
Clear female divine voice, slightly echo effect

Additional Instructions:
No subtitles **or** on-screen text
Focus on facial expressions **and** lip-sync accuracy
Maintain cultural authenticity (Indian traditional style)
Smooth animation, no jerky movements

Add ending: Goddess Lakshmi raising hand **in** blessing (abhaya mudra)
Add slight glow increase at the end (feels divine)

ChatGPT-2 :

prompt

A south indian family celebrating Ugadi **with** Ugadi special chutney full festival environment fu ll josh. Generate the Gemini Prompt to wish **in** Kannada about Ugadi Wishes

response

It will generate the prompt.
That prompt **is** given below
Take the prompt **and** run **in** the "Google Gemini" by selecting "Create Video" option

Google Gemini-2

prompt

A lively South Indian family celebrating Ugadi festival inside a traditional house. The home entrance **is** decorated **with** fresh mango leaf torans, colorful rangoli (muggulu) at the doorstep, **and** oil lamps glowing warmly.

Family members are dressed **in** traditional attire:
Men wearing white dhoti **and** kurta
Women wearing bright silk sarees **with** flowers (mallige) **in** their hair
Children wearing festive traditional dresses

The family **is** happily preparing **and** serving Ugadi special chutney (Bevu-Bella) **in** a decorative bowl, symbolizing different flavors of life. Everyone **is** smiling, laughing, **and** enjoying together **with** full festive energy.

The atmosphere **is** vibrant, joyful, **and** filled **with** positive energy. Background includes soft morning sunlight entering the house, temple bells ringing lightly, **and** festive decorations.

One elder family member (**or** a smiling mother figure) looks at the camera **and** speaks warmly **in** Kannada:

Dialogue (Kannada):
"ನಮಸ್ಕಾ ರ ಸ್ನ ೇಹಿತರೇ, ನಿಮ್ಮೆಲ್ಲ ರಿಗೂ ಮತ್ತು ನಿಮೆ ಕುಟುಂಬದವರಿಗೆ ಉಗಾದಿ ಹಬಬದ ಹಾದಿೆಕ ಶುಭಾಶಯಗಳು!"

(Meaning: Hello friends, wishing you **and** your family a very happy Ugadi)

Lip-sync should be natural **and** accurate. Expression should be joyful, loving, **and** festive.

Camera & Style Details:
Camera: Smooth cinematic movements, slight pan across family → focus on speaker
Lighting: Warm natural sunlight + soft festive glow
Style: Hyper-realistic, cinematic, cultural, 4K quality
Aspect Ratio: 16:9

Audio Details
Background music: Soft South Indian festive music (veena + mridangam)
Light ambient sounds: family chatter, temple bells
Voice: Natural, warm Kannada tone

Additional Instructions
No subtitles **or** text on screen
Focus on emotions, family bonding, **and** festival vibe
Show close-up of Ugadi chutney preparation (Bevu-Bella mix)
Smooth **and** natural animation

Add ending scene: Whole family together saying “Happy Ugadi” **with** smiles
Add slow-motion moment of serving Bevu-Bella (very emotional touch)

ChatGPT-3:

prompt

Explain importance of Ugadi. I want to use this script to generate a video by using notebooklm

response

It will generate the prompt.
Take the prompt **and** run **in** the "NotebookLM"

NotebookLM-3
Create New Notebook

Select the language ==> "Settings ==> Output Language"

"Add Sources" ==> select "copied text" **and** paste the text generated by "ChatGPT"

From that we can generate "Audio, Slide Deck, Video Overview, Mind Map, Reports, FlashCards , Quiz, Infographic, Datatable"

Select "Video Overview" form the options **and** Choose Visual Style "Auto-select" **and** "Generate"



Motivational Video Song

Song Lyrics : ChatGPT
Image **and** Video Prompts **with** consistency: Chatgpt
Image generation by using image prompt: ChatGPT/Gemini
Video generation by using image **and** video prompt: Gemini/Flow
Audio song: SUNO AI
Combining multiple videos into a single video==>Video editor--->Filmora

Step-1 : Song Lyrics : ChatGPT

prompt

My students are worrying like anything about AI.They may lose their job, otherwise they may re quire to struggle because almost all software-related activities performed by AI tools only. Okay, so I want to give bharosa **for** my students. Can you please generate a motivational song l yrics **in** Telugu? I want to create one video song **for** them. I require lyrics **in** Telugu language **w ith** full motivation saying nothing will happen. Yes, you should have a creative mind. You shou ld have full confidence. Can you please provide Telugu lyrics **for** this motivational song?

response

ChatGPT provide the Lyrics
Ask "Style for the Audio song in Suno" ==> ChatGPT

prompt

To generate audio song **in** suno, which style I required to use

Motivational + Energetic Style

Step-2 : Audio : Suno.ai

Create ==> Audio ==> Lyrics
  Paste the above lyrics
    At the top include the following line
      "Theme: Motivating students not to fear AI, confidence, growth mindset"
    Styles ➔
      "Motivational Telugu Rap, Cinematic Inspirational Energetic, Uplifting, Strong Beats, Male Voice"
It will generate 2-versions of audio songs.
Save them as ".mp3" format(audio_1.mp3, audio_2.mp3)

Step-3 : Image and Video Prompts : ChatGPT

Take the “Lyrics generated by ChatGPT” in “Step-1” and add following prompt to the lyrics

prompt

Generate Image **and** Video Prompts **for** this audio song lyrics, consistency of the characters must be required

run

Run the prompt **in** `ChatGPT`
ChatGPT will generate Image, Video prompts **with** Scenes

Step-4 : Scene-1 - Image prompt

Run the below prompt in ==> “Google Gemini” it wll generates image. Save that image

Img1.png

Main Student Hero (Rahul type)
22-year-old Indian male
Medium brown skin
Short black hair
Clean shave
Wearing blue shirt + black jeans
Slightly serious face → later confident smile

AI Visualization
Futuristic glowing hologram interface
Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)
Indian male teacher
Formal shirt
Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity

A worried young Indian student sitting alone at a desk, head down, looking stressed, laptop open showing AI tools **and** code, dim lighting, cinematic mood, 22-year-old male, blue shirt, black jeans, realistic photography, 16:9

Step-5: Scene-1 - Video prompt

Run the below prompt in “Flow” by uploading the image(Img1.png) ==> “which is generated at Step-4”

Video-1

Main Student Hero (Rahul type)
22-year-old Indian male
Medium brown skin
Short black hair
Clean shave
Wearing blue shirt + black jeans
Slightly serious face → later confident smile

AI Visualization
Futuristic glowing hologram interface
Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)
Indian male teacher
Formal shirt
Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity

The student sits silently at a desk, looking worried, slowly touching his head, laptop screen glowing **with** AI interfaces, slight camera push-**in**, low light cinematic tone, emotional mood

It is going to generate the Video ==> save that video as “v1.mp4”

Step-6 : Scene-2 - Image Prompt
Run the below prompt **in** ==> "Google Gemini" by uploading the image(Img1.png) ==> "which is generated at Step-4"

It wll generates image. Save that image ➔ **Img2.png**

prompt

Main Student Hero (Rahul type)
22-year-old Indian male
Medium brown skin
Short black hair
Clean shave
Wearing blue shirt + black jeans
Slightly serious face → later confident smile

AI Visualization
Futuristic glowing hologram interface
Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)
Indian male teacher
Formal shirt
Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity
A young Indian student looking at a futuristic AI hologram **with** glowing blue interface, floating code **and** robotic visuals around him, dark background, fear expression, cinematic lighting, 16:9

Step-7: Scene-2 - Video prompt

Run the below prompt in “Flow” by uploading the image(Img2.png) ==> “which is generated at Step-6”

Video-2

prompt

Main Student Hero (Rahul type)
22-year-old Indian male
Medium brown skin
Short black hair
Clean shave
Wearing blue shirt + black jeans
Slightly serious face → later confident smile

AI Visualization
Futuristic glowing hologram interface
Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)
Indian male teacher
Formal shirt
Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity

Holographic AI visuals appear around the student, floating code **and** digital brain forming, student looks shocked **and** confused, slow motion, blue glow lighting

response

It will generate the Video ==> save that video as "v2.mp4"
Repeat the process **for** all the "Scenes"

Step-8 : Combine all Videos - Filmora

Create "New Project" ==> "16:9" ==> "Save"
Import the files ==> "audio_1.mp3, v1.mp4, v2.mp4"
Bottom part of the Filmora **is** "Timeline"
Drag **and** drop the video files to the "Timeline Video section" ➔ first "v1.mp4" **and** then "v2.mp4"
Actually **in** the videos "audio" also present. Change the audio to "0"(mute)
Drag **and** drop the audio files to the "Timeline Audio section" ==> first "audio_1.mp3"
After that "Export" the entire video **and** save "ai_fear.mp4"



1/7 scale image and video

Generate an AI Image : Google Nano Banana

prompt

I want very cute south indian lady image like sri leela telugu heroine **in** 9:16 **for** my youtube short.

Save the generated image. ➔ “image.png”

Image Prompt : Google Nano Banana

Upload the “generated image” (image.png) from above step and run the following prompt
prompt

Create a 1/7 scale commercialized figurine of (the character **in** the picture),realistic style, **in** 9:16 a real environment.

Figurine placed on a computer desk. has a round transparent acrylic base.

Next to the desk **is** the real person **in** the picture, **in** the real life size **with** the same attire as **in** the picture **and** the figurine, cleaning the figurine carefully **with** a fine brush, **in** a modern styled studio room, brightly lit. With some collection of toys **and** action figures **in** the background

Save the generated image as “image_generated.png”

Video Prompt : Gemini / Flow

Upload the “generated image” (image_generated.png) and provide the following prompt

prompt

A serene video of a girl **in** traditional attire, sitting at a desk, gently painting a miniature doll that resembles herself.

She has a slight, warm smile on her face as she carefully works on the details.

The camera slowly zooms **in** on her expressions **and** the doll, capturing the joy of creating.

Soft, melodious instrumental music plays **in** the background, adding a soothing **and** artistic at mosphere.

No voiceover, only music.

Save the generated video as “video1.mp4”

Video Prompt-2 : Gemini / Flow

Upload the “generated image” (image_generated.png) and provide the following prompt

prompt

A serene video of a girl **in** traditional attire, sitting at a desk, gently painting a miniature doll th at resembles herself.

She has to cry **and** face **with** full irritation **and** she carefully works on the details.

The camera slowly zooms **in** on her expressions **and** the doll, capturing the sad of creating.

Hard, horrible instrumental music plays **in** the background, adding a very rough **and** artistic at mosphere.

No voiceover, only music.

Save the generated video as “video2.mp4”

Dog image

AI image Generation : Google Gemini

prompt

Create image of very cute lady **with** bubbly face. main focus should be on face

Image Prompt : Google Gemini

Run this prompt **in** Google Gemini **and** save the generated image ==> "penguine_generated.png "

Replace "[CHARACTER/SUBJECT]" **with** any animal name / AI generated image ==> "Dog"

prompt

9:16, A hyper-realistic surreal portrait of [CHARACTER/SUBJECT]'s face emerging from a gloss y translucent liquid puddle on a smooth marble or reflective surface.

The liquid looks like clear gel, melted glass, or transparent silicone slowly flowing around the contours of the face and head.

The face appears suspended inside the liquid, with soft ripples, dripping edges, and glossy reflections forming natural fluid shapes around the cheeks and jawline.

The liquid creates realistic transparency & wet highlights with studio lighting reflecting across the surface.

The character's facial expression shows subtle surprise or curiosity, mouth slightly open, eyes looking forward.

Ultra-detailed skin texture, pores, reflections on glasses or eyes if present, cinematic lighting, soft shadows, shallow depth of field, photorealistic rendering, 8K detail, macro portrait photogr aphy, surreal liquid sculpture aesthetic.

Style Tags (optional):

hyperrealistic, surreal liquid sculpture, glossy gel texture, studio lighting, ultra detailed skin te xture, reflective highlights, cinematic portrait, macro photography, 8k, photorealism

Video Prmpt : Google / Flow

upload "generated image" (penguine_generated.png) from above step and run the following prompt

prompt

An oddly satisfying surreal ASMR macro video **in** 9:16 vertical format. A tight centered facial close-up of [CHARACTER/SUBJECT] resting flat on a polished white marble surface. The came ra **is** positioned **in** a direct top-down overhead view, completely locked **in** a macro shot **with** on ly very subtle natural micro-vibrations **for** realism.

The entire face **is** covered **in** a thick layer of crystal-clear elastic gel that **is** dense, glossy, ultra-transparent, highly stretchable, **and** softly wobbling. The gel evenly coats the face **while** still all owing the eyes, nose, lips, **and** skin details to remain clearly visible beneath it.

Soft diffused studio lighting creates beautiful translucent highlights, reflections, **and** refraction s across the gel surface. The background remains a minimal clean studio environment to keep the focus entirely on the face **and** gel texture.

Two realistic hands slowly enter the frame from opposite sides **and** begin interacting **with** the gel. The fingers gently press into the gel, performing slow rhythmic squishing motions. The han ds smoothly stretch **and** fold the gel over the contours of the face, followed by soft kneading mo vements that create
mesmerizing ripples **and** waves across the surface.

As the gel moves, the facial features distort smoothly underneath **in** a surreal but satisfying way. The eyes, nose, lips, **and** cheeks compress, stretch, **and** shift naturally beneath the transp arent gel. The material behaves **with** slow elastic physics, showing thick viscous movement, rea listic wobble, **and** soft rebound after
pressure **is** released.

The camera captures ultra-sharp macro texture details, including skin pores, gel thickness, glossy reflections, **and** subtle surface tension. The overall mood **is** hypnotic, oddly satisfying, ta

ctile ASMR **with** a surreal visual aesthetic, filmed **in** hyper-realistic high-definition macro cinematography

Save the generated video.

Negative Prompting

Negative prompting means telling AI what NOT to generate.
Normal Prompt = What you want
Negative Prompt = What you don't want

Normal Prompt

Generate a photo of a beautiful girl.

Problem: AI may add unwanted things like blur, extra fingers, bad lighting

With Negative Prompt

Run the below prompt in "Google Gemini"

prompt

Generate a photo of a beautiful girl.
Negative Prompt: blurry,low quality, extra fingers, distorted face, bad lighting.

Result ==> Much cleaner and better image

Common Negative prompt words

low quality
blurry
distorted
extra fingers
extra limbs
bad anatomy
deformed face
duplicate
watermark
text
logo
noise
overexposed
underexposed

Examples

Generate a premium real estate layout image, wide roads, green trees, modern entrance gate, clean plots, people walking, bright sunlight, high quality, aerial view.

Negative Prompt: blurry, low quality, empty land, dry grass, garbage, broken roads, no greenery , dark lighting, unrealistic building, distorted perspective, watermark, text, logo

Note

Always use negative prompts when:
Creation of AI images
Generating videos
Doing realistic faces
Product photography

Matchstick Shorts:

Matchstick Eagle :

ChatGPT : Explore GPTs
Search **for** "Matchstick shorts" : "By aicenturies.com"
Here we have to provide the object then it will generate.Eg: Eagle
Text-to-Image Prompt
Text-to-Video Prompt

Text-to-Image Prompt – Matchstick Eagle : Google Gemini

prompt

9:16 image strictly required, A highly detailed centered full-object shot of a giant eagle sculpture made entirely of red-tipped matchsticks, carefully handcrafted **and** realistic, placed on a clean white pedestal **in** a bright modern studio, the entire object fully visible **in** frame, a blurred person standing **in** the background behind it, bright high-key studio lighting, evenly illuminated scene, vibrant colors, soft shadows, shallow depth of field, sharp focus on the matchstick sculpture, well-lit environment,clean background, commercial product photography style, high exposure, crisp details

Save the generated image.

Text-to-Video Prompt – Burning Matchstick Eagle : Google Gemini/ Flow

Select the image which saved in previous step and run the following prompt

prompt

The shot begins **with** a hand pressing a lit match against a eagle sculpture made entirely of red-tipped matchsticks, causing instant ignition.

Flames spread rapidly across the sculpture **with** glowing embers **and** smoke.

The full object **is** visible **with** a blurred person **in** the background.



Mahabharath war

ChatGPT : Explore GPTs
- Search **for** "Time Travel Vlog" : "By aicenturies.com"
- Start Chat **and** upload "AI generated Image" **and** give "Mahabharath war"
  Text-to-Image Prompt
  Text-to-Video Promt

TEXT-TO-IMAGE PROMPTS : Google Gemini

Standing on the Battlefield Before the War Begins

upload the AI generated Image and runt he below prompt

prompt

9:16 strict image required,A realistic cinematic vlog-style front-facing phone shot of the charact er from the reference image **in** ancient India during the Mahabharata war at Kurukshetra battlefield.

Here **is** the reference image of the character.

The character **is** wearing the same modern outfit, clothing style, **and** overall appearance as **in** the reference image, as a believable time traveler inside the scene.

The shot **is** captured from the character's front-facing phone camera at arm's length, **with** the recording device **not** visible **in** frame **and** no separate camera shown **in** the hand.

The character **is** facing the lens **with** a tense,amazed expression,reacting to the massive armies assembling behind her.

The surrounding world must be accurate to the setting, **with** chariots, armored warriors, elephants, **and** war banners filling the horizon. In the background, show soldiers preparing we apons, horses being lined up, **and** dust rising from marching troops.

The character must feel naturally integrated into the environment **with** matching warm sunligh t, dusty atmosphere, consistent shadows, realistic skin tones, **and** cinematic color grading. Full color, immersive realism, high detail.

Save the generated image

Walking Among War Elephants and Soldiers

upload the "AI generated Image" and runt he below prompt

prompt

9:16 strict image required,A realistic cinematic vlog-style front-facing phone shot of the charact er from the reference image **in** ancient India during the Mahabharata war at a crowded military encampment.

Here **is** the reference image of the character.

The character **is** wearing the same modern outfit **and** appearance as a time traveler.

The shot **is** captured from the character's front facing phone camera at arm's length.

The character **is** facing the lens **with** a shocked yet curious expression, reacting to massive war elephants being prepared behind her.

The environment includes warriors **in** traditional armor, chariots, **and** campfires.

In the background, show handlers guiding elephants, soldiers sharpening weapons, **&** banners fluttering.

Natural golden-hour lighting, dusty air, **and** cinematic realism **with** detailed textures **&** depth.

Save the generated image

Arjuna and Krishna Preparing for Battle

upload the "AI generated Image" and runt he below prompt

prompt

9:16 strict image required,A realistic cinematic vlog-style front-facing phone shot of the character from the reference image **in** ancient India during the Mahabharata war near Arjuna's chariot.

Here **is** the reference image of the character.
The character **is** wearing the same modern outfit **and** appearance.

The shot **is** captured from the character's front-facing phone camera at arm's length.

The character **is** facing the lens **with** awe **and** disbelief, reacting to seeing Arjuna **and** Krishna behind her preparing **for** battle.

The background shows a divine chariot, warriors assembling, **and** a glowing early morning sky.

Subtle divine aura lighting, atmospheric depth, realistic integration, cinematic color grading, **&** high detail.

Save the generated image

Chaos as the War Begins

upload the “AI generated Image” and runt he below prompt

prompt

9:16 strict image required,A realistic cinematic vlog-style front-facing phone shot of character from the reference image **in** ancient India during the Mahabharata war at the moment battle erupts.

Here **is** the reference image of the character.

The character **is** wearing the same modern outfit.

The shot **is** captured from the character’s front-facing phone camera at arm’s length.

The character **is** facing the lens **with** fear **and** adrenaline, reacting to chaos unfolding behind her.

The background shows warriors charging, arrows flying, chariots colliding, **and** dust clouds rising.

Motion blur, dynamic lighting, smoke, **and** intense cinematic realism **with** immersive action.

Save the generated image

Aftermath of the Battlefield

upload the “AI generated Image” and runt he below prompt

prompt

9:16 strict image required,A realistic cinematic vlog-style front-facing phone shot of the charact er from the reference image **in** ancient India during the Mahabharata war at sunset after battle

Here **is** the reference image of the character.
The character **is** wearing the same modern outfit.
The shot **is** captured from the character’s front-facing phone camera at arm’s length.

The character **is** facing the lens **with** a somber, emotional expression, reacting to the aftermath behind her.

The background shows fallen warriors, broken chariots, **&** a quiet battlefield under an orange s ky. Smoke drifts slowly, **and** distant survivors move silently. Cinematic lighting, emotional tone , **and** realistic detail.

Save the generated image

TEXT-TO-VIDEO PROMPTS

The Armies Gather

prompt

Spoken Script:
"Okay… I don't think I was supposed to be here. This is the Mahabharata war… and those are e ntire armies behind me!"

A realistic cinematic vlog-style front-facing phone video of a character **in** ancient India during the Mahabharata war at Kurukshetra battlefield.

The character **is** speaking **with** a nervous, stunned expression **while** reacting to thousands of soldiers assembling.

Behind her, massive armies line up **with** chariots, elephants, **and** flags.

Dust rises as troops march, banners wave, **and** sunlight filters through the haze.

The camera subtly shakes as she turns slightly to show the scale of the battlefield before refocusing on her face.

Among War Elephants

prompt

Spoken Script:
"You guys… those are actual war elephants. Like real ones… and they're getting ready for battle right now."

A realistic cinematic vlog-style front-facing phone video of a character **in** ancient India at a war camp.

The character speaks **with** awe **and** disbelief **while** reacting to elephants being prepared behin d her.

Background movement includes handlers guiding elephants, soldiers loading weapons, **&** fires flickering.

The camera pans briefly to reveal the massive animals before returning to her face.

Dust particles float **in** warm sunlight, creating cinematic atmosphere.

Witnessing Legends

prompt

Spoken Script:
"I think… that's Arjuna… and Krishna is right there. This feels unreal… like I'm inside history."

A realistic cinematic vlog-style front-facing phone video of a character near Arjuna's chariot.

The character speaks **in** awe **while** reacting to legendary figures behind her.

Krishna stands calmly **while** Arjuna prepares his bow. Subtle divine glow surrounds them.

Background includes warriors organizing **and** horses shifting.

The camera slightly turns to reveal the chariot,then back to her face, capturing her amazement

Battle Erupts

prompt

Spoken Script:
"It's starting! Oh my god—arrows everywhere—this is insane, I need to get out of here!"

A realistic cinematic vlog-style front-facing phone video during the start of battle.

The character speaks **with** panic **and** adrenaline as chaos erupts. Behind her, warriors charge, arrows fly, **and** chariots crash.

Dust clouds **and** smoke fill the air.

The camera shakes **with** movement as she turns to show the action before snapping back to her face.

Intense motion **and** sound implied visually.

Aftermath Reflection

prompt

Spoken Script:
"I… I don't even know what to say. This place was full of life hours ago… and now it's just silence."

A realistic cinematic vlog-style front-facing phone video after the battle at sunset.

The character speaks softly **with** a somber expression **while** reacting to the aftermath.

Behind her, broken chariots, fallen warriors, **and** drifting smoke fill the quiet battlefield.

Survivors move slowly **in** the distance. The camera gently pans to show the devastation before returning to her emotional face.

Warm sunset light casts long shadows, enhancing the cinematic mood.

Sequence of execution
First run the "Text-to-image" prompt ==> save the image
Next run the "Text-to-video" prompt ==> by uploading the image generated **in** previoue step

Building Timelapse:

Prompt1 (Image) : Google Gemini

Upload an image of any building ==> “Eg: Burj AI Arab”

prompt

Remove all existing structures from the image so that no part of the building remains.

The area should be transformed into a vacant, undeveloped land parcel.

The surface must look even **and** flattened, showing natural earth textures such as soil, sand, **a nd** scattered gravel, similar to a plot prepared **for** future construction.

Ensure the scene contains no architectural features – no walls, roofs, windows, pillars, **or** built elements of any kind.

Save the generated image ==> before.png

Prompt2 (Video) : Gemini/ Flow

Upload the previously generated image “before.png” and “original image”

prompt

Use the first image as the starting frame: empty artificial island **with** only the base platform.

Create a realistic construction time-lapse of a Burj Al Arab-style tower built step by step: found ations **and** columns → central core rising floor by floor → floors slabs added level by level → white arches assembled **in** segments → glass facade **and** cladding installed from bottom to top → helipad **and** rooftop details → palm trees, paths, lights, cars, people.

No instant popping **or** morphing of the building – every stage should look like fast real construction **with** moving cranes **and** workers.

Add animated ocean waves around the island **and** a slow, subtle zoom-**in** of the aerial camera.

Keep cinematic, photoreal, high detail **and** smooth natural light changes over time.

Save the generated video

Floor Transformation GPT

ChatGPT : Explore GPTs

Search **for** "Floor Transformation" : "By aicenturies.com"

Transform home interiors into cinematic timelapse prompts featuring artistic luxury epoxy flo ors.

Open the GPT **and** give the prompt ==> `How to use it`

Choose your room

- Kitchen
- Living Room
- Garage
- Bedroom
- Bathroom
- Dining Area
- Home Office
- Studio Apartment
- Retail Interior
- Luxury Showroom

Select “Studio Apartment” It will genereate “Image Prompts” and “Video Prompts”

Image Prompts(Same Camera, Same Layout)

Run all the prompts ==> “Gemini”

Image-1 : EMPTY / UNDER CONSTRUCTION

prompt

9:16 image strictly required, A wide-angle cinematic view of a compact studio apartment under construction.

The space has raw concrete flooring, unfinished walls **with** exposed plaster, **and** construction dust scattered across the ground.

Small debris, buckets, **and** tools lie around. Natural light enters from a single window. No furniture.

Static camera, consistent framing **for** all stages.

Save the generated image as “img1.jpeg”

Image-2 : MID-CONSTRUCTION

prompt

9:16 image strictly required, The same studio apartment from the exact same camera angle.

Workers wearing PPE are actively preparing the floor, using grinding machines **and** tools.

The concrete floor **is** partially smoothed **with** visible preparation marks.

Ladders, equipment, **and** buckets are present.

Walls are semi finished. Natural daylight fills the room.
Realistic construction activity frozen **in** time.

Save the generated image as “img2.jpeg”

Image-3 : COMPLETED EPOXY FLOOR

prompt

9:16 image strictly required, The same studio apartment **with** a completed luxury metallic epoxy floor.

The floor features a high-gloss marble swirl design **with** deep charcoal, silver, gold, **and** hints of electric blue pigments flowing naturally.

The surface **is** mirror-like **and** reflective.

The room **is** empty, clean, **and** minimal.

Soft cinematic lighting enhances reflections. No furniture. Same camera position.

Save the generated image as “img3.jpeg”

Image-4 : FURNISHED FINAL ROOM

prompt

9:16 image strictly required, The same studio apartment fully furnished **in** a modern luxury style.

A compact sofa, minimalist bed, stylish coffee table, indoor plants, **and** warm ambient lighting are added.

The glossy metallic epoxy floor remains clearly visible beneath the furniture, reflecting light **&** decor.

Cozy, high-end atmosphere. Same exact camera angle **and** layout.

Save the generated image as “img4.jpeg”

Video Prompts

Run all the prompts ==> “Flow”

Video 1 — Construction Begins (Image 1 → 2)

Upload “img1.jpeg” and “img2.jpeg” and run the prompt

prompt

A cinematic time-lapse of a studio apartment under construction from a fixed camera angle.

Workers enter the space wearing PPE **and** begin grinding the concrete floor using heavy machin es.

Dust moves across the floor as tools operate.

Buckets **and** ladders shift position as workers actively prepare the surface.

Natural lighting subtly changes over time. Real human motion, no teleportation.

Save the generated video ==> “v1.mp4”

Video 2 — Epoxy Transformation (Image 2 → 3)

Upload “img2.jpeg” and “img3.jpeg” and run the prompt

prompt

A realistic transformation sequence where workers pour metallic epoxy onto the prepared floor.

Thick liquid pigments **in** charcoal, silver, gold, **and** blue are spread **and** manipulated using tool s.

The epoxy flows **and** self-levels, forming natural marble-like swirls.

Workers smooth the surface, then gradually leave the frame.

The floor becomes glossy **and** reflective. Static camera, cinematic realism.

Save the generated video ==> “v2.mp4”



Video 3 — Human Furnishing Timelapse (Image 3 → 4)

Upload “img3.jpeg” and “img4.jpeg” and run the prompt

prompt

A fully realistic human-driven timelapse inside the completed studio apartment.

Workers **and** designers walk into the empty space carrying furniture pieces.

They assemble a bed frame, place a mattress, position a sofa, adjust a coffee table, **&** arrange decor items like plants **and** lamps.

Each item **is** physically carried, rotated, **and** carefully positioned by hand. Lighting **is** turned on manually.

No objects appear instantly.

The epoxy floor reflects every movement. Camera remains completely static.

Save the generated video ==> “v3.mp4”

Combine all Videos - Filmora

Create "New Project" ➔ "9:16" ➔ "Save"
Import the files ➔ "v1.mp4, v2.mp4, v3.mp4"
Bottom part of the Filmora **is** "Timeline"
Drag **and** drop the video files to the "Timeline Video section" ➔ first "v1.mp4" then "v2.mp4" **&** " v3.mp4"
Actually **in** the videos "audio" also present. Change the audio to "0"(mute)
After that "Export" the entire video **and** save "floor.mp4"

Bunker Timelapse AI Studio GPT

ChatGPT ==> Explore GPTs ==> search "Bunker Timelapse AI Studio"
Procedural bunker timelapse engine + viral strategist
By Zapiwala AI
Explore how to use this

PAL Prompting

Most people are asking AI like this...
"Give me answer..."

But smart people like you ask like this...
"Write code and solve..."

This **is** called PAL prompting.



Core Idea

PAL mean ➔ Program-Aided Language Prompting
Meaning : Dont ask AI to directly Answer
Ask AI to write program **and** solve
Language ➔ Code ➔ Answer

Examples

Example-1

Normal Prompt

What **is** 25 x 48 + 120?

AI may guess

PAL Prompt

Write python program to calculate 25 x 48 + 120?

No guessing
Full Logic
Accurate

Example-2

Problem:
Buy price = Rs 30000 per square yard
Sell price = Rs 38000 per square yard
Size = 200 Sq Yards
Write python program to calculate total profit?

Where to use PAL

Calculations
Finance
Real Estate
Data Analysis
Business Planning
Automation

Note:

Where ever maths **is** there, where ever calculations are required , there we can use PAL prompting

Where NOT to use

Stories
Motivations
Marketing Content



Note:

If no maths **or** calculations then **not** recommended to use PAL prompting

PAL prompting structure:

Basic:
Write a Python program to + [your problem]

Advanced
Write a Python program to solve the following problem
[details]
Show step-by-step output

Examples

Example-1

Normal Prompt

Problem:
Loan = Rs 1000000
Interest = 10% yearly
Time = 3 years
What **is** EMI?

PAL prompt

Write a python program to calculate EMI for a loan of Rs. 1000000 with 10% annual interest for 3 years

Example-2 : Business monthly revenue
Problem:
You earn:
Jan : 1 Lakh
Feb : 1.2 Lakh
Mar : 1.5 Lakh

Normal Prompt

What **is** the total **and** average revenue?

PAL prompt:

Write a python program to calculate total **and** average revenue
for monthly incomes [100000,120000,150000]

Example-3 : Student Ranking System

Marks:
Rahul = 85
Asha = 70
Aravind = 90
Find topper

PAL prompt

Write a python program to find the student **with** highest marks from a dictionary?

Note:

General prompt **is not** scalable where as PAL prompt **is** scalable

Example-4 : Investment growth

Problem:
Rs 2 Lakh
15% returns
30 years
What **is** the final amount

PAL prompts

Write a python program to calculate compound growth of 2 lakh at 15% **for** 38 years

Write a python program to calculate compound growth of 2 lakh at 15% **for** 38 years.
Explain step by step

Micro Camera Animals GPT

ChatGPT : Explore GPTs

Search **for** "Micro Camera Animals" : "By aicenturies.com"
Create ultra-realistic wildlife documentary prompts **for** tiny animals wearing micro-cameras, **with** instant image, video, **and** angle variations.
Open the GPT **and** give the prompt ==> `Start`

Choose a tiny animal **for** the mounted micro-camera experiment:
Ant
Termite
Field mouse
Shrew
Mole cricket
Beetle
Vole
Harvest mouse
Burrowing spider

Centipede
Millipede
Pygmy gerbil
Naked mole-rat
Dung beetle
Juvenile ground squirrel
Reply **with** the number of the animal you want, **or** type more **for** 15 new animals.

Select “snake” It will generate Prompts

Surface Setup Image : Gemini

prompt

9:16 strict image, Ultra-realistic macro wildlife Indian research photograph of a field researcher crouched beside a narrow soil crevice **in** a dry grassland, carefully holding a small burrowing snake **while** attaching a tiny micro-camera securely onto the upper body just behind the head using a miniature soft harness, the camera aligned exactly **with** the snake’s forward direction, natural scale accuracy, detailed snake scales, subtle muscle tension **in** the body, dry soil texture, sparse, small rocks, natural daylight, no underground view yet, no fantasy, no cinematic styling, raw scientific documentation realism

Save the generated image as “simg.png”

Mounted POV Entering the Burrow : Flow

Upload the image “simg.png” and run the following prompt

prompt

Ultra-realistic mounted micro-camera POV from a small burrowing snake slithering forward into a narrow underground crevice, the camera **is** physically strapped just behind the head **and** faces exactly where the snake faces, 5 to 10 percent of the snake’s scaled body visible at the bottom of frame, natural side-to-side slithering motion causing subtle lateral camera sway, no stabilization, the snake pushes into the tunnel, daylight rapidly disappears, a tiny mounted LED becomes the only light source, narrow harsh beam revealing compact soil walls, fine roots, shifting dust particles, slight scraping sounds as scales contact dirt, darkness beyond the bea m, raw scientific field footage, no music, no narration, only natural sounds.

Save the generated video ➔ “sv1.mp4”

Tight Tunnel Navigation : Flow

prompt

Ultra-realistic mounted micro-camera POV from a small burrowing snake navigating a tight wi nding tunnel, camera fixed behind the head, moving exactly **with** each muscular contraction, 5 to 10 percent of the snake’s body visible, frame subtly compresses **and** shifts as the snake pres ses against tunnel walls, LED light flickers slightly over uneven soil, roots brushing past lens,

occasional dirt particles falling, slight body friction sounds, no stabilization, darkness surrounding the narrow beam, realistic underground structure **with** branching paths, raw field recording, no music, no narration

Save the generated video ➔ “sv2.mp4”

Encounter With Other Burrow Occupants : Flow

prompt

Ultra-realistic mounted micro-camera POV from a small burrowing snake entering a shared underground passage, camera fixed to body, forward-facing, slight body sway motion, LED beam reveals insects **and** small burrowing organisms reacting **and** moving away, tunnel walls a live **with** subtle movement, fine debris shifting as the snake advances, 5 to 10 percent of the snake visible at the bottom, slight jolts as the body brushes against walls, natural micro sound s of soil friction **and** tiny creatures, no stabilization, no artificial lighting beyond the mounted LED, raw scientific realism

Save the generated video ➔ “sv3.mp4”

Chamber Exploration : Flow

prompt

Ultra-realistic mounted micro-camera POV from a small burrowing snake entering a slightly wider underground chamber, camera remains fixed behind the head, 5 to 10 percent of body visible, the snake pauses **and** lifts slightly causing a gentle upward tilt, LED beam reveals uneven chamber walls, compacted soil, small cavities, organic debris, possible prey traces, dust suspended **in** the air, subtle body movements cause micro vibrations, no stabilization, darkness beyond the beam, natural ambient underground sounds only

Save the generated video ➔ “sv4.mp4”

Deep Tunnel Descent : Flow

prompt

Ultra-realistic mounted micro-camera POV from a small burrowing snake descending deeper into a vertical sloping tunnel, camera fixed to the body **and** moving **with** each contraction, slight downward tilt as the snake angles its head, LED beam illuminating loose soil **and** falling particles, walls becoming more compact **and** moist, increased resistance causing subtle camera jolts, 5 to 10 percent of snake visible, no stabilization, no external light, raw underground exploration **with** natural soundscape only

Save the generated video ➔ “sv5.mp4”

Combine all Videos - Filmora

Create "New Project" ➔ "9:16" ➔ "Save"

Import the files ==> "sv1.mp4, sv2.mp4, sv3.mp4, sv4.mp4, sv5.mp4"
Bottom part of the Filmora **is** "Timeline"
Drag **and** drop the video files to the "Timeline Video section" ➔ "sv1.mp4, sv2.mp4, sv3.mp4, sv4.mp4 and sv5.mp4" ➔ **in** the same order
After that "Export" the entire video **and** save "snake.mp4"

Project

ChatGPT
I Want to create a video about IPL 2026 just 50 seconds **in** telugu.
Provide audio script **for** this

Generated Script

IPL 2026 – 50 సెకన్ల తెలుగు ఆడియో స్క్రిప్ట్
ఇది మీ వీడియో కోసం సరైన, ఎనర్జిటిక్ వాయిస్‌ఓవర్ స్క్రిప్ట్:

"ఇది క్రికెట్ పండుగ... ఇది ఇండియన్ ప్రీమియర్ లీగ్ 2026!"

స్టేడియాలలో గోల, అభిమానుల కేకలు...
ప్రతి మ్యాచ్ ఒక థ్రిల్... ప్రతి బంతి ఒక కథ!

ఈ సీజన్‌లో స్టార్ ప్లేయర్లు, యువ టాలెంట్...
అందరూ ఒక్కచోటే లక్ష్యం — ట్రోఫీ!

చెన్నై, ముంబై, బెంగళూరు, హైదరాబాద్...
ప్రతి టీమ్ తమ శక్తిని చూపించడానికి సిద్ధంగా ఉంది.

సిక్సర్లు, ఫోర్లు, అద్భుత క్యాచ్‌లు...
క్రికెట్ అభిమానుల కోసం పూర్తి ఎంటర్‌టైన్‌మెంట్!

ఈ IPLలో ఎవరు హీరో అవుతారు?
ఏ టీమ్ ఛాంపియన్ అవుతుంది?

ఇది కేవలం గేమ్ కాదు... ఇది ఒక భావోద్వేగం!
IPL 2026...

మిస్ అవ్వకండి!

Flow

select image of "csk jersy" from internet ==> save it as "csk.png"
Generate an "AI image" ==> save it as "ai_image.png"
upload the images "csk.png and ai_image.png" **and** run the below script(prompt)

change the outfit to CSK jersey 2026 **and** add mike **in** the hand

Save the generated image ➔ "csk2.png"

Heygen
https://www.heygen.com/en-**in**
Generate the video by providing "csk2.png and the script generated by ChatGPT"